DISCRETE STRUCTURES & THEORY OF LOGICS (BCS303)

UNIT -1 : (i) Set Theory & Relations

(ii) POSET & Lattices

Lec-1

Today's Target

- What is set?
- Standard sets
- Number system

Gateway Classes : 7455 961284

Course Details(Paid) : All Subjects

1	Recorded Video Lectures (100 % Syllabus Coverage)
2	Pdf Notes
3	Lecture wise DPP
4	Unit wise set of PYQs
5	Course will be completed by 'Last week of Nov 2023'
	Tentative AKTU Semester Exam- Jan/Feb

Gateway Classes: 7455 961 284

Paid Courses are available in Gateway Classes Application

Link in Description

SET

SET: A Set is a Well-defined collection of objects

Well-defined collection: Collection does not change person to person

Not Well-defined collection: Collection can be change person to person

Gateway Classes : 7455 9612 84

Example-1 The collection of all odd natural number. less than 10

Abhishek - 1, 3, 5, 7, 9
Sachin - 1, 3, 5, 7, 9
Wilson - 1, 3, 5, 7, 9

So, this collection is well defined and therefore, it is a set

Example-2 The collection of top 3 batsmen in the World

Abhishek - Sachin, Rahul Dravid, Ponting
Sachin - Sachin, Dhoni, Smith
Wilson - Virat, Dhoni, Devilliers

So, this collection is not well defined and therefore, it is not a set.

NOTE- Every set is a collection but every collection is not a set.

Example- Group of good cricket players is a collection but it is not a set.

Gateway Classes : 7455 9612 84

$$A = \{a, e, i, o, u\}$$

Name of Set

Elements or member or objects of set

Sets are usually denoted by capital letters, A, B, C, X, Y, Z etc.

The elements of a set are represented by small letters a, b, c, y, z etc.

NOTE

(i) a belongs to A

$a \in A$

(ii) b does not belong to A

$b \notin A$

Where ε = epsilon

Gateway Classes : 7455 9612 84

Some Standard Sets

N: The set of all natural numbers

W : Set of whole numbers

Z or I : The set of all integers

Q: The set of all rational numbers

R: The set of real numbers

C: The set of all complex number

I^+ **or** Z^+**:** The set of positive integers

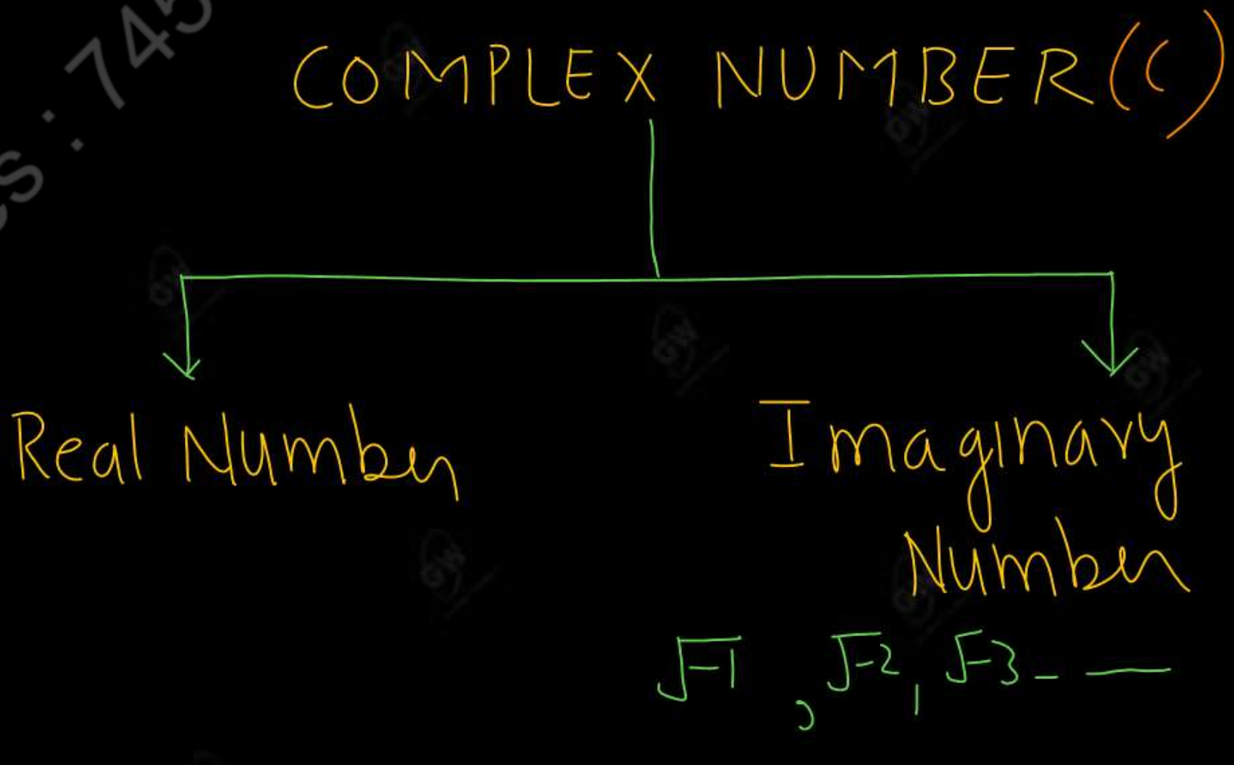
Q^+**:** The set of positive rational numbers.

R^+**:** The set of positive real numbers.

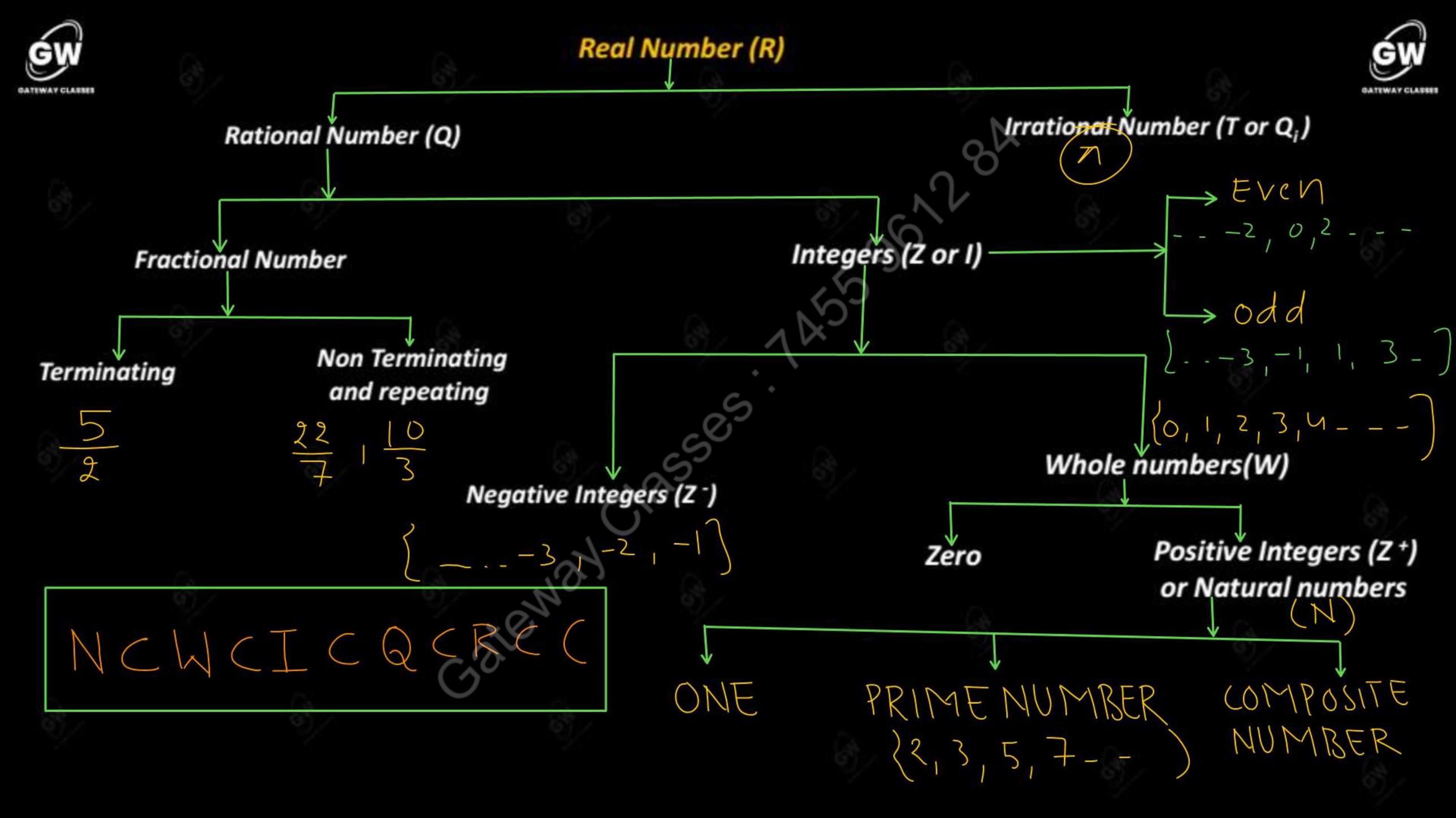
$N = \{1, 2, 3, 4 ___ \}$

$W = \{0, 1, 2, 3 ___ \}$

$I = \{___ -3, -2, -1, 0, 1, 2 ___ \}$



Gateway Classes : 7455 961284



Gateway Classes : 7455 9612 84

Rational Numbers: - These are the real numbers which can be expressed in the form of $\frac{p}{q}$, where p and q are integers and $q \neq 0$.

Example-

Irrational Numbers: - (Non-terminating and non repeating decimal) : These are the real numbers which can not be expressed in the form of $\frac{p}{q}$, where p and q are integers and $q \neq 0$.

Example- $\pi, \sqrt{2}, \sqrt{3}, \sqrt{5}, \sqrt{6}, \ldots$

Non Positive Integers : $0, -1, -2, -3, \ldots$

Non Negative Integers : $0, 1, 2, 3, \ldots$

©Gateway Classes 7455 9612 84

GATEWAY CLASSES

AKTU Full Courses (Paid)

Download Gateway Classes Application

From Google Play store

All Subjects

Link in Description

Gateway Classes : 7455961284

Thank You

DISCRETE STRUCTURES & THEORY OF LOGICS (BCS303)

UNIT -1 : (i) Set Theory & Relations

(ii) POSET & Lattices

Lec-2

Today's Target

- *Representation of Set*
- Types of sets

Gateway Classes : 7455 961284

Course Details(Paid) : All Subjects

1	Recorded Video Lectures (100 % Syllabus Coverage)
2	Pdf Notes
3	Lecture wise DPP
4	Unit wise set of PYQs
5	Course will be completed by 'Last week of Nov 2023'
	Tentative AKTU Semester Exam- Jan/Feb

Gateway Classes: 7455 961284

Paid Courses are available in Gateway Classes Application

Link in Description

Representation of Set

- Roster Form Or Tabular Form
- Set – builder Form or Rule Method

Gateway Classes: 7455 9612 84

Roster Form or Tabular Form

In roster form, all the elements are being separated by commas and are enclosed with in curly brackets { }

Examples

(1) The set of all natural numbers which divide 42

$N = \{1, 2, 3, 6, 7, 14, 21, 42\}$

(2) The set of all prime number less than 20

$A = \{2, 3, 5, 7, 11, 13, 17, 19\}$

(3) The set of odd natural number .

$N = \{1, 3, 5, 7 \ldots\}$

(4) The set of Binary digits.

$A = \{0, 1\}$

Note: (i) All infinite sets can not be written in roster form

Example: The set of real number can not be written in roster form.

(ii) In roster form the order in Which the elements of a set are written is not important .

Example: $A = \{1, 2, 3\}$ $B = \{2, 1, 3\}$ Here A and B are same set

(iii) In roster form, an element is not generally repeated, i.e all elements are taken as distinct

Example: $A = \{1, 2, 3\}$

$B = \{1, 2, 3, 2, 1\}$ Here set A and set B are same

Gateway Classes : 7455 9612 84

Set – Builder Form

In Set – builder form, all the elements of a set possess a single common property which is not possessed by any element out side the set .

such that

$$A = \{x : p(x)\}$$

$N = \{1, 2, 3 - - -\}$

Where $P(x)$ is the property by which $x \in A$

Note:- *A vertical bar is also used in place of colon (:)*

Examples

(i) The set of all natural number less than 10 .

$N = \{x : x \in N \text{ and } x < 10\}$

(ii) The set of Real number .

$R = \{x : x \text{ is a real number}\}$

(iii) The set of natural number .

$N = \{x : x \text{ is a natural number}\}$

Gateway Classes : 7455 9612 84

Types of Set

1. Empty Set / Null Set/ Void Set

- A Set having no element in it.
- It is denoted by { } or ϕ

Examples

① $A = \{x : x \in N, 4 < x < 5\} = \phi$

② $A = \{x : x^2 = 9, x \in \text{Even number}\} = \phi$

2. Singleton Set

A Set containing exactly one element is called a singleton set.

Examples

① $A = \{5\}$

② $B = \{x : x^2 = 4, x \in N\} = \{2\}$

Gateway Classes 7455 9612 84

Cardinal Number of a Set

OR

Cardinality of a set

The number of distinct elements containing in any set A is called cardinal number of set A

It is denoted by n (A) or |A|

Examples

① $A = \{2, 4, 6, 8\}$

$n(A) = 4$

② $A = \{\phi\}$

$n(A) = 1$

$B = \{1, 2, 3, 2, 4, 1\}$

$n(B) = 4$

Gateway Classes : 7455 961284

Finite and infinite Set

3. Finite Set - A Set is said to be finite if it contains finite number of elements .

Examples ① $A = \{1, 2, 3\}$

② The set students in a class

4. Infinite Set - A Set is said to be infinite if it contains infinite number of elements .

Examples

(a) The Set of natural number

$N = \{1, 2, 3, 4 - - - \}$

(b) The Set of integers

$Z = \{- - - -3, -2, -1, 0, 1, 2, 3 - - - - \}$

Gateway Classes : 7455 9612 84

5. Equal Set : *Two Sets A and B are said to be equal, if they have exactly the same elements and we Write*

$$A = B$$

Examples $A = \{1, 4, 6, 7\}$

$B = \{4, 1, 6, 7\}$

6.Equivalent Set : *Two finite Sets A and B are said to be equivalent if*

$$n(A) = n(B)$$

Examples

$A = \{2, 4, 6, 8\}$

$B = \{a, b, c, d\}$

Gateway Classes 7455 9612 84

Subset and Super Set

7. Subset : a set A is said to be a subset of set B, if every element of A is also an element of B

$$A \subseteq B$$

Examples $A = \{2, 4, 6\}$

$B = \{1, 2, 3, 4, 5, 6\}$

$A \subseteq B$

Note: (i) If A is not a subset of B $A \not\subseteq B$

(ii) Null set is the subset of all set $\phi \subseteq A$

(iii) Every set is a subset of itself $A \subseteq A$

The total number of subset of a finite set containing n elements = 2^n

Gateway Classes : 7455 9612 84

Q.1 Write down all possible subsets of

(i) $A = \{a, b\}$

(ii) $B = \{1, 2, 3\}$

(i) $A = \{a, b\}$

Subsets of A

$\{a\}, \{b\}, \{\}, \{a, b\}$

(ii) $B = \{1, 2, 3\}$

Subsets of B

$\{\}, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{2, 3\}, \{1, 3\}, \{1, 2, 3\}$

Gateway Classes : 7455 9612 84

8. Super Set : *If $A \subseteq B$, the B is called a superset of A and we write*

$$B \supseteq A$$

Examples

$A = \{a, b, c\}$

$B = \{a, b, c, d\}$

$A \subseteq B$

$B \supseteq A$

Gateway Classes : 7455 9612 84

9. Proper subset: If $A \subseteq B$ and $A \neq B$, then A is called a proper subset of B and we write

$$A \subset B$$

Examples $A = \{1, 2\}$

$B = \{1, 2, 3\}$

Note (1) $\boxed{N \subset W \subset Z \subset Q \subset R}$

(2) Set A is not proper subset of A

The total number of proper subset of a finite set containing n elements $= 2^n - 1$

Gateway Classes : 7455 9612 84

10. Power Set

- The collection of all subsets of a set A is called the power set of A
- It is denoted by P(A)

$m = 3$

In P(A), every element is a set

If $n(A) = m$ then $n[P(A)] = 2^m$

Q.1. Write down the power set of A = {1, 2, 3}

$P(A) = \{ \{\}, \{1\}, \{2\}, \{3\}, \{1,2\}, \{2,3\}, \{1,3\}, \{1,2,3\} \}$

$n[P(A)] = 8$

Gateway Classes : 7455 9612 84

11. Universal Set:- Collection of all elements under consideration is called universal set

Example

$A = \{1, 2, 3\}$

$B = \{2, 3, 4, 5\}$

$C = \{4, 5, 6, 7\}$

$U = \{1, 2, 3, 4, 5, 6, 7\}$

Gateway Classes : 7455 9612 84

Interval as subsets of R: If a, b are real numbers such that $a < b$

(1) Closed interval

$$[a,\ b] = \{x : x \in R \ \ and \ \ a \leq x \leq b\}$$

(2) Open interval

$$(a, b) =]\,a, b\,[= \{x : x \in R \ \ and \ \ a < x < b\}$$

(3) Semi-open or semi closed interval

(i) $[a,\ b) = \{x : x \in R \ \ and \ \ a \leq x < b\}$

(ii) $(a,\ b] = \{x : x \in R \ \ and \ \ a < x \leq b\}$

Gateway Classes : 7455 9612 84

GATEWAY CLASSES

GATEWAY CLASSES

AKTU Full Courses (Paid)

Download Gateway Classes Application

From Google Play store

All Subjects

Link in Description

Gateway Classes : 7455 961 284

Thank You

$\supseteq \subset \not\supseteq$

DISCRETE STRUCTURES & THEORY OF LOGICS (BCS303)

UNIT -1 : (i) Set Theory & Relations

(ii) POSET & Lattices

Lec- 3

Today's Target

Operations on Sets

Gateway Classes : 7455 961284

Course Details(Paid) : All Subjects

1	Recorded Video Lectures (100 % Syllabus Coverage)
2	Pdf Notes
3	Lecture wise DPP
4	Unit wise set of PYQs
5	Course will be completed by 'Last week of Nov 2023'
	Tentative AKTU Semester Exam- Jan/Feb

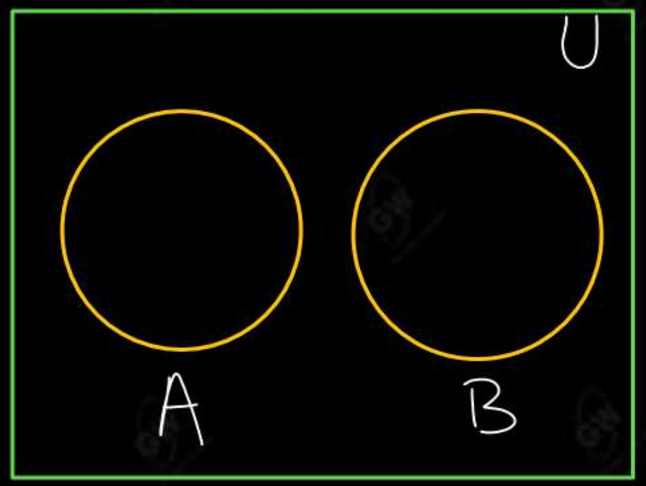
Gateway Classes: 7455 961284

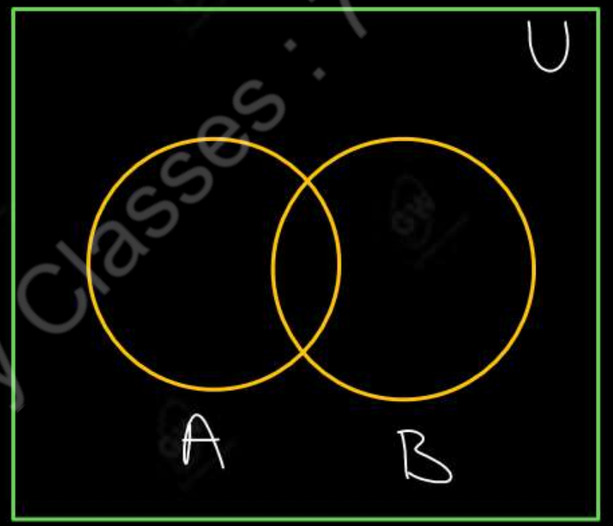
Paid Courses are available in Gateway Classes Application

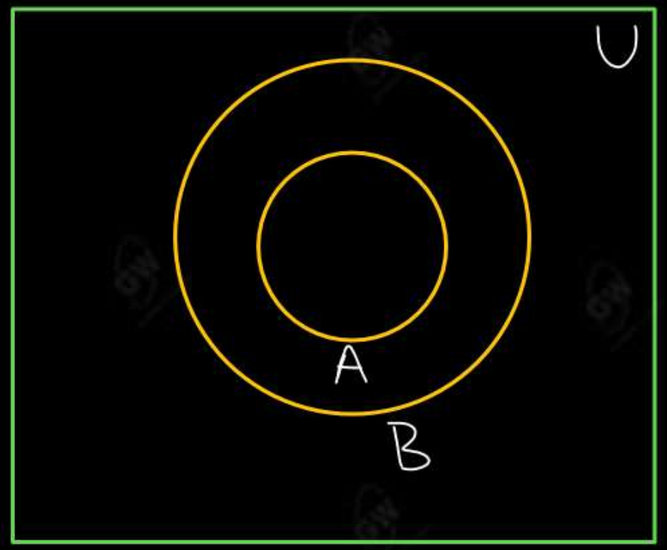
Link in Description

Venn Diagram

- Most of the relationships between sets can be represented by means of diagram which are known as Venn Diagram
- The universal set is represented by the interior of a rectangle and its subsets are represented by circular areas drawn within the rectangle.







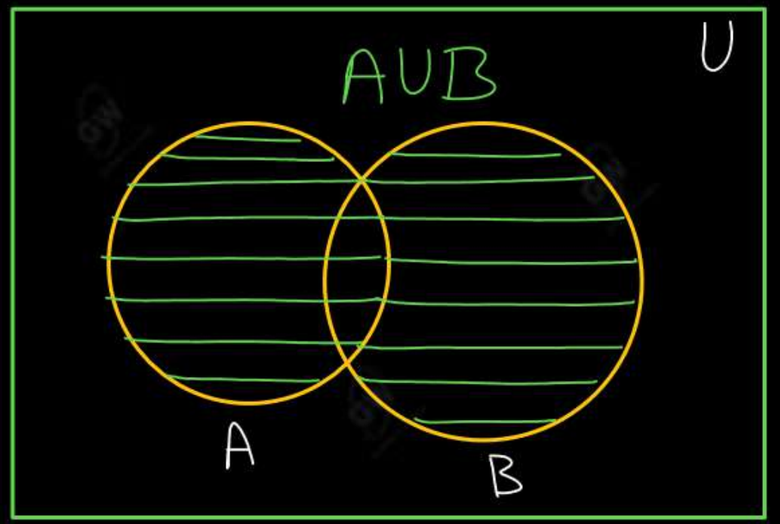
Gateway Classes : 7455 961 284

Operations on Sets

1. Union of Set:- The union of two sets A and B is the set of all elements which are either in A or in B or in both A and B.

It is denoted by $A \cup B$

$$A \cup B = \{x : x \in A \text{ or } x \in B\}$$



Note:

(1) The common elements being taken only once

(2) $x \in A \cup B \Rightarrow$ $x \in A$ or $x \in B$

(3) $x \notin A \cup B \Rightarrow$ $x \notin A$ and $x \notin B$

Gateway Classes : 7455 9612 84

Example: Let $A = \{1, 2, 3, 4\}$ and $B = \{2, 4, 6, 8\}$ Find $A \cup B$.

$A \cup B = \{1, 2, 3, 4, 6, 8\}$

Example: If $A = \{x : x \text{ is a prime number less than } 15\}$

and $B = \{x : x \in N, x \text{ is a factor of } 12\}$ Find $A \cup B$.

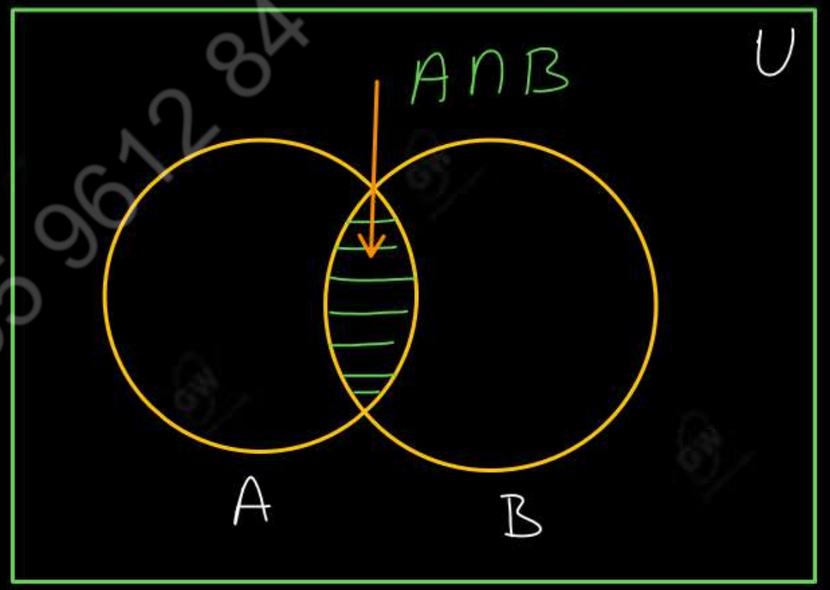
$A = \{2, 3, 5, 7, 11, 13\}$ and $B = \{1, 2, 3, 4, 6, 12\}$

$A \cup B = \{1, 2, 3, 5, 6, 7, 11, 12, 13\}$

Gateway Classes : 7455 9612 84

2. Intersection of sets:- *The intersection of two sets A and B is the set of all elements which are common to both A and B.*

- *It is denoted by* $A \cap B$
- $A \cap B = \{x : x \in A \text{ and } x \in B\}$



Note: (1) $x \in A \cap B \Rightarrow$ $x \in A$ and $x \in B$

(2) $x \notin A \cap B \Rightarrow$ $x \notin A$ or $x \notin B$

Example :- *If* $A = \{2, 4, 6, 8, 10\}$ *and* $B = \{1, 2, 3, 4\}$ *Find* $A \cap B$.

$A \cap B = \{2, 4\}$

Gateway Classes : 7455 9612 84

3. Disjoint Sets:- Two sets A and B are said to be disjoint if

$$A \cap B = \emptyset$$

Null set

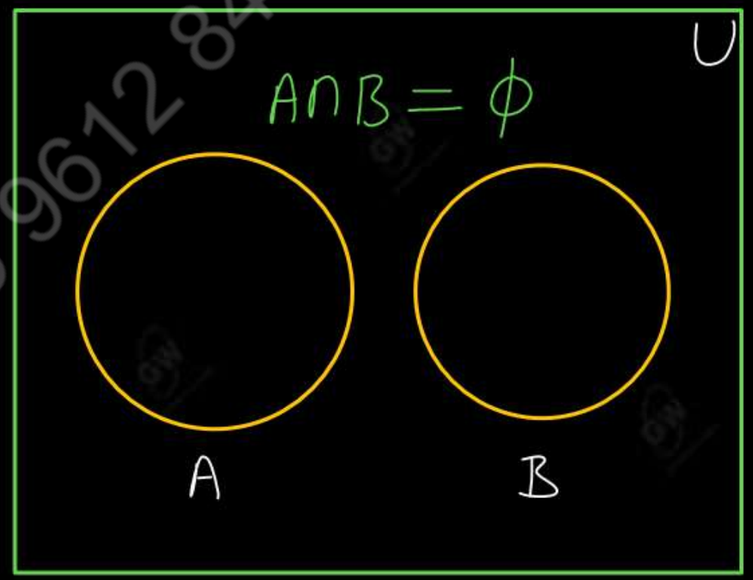
Example :-

If $A = \{2, 4, 6, 8, 10\}$

and $B = \{1, 3, 5, 7, 9\}$

$A \cap B = \phi$

Here, A and B are disjoint sets



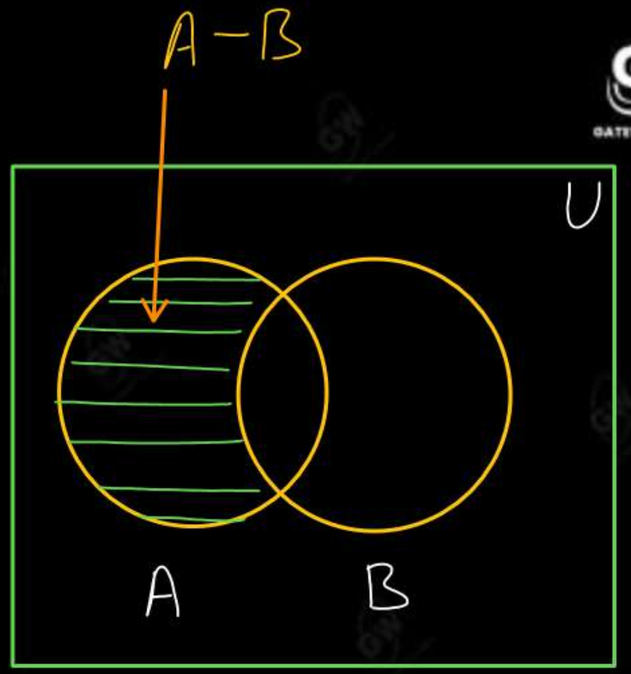
GatewayClasses : 7455 9612 84

4. Difference of Sets : For any two sets A and B

(i) A - B =

It contains all the elements of set A which are not present in set B.

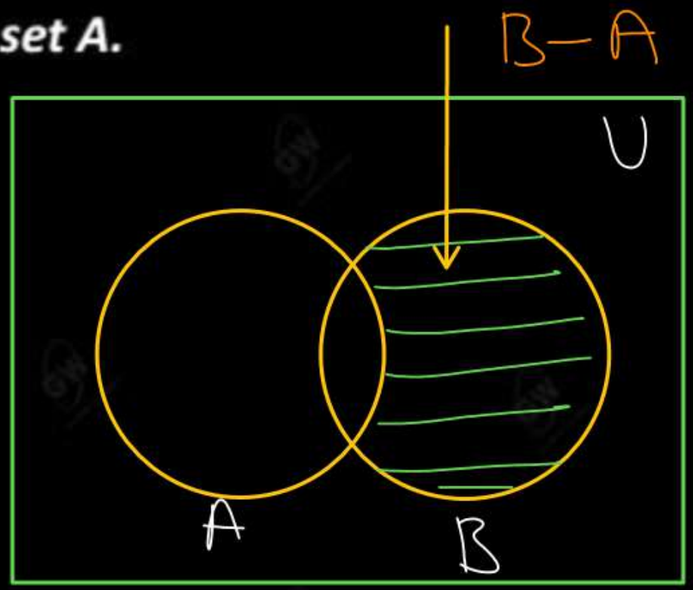
$$A - B = \{x : x \in A \text{ and } x \notin B\}$$



(ii) B - A

It contains all the elements of set B which are not present in set A.

$$B - A = \{x : x \in B \text{ and } x \notin A\}$$



GatewayClasses : 7455 961284

Example:- If $A = \{3, 5, 6, 8, 9\}$ and $B = \{5, 8\}$ Find $A - B$

$A - B = \{3, 6, 9\}$

Example :- If $A = \{x : x \in N, x \text{ is a factor of } 12\}$ and

$B = \{x : x \in N, x \text{ is a factor of } 15\}$ then Find (i) $A-B$ (ii) $B-A$.

$A = \{1, 2, 3, 4, 6, 12\}$ and $B = \{1, 3, 5, 15\}$

$A - B = \{2, 6, 12\}$

$B - A = \{5, 15\}$

Gateway Classes 7455 9612 84

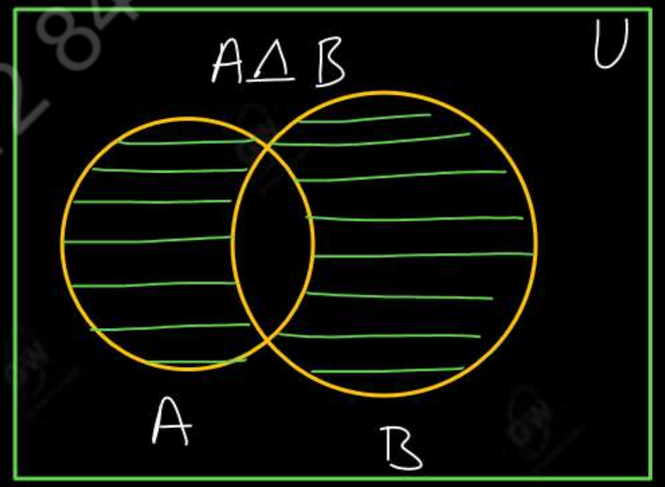
(5) Symmetric difference of sets

The symmetric difference of any two sets A and B is denoted by $A \triangle B$ and defined as

$$A \triangle B = (A - B) \cup (B - A)$$

OR

$$A \triangle B = (A \cup B) - (B \cap A)$$



Example:- *If* $A = \{1, 2, 3, 6\}$ *and* $B = \{3, 6, 9, 12\}$, *then find* $A \triangle B$.

$A - B = \{1, 2\}$

$B - A = \{9, 12\}$

$A \triangle B = \{1, 2, 9, 12\}$

$A \cup B = \{1, 2, 3, 6, 9, 12\}$

$A \cap B = \{3, 6\}$

$A \triangle B = \{1, 2, 9, 12\}$

Gateway Classes : 7455 9612 84

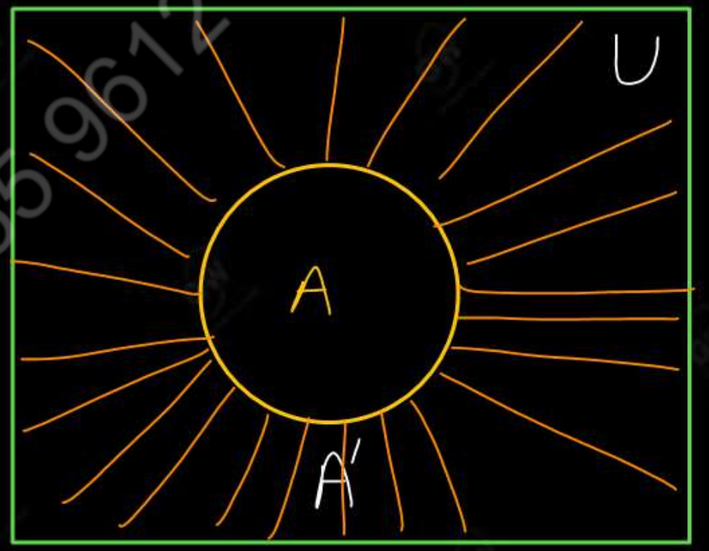
(6) Complement of set :- Let ∪ be the universal set and A be any subset of ∪.

The complement of A, denoted by A' or A^c is the set of elements which belongs to ∪ but which does not belongs to A.

$$A' = \cup - A$$

$$\boxed{A' = \{x : x \in \cup \text{ and } x \notin A\}}$$

$$x \in A^c \Rightarrow x \notin A$$



Note:-

$$A \cap A^c = \emptyset$$

$$A \cup A^c = \cup$$

$$(A^c)^c = A$$

Gateway Classes : 7455 9612 84

Example:- *If* $\cup = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$ *and*

$A = \{1, 3, 5, 7, 9, 12\}$

Find

(i) A^c

(ii) $A \cap A^c$

(iii) $A \cup A^c$

(iv) $(A^c)^c$

(i) $A' = \{2, 4, 6, 8, 10, 11\}$

(ii) $A \cap A' = \phi$

(iii) $A \cup A' = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$

(iv) $(A')' = ?$

Gateway Classes: 7455 9612 84

GATEWAY CLASSES

GATEWAY CLASSES

AKTU Full Courses (Paid)

Download Gateway Classes Application

From Google Play store

All Subjects

Link in Description

Gateway Classes : 7455 961 284

Thank You

$\supseteq \subset \not\supseteq$

DISCRETE STRUCTURES & THEORY OF LOGICS (BCS303)

UNIT -1 : (i) Set Theory & Relations

(ii) POSET & Lattices

Lec-4

Today's Target

- *Algebra of sets*

Gateway Classes 7455 5612 84

Course Details(Paid) : All Subjects

1	Recorded Video Lectures (100 % Syllabus Coverage)
2	Pdf Notes
3	Lecture wise DPP
4	Unit wise set of PYQs
5	Course will be completed by 'Last week of Nov 2023'
	Tentative AKTU Semester Exam- Jan/Feb

Gateway Classes: 7455 961284

Paid Courses are available in Gateway Classes Application

Link in Description

Revise

Operations on Sets

1. Union of Set:

$$A \cup B = \{x : x \in A \text{ or } x \in B\}$$

Note:

(a) $x \in A \cup B \Rightarrow x \in A$ or $x \in B$

(b) $x \notin A \cup B \Rightarrow x \notin A$ and $x \notin B$

2. Intersection of sets:

$$A \cap B = \{x : x \in A \text{ and } x \in B\}$$

Note:

(a) $x \in A \cap B \Rightarrow x \in A$ and $x \in B$

(b) $x \notin A \cap B \Rightarrow x \notin A$ or $x \notin B$

GatewayClasses : 7455 9612 84

GW GATEWAY CLASSES

3. Difference of Sets : For any two sets A and B

$$A - B = \{x : x \in A \text{ and } x \notin B\}$$

$$B - A = \{x : x \in B \text{ and } x \notin A\}$$

$x \in (A - B) = x \in A$ and $x \notin B$

$x \in (B - A) = x \in B$ and $x \notin A$

4. Symmetric difference of sets

$$A \triangle B = (A - B) \cup (B - A)$$

OR

$$A \triangle B = (A \cup B) - (B \cap A)$$

A' OR A^c

5. Complement of set :-

$$A' = \cup - A$$

$$A' = \{x : x \in \cup \text{ and } x \notin A\}$$

$$x \in A^c \Rightarrow x \notin A$$

$x \in A \Rightarrow x \notin A^c$

Note:-

$$A \cap A^c = \emptyset$$

$$A \cup A^c = \cup$$

$$(A^c)^c = A$$

Gateway Classes : 7455 9612 82

Algebra of sets

1. Idempotent Laws

(a) $A \cup A = A$

(b) $A \cap A = A$

2. Commutative Laws

(a) $A \cup B = B \cup A$

(b) $A \cap B = B \cap A$

3. Associative Laws

(a) $(A \cup B) \cup C = A \cup (B \cup C)$

(b) $(A \cap B) \cap C = A \cap (B \cap C)$

4. Distributive Laws

(a) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

(b) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

5. DeMorgan's Laws

(a) $(A \cup B)' = A' \cap B'$

(b) $(A \cap B)' = A' \cup B'$

Gateway Classes : 7455 96 1284

6. Identity Laws

(a) $A \cup \phi = A$

(b) $A \cap U = A$

(c) $A \cup U = U$

(d) $A \cap \phi = \phi$

7. Involution Laws

$(A')' = A$

8.Complement Laws

(a) $A \cup A' = U$

(b) $A \cap A' = \phi$

(a) $U' = \phi$

(b) $\phi' = U$

$A = \{1, 2, 3\}$

$B = \{1, 2, 3\}$

$A \subseteq B$

$B \subseteq A$

$\boxed{A = B}$

9. Absorption Laws

(a) $A \cup (A \cap B) = A$

(b) $A \cap (A \cup B) = A$

Methods

① By definition

② By venn diagram ✗

③ For any two sets A and B

$A \subseteq B$ —①

$B \subseteq A$ —②

$\Rightarrow A = B$

Gateway Classes 7455 961984

1. Idempotent Laws : For any set A, prove that

(a) $A \cup A = A$

Proof :

Let $x \in A$

$\Rightarrow x \in A$ or $x \in A$

$\Rightarrow x \in A \cup A$

$\Rightarrow A \subseteq A \cup A$ — ①

Conversely if $x \in A \cup A$

$\Rightarrow x \in A$ or $x \in A$

$\Rightarrow x \in A$

$\Rightarrow A \cup A \subseteq A$ — ②

From ① and ②

$\boxed{A \cup A = A}$ Hence, proved

2nd Method

$A \cup A = \{x : x \in A \text{ or } x \in A\}$

$= \{x : x \in A\}$

$\boxed{A \cup A = A}$ Hence proved

Gateway Classes 7455 9612 84

(b) $A \cap A = A$

Proof :

Let $x \in A$

$\Rightarrow x \in A$ and $x \in A$

$\Rightarrow x \in A \cap A$

$\Rightarrow A \subseteq A \cap A$ — ①

Conversely if $x \in A \cap A$

$\Rightarrow x \in A$ and $x \in A$

$\Rightarrow x \in A$

$\Rightarrow A \cap A \subseteq A$ — ②

From ① and ②

$$\boxed{A \cap A = A}$$

2nd Method

$$A \cap A = \{x : x \in A \text{ and } x \in A\}$$

$$= \{x : x \in A\}$$

$$\boxed{A \cap A = A}$$

Gateway Classes : 7455 9612 84

2. Commutative Laws: For any two sets A and B, Prove that

(a) $A \cup B = B \cup A$

Proof

Let $x \in A \cup B$

$\Rightarrow x \in A$ or $x \in B$

$\Rightarrow x \in B$ or $x \in A$

$\Rightarrow x \in B \cup A$

$\Rightarrow A \cup B \subseteq B \cup A$ — (1)

Conversely, if

$x \in B \cup A$

$\Rightarrow x \in B$ or $x \in A$

$\Rightarrow x \in A$ or $x \in B$

$\Rightarrow x \in A \cup B$

$\Rightarrow B \cup A \subseteq A \cup B$ — (2)

From (1) and (2)

$A \cup B = B \cup A$

2nd Method

$A \cup B = \{x : x \in A \text{ or } x \in B\}$

$= \{x : x \in B \text{ or } x \in A\}$

$= B \cup A$

$A \cup B = B \cup A$

Gateway Classes : 7455 961284

(b) $A \cap B = B \cap A$

Proof

Let $x \in A \cap B$

$\Rightarrow x \in A$ and $x \in B$

$\Rightarrow x \in B$ and $x \in A$

$\Rightarrow x \in B \cap A$

$\Rightarrow A \cap B \subseteq B \cap A$ — (1)

Conversely, if

$x \in B \cap A$

$\Rightarrow x \in B$ and $x \in A$

$\Rightarrow x \in A$ and $x \in B$

$\Rightarrow x \in A \cap B$

$\Rightarrow B \cap A \subseteq A \cap B$ — (2)

From (1) and (2)

$$A \cap B = B \cap A$$

Gateway Classes : 7455961284

3. Associative Laws: *For any three sets A, B, C, prove that*

(a) $(A \cup B) \cup C = A \cup (B \cup C)$

Proof

Let $x \in (A \cup B) \cup C$

$\Rightarrow x \in (A \cup B)$ or $x \in C$

$\Rightarrow (x \in A$ or $x \in B)$ or $x \in C$

$\Rightarrow x \in A$ or $(x \in B$ or $x \in C)$

$\Rightarrow x \in A$ or $x \in (B \cup C)$

$\Rightarrow x \in A \cup (B \cup C)$ — ①

$\Rightarrow (A \cup B) \cup C \subseteq A \cup (B \cup C)$ — ①

Similarly

$A \cup (B \cup C) \subseteq (A \cup B) \cup C$ — ②

From ① and ②

$$\boxed{(A \cup B) \cup C = A \cup (B \cup C)}$$

2nd Method

$(A \cup B) \cup C = \{x : x \in (A \cup B)$ or $x \in C\}$

$= \{x : (x \in A$ or $x \in B)$ or $x \in C\}$

$= \{x : x \in A$ or $(x \in B$ or $x \in C)\}$

Gateway Classes : 7455961284

$$(A \cup B) \cup C = \{x : x \in A \text{ or } (x \in B \text{ or } x \in C)\}$$

$$= \{x : x \in A \text{ or } x \in (B \cup C)\}$$

$$= \{x : x \in A \cup (B \cup C)\}$$

$$= A \cup (B \cup C)$$

$$\boxed{(A \cup B) \cup C = A \cup (B \cup C)}$$

Gateway Classes : 7455 9612 84

(b) $(A \cap B) \cap C = A \cap (B \cap C)$

Gateway Classes : 7455 9612 84

4. Distributive Laws: For any three sets A, B, C, prove that

(a) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

Proof

Let $x \in A \cup (B \cap C)$

$\Rightarrow x \in A$ or $x \in (B \cap C)$

$\Rightarrow x \in A$ or $(x \in B$ and $x \in C)$

$\Rightarrow x \in A$ or $x \in B$ and $x \in A$ or $x \in C$

$\Rightarrow x \in (A \cup B)$ and $x \in (A \cup C)$

$\Rightarrow x \in (A \cup B) \cap (A \cup C)$

$\Rightarrow A \cup (B \cap C) \subseteq (A \cup B) \cap (A \cup C)$ — (1)

conversely if

$x \in (A \cup B) \cap (A \cup C)$

$\Rightarrow (x \in A$ or $x \in B)$ and $(x \in A$ or $x \in C)$

$\Rightarrow x \in A$ or $(x \in B$ and $x \in C)$

$\Rightarrow x \in A$ or $x \in (B \cap C)$

$\Rightarrow x \in A \cup (B \cap C)$

$(A \cup B) \cap (A \cup C) \subseteq A \cup (B \cap C)$ — (2)

From (1) and (2)

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

GW GATEWAY CLASSES
Gateway Classes : 7455 961284

(b) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

Gateway Classes : 7455 9612 84

5. DE Morgan's Laws: For any two sets A and B, Prove that

(a) $(A \cup B)' = A' \cap B'$

Proof:

Let $x \in (A \cup B)'$

$\Rightarrow x \notin (A \cup B)$

$\Rightarrow x \notin A$ and $x \notin B$

$\Rightarrow x \in A'$ and $x \in B'$

$\Rightarrow x \in A' \cap B'$

$\Rightarrow (A \cup B)' \subseteq A' \cap B'$ — (1)

conversely if

$x \in A' \cap B'$

$\Rightarrow x \in A'$ and $x \in B'$

$\Rightarrow x \notin A$ or $x \notin B$

$\Rightarrow x \notin (A \cup B)$

$\Rightarrow x \in (A \cup B)'$

$\Rightarrow A' \cap B' \subseteq (A \cup B)'$ — (2)

From (1) and (2)

$$(A \cup B)' = A' \cap B'$$

Gateway Classes : 7455 9612 84

(b) $(A \cap B)' = A' \cup B'$

Proof

$$A' \cup B' = \{x : x \in A' \text{ or } x \in B'\}$$

$$= \{x : x \notin A \text{ and } x \notin B\}$$

$$= \{x : x \notin (A \cap B)\}$$

$$= \{x : x \in (A \cap B)'\}$$

$$= (A \cap B)'$$

$$A' \cup B' = (A \cap B)'$$

$$\boxed{(A \cap B)' = A' \cup B'}$$

Gateway Classes : 7455 961284

GATEWAY CLASSES

GATEWAY CLASSES

AKTU Full Courses (Paid)

Download Gateway Classes Application

From Google Play store

All Subjects

Link in Description

Gateway Classes : 7455 961 284

Thank You

$\supseteq \subset \not\supseteq$

DISCRETE STRUCTURES & THEORY OF LOGICS (BCS303)

UNIT -1 : (i) Set Theory & Relations

(ii) POSET & Lattices

Lec-5

Today's Target

- *Question Based on Set theory*
- *PYQ*
- *DPP*

Gateway Classes 7455 5612 84

Course Details(Paid) : All Subjects

1	Recorded Video Lectures (100 % Syllabus Coverage)
2	Pdf Notes
3	Lecture wise DPP
4	Unit wise set of PYQs
5	Course will be completed by 'Last week of Nov 2023'
	Tentative AKTU Semester Exam- Jan/Feb

Gateway Classes: 7455 961 284

Paid Courses are available in Gateway Classes Application

Link in Description

Algebra of sets

General Identities

1. Idempotent Laws

(a) $A \cup A = A$

(b) $A \cap A = A$

2. Commutative Laws

(a) $A \cup B = B \cup A$

(b) $A \cap B = B \cap A$

3. Associative Laws

(a) $(A \cup B) \cup C = A \cup (B \cup C)$

(b) $(A \cap B) \cap C = A \cap (B \cap C)$

4. Distributive Laws

(a) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

(b) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

5. DeMorgan's Laws

(a) $(A \cup B)' = A' \cap B'$

(b) $(A \cap B)' = A' \cup B'$

Gateway Classes : 7455 961284

6. Identity Laws

(a) $A \cup \phi = A$

(b) $A \cap U = A$

(c) $A \cup U = U$

(d) $A \cap \phi = \phi$

7. Involution Laws

$(A')' = A$

8.Complement Laws

(a) $A \cup A' = U$

(b) $A \cap A' = \phi$

(a) $U' = \phi$

(b) $\phi' = U$

9. Absorption Laws

(a) $A \cup (A \cap B) = A$

(b) $A \cap (A \cup B) = A$

Gateway Classes : 7455 961 984

Q. 1 *Let $A = \{a, b\}$ then calculate $A \cup P(A)$, where $P(A)$ is a power set of A.*

$A = \{a, b\}$

Subset of A

$\phi, \{a\}, \{b\}, \{a, b\}$

$P(A) = \{\phi, \{a\}, \{b\}, \{a, b\}\}$

$A \cup P(A) = \{a, b, \phi, \{a\}, \{b\}, \{a, b\}\}$

Gateway Classes 7455 9612 84

Q.2 ***For any set A and B , show that*** $A - B = A \cap B^c$. V. Imp

To Prove

$A - B = A \cap B^c$

Let $x \in (A - B)$

$\Rightarrow x \in A$ and $x \notin B$

$\Rightarrow x \in A$ and $x \in B^c$

$\Rightarrow x \in A \cap B^c$

$\Rightarrow A - B \subseteq A \cap B^c$ — (1)

conversely if

$x \in A \cap B^c$

$\Rightarrow x \in A$ and $x \in B^c$

$\Rightarrow x \in A$ and $x \notin B$

$\Rightarrow x \in (A - B)$

$\Rightarrow A \cap B^c \subseteq A - B$ — (2)

From (1) and (2)

$$A - B = A \cap B^c$$

Gateway Classes : 7455 961284

Q.3 Show that for any two sets A and B : $A-(A\cap B)=A-B$ AKTU-2021-2022

To Prove

$A-(A\cap B)=A-B$

LHS

$A-(A\cap B)$

$=A\cap(A\cap B)^{c}$ $\{A-B=A\cap B^{c}\}$

$=A\cap(A^{c}\cup B^{c})$

$=(A\cap A^{c})\cup(A\cap B^{c})$ (By Distributive law)

$=\phi\cup(A\cap B^{c})$

$=A\cap B^{c}$

$=A-B$ $\{\because A-B=A\cap B^{c}\}$

LHS = RHS

Hence proved

Gateway Classes · 7455 9612 84

Q.4 *If A, B, C are three sets then show that* $A \cup (B - A) = A \cup B$

To Prove

$A \cup (B - A) = A \cup B$

LHS

$A \cup (B - A)$

$= A \cup (B \cap A')$ $\{\because A - B = A \cap B'\}$

$= (A \cup B) \cap (A \cup A')$

$= (A \cup B) \cap U$

$= A \cup B$

LHS = RHS

Hence proved

Gateway Classes : 7455 961 284

Q.5 *If A, B, C are three sets then prove that* $A - (B \cup C) = \overline{(A - B)} \cap \overline{(A - C)}$

To Prove

$A - (B \cup C) = (A - B) \cap (A - C)$

LHS

$A - (B \cup C)$

$= A \cap (B \cup C)'$ { $\because A - B = A \cap B'$ }

$= A \cap (B' \cap C')$ { By Demorgan's law }

$= A \cap A \cap (B' \cap C')$ { By idempotent law }

$= (A \cap B') \cap (A \cap C')$

$= (A - B) \cap (A - C)$ { $\because A - B = A \cap B'$ }

LHS = RHS

Hence proved

GW GATEWAY CLASSES
Gateway Classes : 7455 912 884

Q. 6 *If A, B, C are three sets then show that* $A - (B \cap C) = (A - B) \cup (A - C)$.

AKTU-2021-2022

To Prove

$A - (B \cap C) = (A - B) \cup (A - C)$

LHS

$A - (B \cap C)$

$= A \cap (B \cap C)'$ $\{A - B = A \cap B'\}$

$= A \cap (B' \cup C')$ {By Demorgan's law}

$= (A \cap B') \cup (A \cap C')$ {By Distributive law}

$= (A - B) \cup (A - C)$ $\{\because A - B = A \cap B'\}$

LHS = RHS

Hence proved

Gateway Classes : 7455 961284

Q.7 *If A, B, C are three sets then show that* $(A \cap B) \cup (A - B) = A$

To Prove

$(A \cap B) \cup (A - B) = A$

LHS

$(A \cap B) \cup (A - B)$

$= (A \cap B) \cup (A \cap B')$ $\{\because A - B = A \cap B'\}$

$= A \cap (B \cup B')$ {By Distributive law}

$= A \cap U$ $\{\because A \cup A' = U\}$

$= A$

LHS = RHS

Hence proved

Gateway Classes : 7455 961284

Q.8 Prove that $A \subseteq B$, then $B' \subseteq A'$

Given

$A \subseteq B$

If $x \in A \implies x \in B$

$x \notin B \implies x \notin A$

$x \in B^c \implies x \in A^c$

$\implies \boxed{B^c \subseteq A^c}$

Hence proved

Gateway Classes : 7455 9612 84

DSTL (by Gulshan Sir)

DPP-1 to 5

Topic : SET Theory

Q .1 *State De Morgan's law and Absorption law* *AKTU-2021-22*

Q .2 Prove for any two sets A and B $(A \cup B)' = A' \cap B'$ *AKTU-2014-15*

Q .3 *Let* $A = \{a, \phi\}$ *then calculate* $A \cup P(A)$, *where* $P(A)$ *is a power set of* A *AKTU-2022-23*

Q .4 *List the subsets of the set* $A = \{0, 1, 2\}$

Q .5 *List the subset of the set* $B = \{1, \{2, 3\}\}$

Q .6 *If* $A = \{1, 2, 3, 4\}, B = \{2, 3, 4, 5\}$ *and* $C = \{3, 4, 5, 6\}$, *find* $A - B$, $B - C$, $A - (B - C)$, $(A - B) - C$

Q .7 *If* $A = \{1, 2, 3, 4\}$, $B = \{2, 4, 6, 8\}$ *and* $C = \{3, 4, 5, 6\}$ *find* *(i)* $(A \cap B) \cap C$ *(ii)* $A \cap (B \cap C)$

Q .8 *Prove that* $A \cap (B - C) = (A \cap B) - (A \cap C)$

Q .9 *Prove that* $A \cap (B - C) = (A \cap B) - (A \cap C)$

Q .10 *Prove that* $A - (B - C) = (A - B) \cup (A \cap C)$

Gateway Classes : 7455 9612 84

GATEWAY CLASSES

AKTU Full Courses (Paid)

Download Gateway Classes Application

From Google Play store

All Subjects

Link in Description

Gateway Classes : 7455961284

Thank You

DISCRETE STRUCTURES & THEORY OF LOGICS (BCS303)

UNIT -1 : (i) Set Theory & Relations

(ii) POSET & Lattices

Lec-6

Today's Target

- *Ordered Pair*
- *Cartesian product of two sets*
- *Relation or Binary Relation, Relation on a set*
- *Domain and Range of relation*
- *Operation on relation*

Gateway Classes : 7455 961 284

Course Details(Paid) : All Subjects

1	Recorded Video Lectures (100 % Syllabus Coverage)
2	Pdf Notes
3	Lecture wise DPP
4	Unit wise set of PYQs
5	Course will be completed by 'Last week of Nov 2023'
	Tentative AKTU Semester Exam- Jan/Feb

Gateway Classes: 7455 961 284

Paid Courses are available in Gateway Classes Application

Link in Description

Ordered Pair:- An ordered pair consist of two elements grouped together in a particular order

Example :- (1, 4)

In the ordered pair (a, b)

(i) a is called first element

(ii) b is called second element

Universal set

Cartesian product of two sets :- The set of all possible ordered pairs (a, b), where $a \in A$ and $b \in B$ is called the Cartesian product of set A and B and is denoted by $A \times B$

usually $\boxed{A \times B \neq B \times A}$

Note :- If $n(A) = m$ and $n(B) = n$ then

$$n(A \times B) = mn$$

Gateway Classes : 7455 9612 84

Example :- If $A = \{1, 2\}$ and $B = \{a, b, c\}$ then find

(i) $A \times B = \{(1, a), (1, b), (1, c), (2, a), (2, b), (2, c)\}$ $\quad n(A \times B) = 6$

(ii) $B \times A = \{(a, 1)(a, 2)(b, 1)(b, 2)(c, 1)(c, 2)\}$ $\quad n(B \times A) = 6$

(iii) $A \times A = \{(1, 1)(1, 2)(2, 1)(2, 2)\}$ $\quad n(A \times A) = 4$

(iv) $B \times B = \{(a, a)(a, b)(a, c)(b, a)(b, b)(b, c)(c, a)(c, b)(c, c)\}$

$n(B \times B) = 9$

Gateway Classes : 7455961284

Relation or Binary relation :- Let A and B be two non-empty sets, then any subset R of the Cartesian product $A \times B$ is called a relation from set A to set B

$$R \subseteq A \times B$$

$$R = \{(x,\ y):\ x \in A, y \in B\ \ and\ \ x\ R\ y\}$$

Note :-

(i) If $(x,\ y) \in R$ then we say that 'x is related to y' and we write $x\ R\ y$

(ii) If $(x,\ y) \notin R$ then we say that 'x is not related to y' and we write $x\ \not{R}\ y$

(iii) If $n(A) = m$ and $n(B) = n$, then $n(A \times B) = mn$

$$\text{Number of relations from } A \text{ to } B = \text{Total number of subsets of } A \times B = 2^{mn}$$

GatewayClasses: 7455 9612 84

Example : If $A = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$ and $B = \{1, 2, 3, 4, 5\}$ and $R = \{(a, b): a \in A, b \in B \text{ and } a = 2b\}$ then find R

Given

$a = 2b$

$R = \{(2, 1), (4, 2), (6, 3), (8, 4)\}$

Gateway Classes : 7455 9612 84

Relation on a set:- *A Relation R from set A to the set A itself is called a relation on set A*

$$R \subseteq A \times A$$

$$R = \{(x, y): x \in A, y \in B \text{ and } x\,R\,y\}$$

Example :- *If* $A = \{1, 2, 3\}$. *Find relation on A defined by " is less than equal to "*

$A = \{1, 2, 3\}$, $A = \{1, 2, 3\}$

$A \times A = \{(1,1)(1,2)(1,3)(2,1),(2,2),(2,3)(3,1)(3,2)(3,3)\}$

$1 = 1 \Rightarrow (1,1) \in R$

$1 < 2 \Rightarrow (1,2) \in R$

$1 < 3 \Rightarrow (1,3) \in R$

$2 \nless 1 \Rightarrow (2,1) \notin R$

$2 = 2 \Rightarrow (2,2) \in R$

$2 < 3 \Rightarrow (2,3) \in R$

$3 \nless 1 \Rightarrow (3,1) \notin R$

$3 \nless 2 \Rightarrow (3,2) \notin R$

$3 = 3 \Rightarrow (3,3) \in R$

GatewayClasses : 7455961284

$$R = \{(1,1)(1,2)(1,3)(2,2)(2,3)(3,3)\}$$

Gateway Classes : 7455 961284

Domain and Range of Relation :-

Let $R = \{(x, y): x \in A, \ y \in B \ \text{and} \ x\,R\,y\}$

Domain :- The set of all first elements of the ordered pairs belonging to R is called the domain of R.
it is denoted by Dom(R) or d(R)

$$d(R) = \{x : x \in A \text{ and } (x, y) \in R\}$$

Range :- The set of all second elements of the ordered pairs belonging to R is called the range of R.

it is denoted by Ran(R) or r(R)

$$r(R) = \{y : y \in B \text{ and } (x, y) \in R\}$$

Gateway Classes 7455 5612 84

Example :- Let $A = \{2, 4, 6\}$ and $B = \{1, 4, 5, 6\}$ then find out the relation from A to B defined by "is less than or equal to". Find out the domain and range of the relation

$A \times B = \{(2,1)(2,4)(2,5)(2,6)(4,1)(4,4)(4,5)(4,6)(6,1)(6,4)(6,5)(6,6)\}$

Relation from A to B "is less than or equal to"

$2 \not\leq 1 \Rightarrow (2,1) \notin R$

$2 < 4 \Rightarrow (2,4) \in R$

$2 < 5 \Rightarrow (2,5) \in R$

$2 < 6 \Rightarrow (2,6) \in R$

$4 \not\leq 1 \Rightarrow (4,1) \notin R$

$4 = 4 \Rightarrow (4,4) \in R$

$4 < 5 \Rightarrow (4,5) \in R$

$4 < 6 \Rightarrow (4,6) \in R$

$6 \not\leq 1 \Rightarrow (6,1) \notin R$

$6 \not\leq 4 \Rightarrow (6,4) \notin R$

$6 \not\leq 5 \Rightarrow (6,5) \notin R$

$6 = 6 \Rightarrow (6,6) \in R$

$R = \{(2,4)(2,5)(2,6)(4,4)(4,5)(4,6)(6,6)\}$

$d(R) = \{2, 4, 6\}$

$r(R) = \{4, 5, 6\}$

Gateway Classes 7455 961284

Operation on relation

(a) Complement of Relation

(b) Inverse relation

(5) Intersection and Union of Relation

Complement of a Relation

Consider a relation R from set the set A to B. The complement of relation R denoted by $\bar{R}$ or R' is a relation from A to B such that

$$\bar{R} = \{(a, b): (a, b) \notin R\}$$

Gateway Classes : 7455 9612 84

Example : *Let R be a relation from X to Y, where $X = \{1, 2, 3\}$ and $Y = \{8, 9\}$ and $R = \{(1,8), (2,8)(1,9), (3,9)\}$ find the complement of relation R*

$R = \{(1,8), (2,8)(1,9)(3,9)\}$

$X \times Y = \{(1,8)(1,9)(2,8)(2,9)(3,8)(3,9)\}$ Universal set

$\overline{R} = \{(2,9)(3,8)\}$

Gateway Classes : 7455 9612 84

Inverse Relation

Let R be relation from a set A to B. The inverse of relation R denoted by R^{-1} is a relation from B to A such that

$$R^{-1} = \{(b, a): (a, b) \in R\}$$

Thus , to find R^{-1} we write in reverse order all ordered pairs belonging to R.

Example: *Find the R^{-1} to the relation R on A defined "$x + y$" divisible by 2.*

For $A = \{1, 2, 3, 4, 6\}$.

$R = \{(1,1)\ (1,3)\ (2,2)\ (2,4)\ (2,6)\ (3,1)\ (3,3)\ (4,2)\ (4,4)\ (4,6)\ (6,2)\ (6,4)\ (6,6)\}$

$R^{-1} = \{(1,1)\ (3,1)\ (2,2)\ (4,2)\ (6,2)\ (1,3)\ (3,3)\ (2,4)\ (4,4)\ (6,4)\ (2,6)\ (4,6)\ (6,6)\}$

Gateway Classes : 7455 9612 84

Intersection and Union of Relation

If R and S are the two relations then intersection of R and S denoted by $R \cap S$ and the union of R and S denoted by $R \cup S$ are two new relation that can be formed from R and S.

Thus $R \cup S = \{(x, y): xRy \text{ or } xSy\}$

$R \cap S = \{(x, y): xRy \text{ and } xSy\}$

Example : *Let* $R = \{(1,1), (2, 2), (3, 3), (4, 4), (3, 4), (4, 3)\}$

and $S = \{(1,1), (2, 2), (3, 3), (4, 4), (1, 2), (2, 1)\}$. *Find* $R \cup S$ *and* $R \cap S$

$R \cup S = \{(1,1)(2,2)(3,3)(4,4)(3,4)(4,3)(1,2)(2,1)\}$

$R \cap S = \{(1,1)(2,2)(3,3)(4,4)$

Gateway Classes : 7455 9612 84

GATEWAY CLASSES

AKTU Full Courses (Paid)

Download Gateway Classes Application

From Google Play store

All Subjects

Link in Description

Gateway Classes : 7455961284

Thank You

DISCRETE STRUCTURES & THEORY OF LOGICS (BCS303)

UNIT -1 : (i) Set Theory & Relations

(ii) POSET & Lattices

Lec-7

Today's Target

- *Properties of Relation*

Gateway Classes : 7455 9012 84

Course Details(Paid) : All Subjects

1	Recorded Video Lectures (100 % Syllabus Coverage)
2	Pdf Notes
3	Lecture wise DPP
4	Unit wise set of PYQs
5	Course will be completed by 'Last week of Nov 2023'
	Tentative AKTU Semester Exam- Jan/Feb

Gateway Classes: 7455 961284

Paid Courses are available in Gateway Classes Application

Link in Description

Properties of Relation (Types of Relation)

1. Void Relation:- A Relation R in a set A is called a void relation or an empty relation, if no element of A is related to any element of A and we denote such a relation by ϕ

Thus

$$R = \phi$$

where $\phi \subseteq A \times A$

Example :- Consider a relation R on the set $A = \{1, 2, 3, 4\}$ and defined by $R = \{(a, b): a - b = 12\}$

$R = \{\ \} = \phi$

Gateway Classes : 7455 961284

2. Universal Relation:- A Relation R in a set A is called a universal relation, if each element of A is related to every element of A

Thus

$$R = A \times A$$

where $A \times A \subseteq A \times A$

Example :- Consider a relation R on the set $A = \{2, 3, 4\}$ and $R = \{(a, b): a + b < 9\}$

$A \times A = \{(2,2)(2,3)(2,4)(3,2)(3,3)(3,4)(4,2)(4,3)(4,4)\}$

$R = \{(2,2)(2,5)(2,4)(3,5)(3,3)(3,7)(4,2)(4,3)(4,8)\}$

Gateway Classes 7455 9612 84

3. Identify Relation:- A Relation R in a set A is called an identify relation, if every element of A is related to it self only .

Example :- Consider the relation R on set $A = \{1, 2, 3, 4\}$ and $R = \{(a, b): a \div b = 1\}$

$$R = \{(1,1)(2,2)(3,3)(4,4)\}$$

Gateway Classes 7455 9612 84

4. Reflexive Relation:- *A relation R on set A is reflexive if every element of A is related to itself.*

Imp

$$\text{Thus } R \text{ is Reflexive} \iff (a, a) \in R \text{ for all } a \in A$$

aRa for all $a \in A$

Example : *Consider the relation R on the set* $A = \{1, 2, 3, 4\}$ *and* $R = \{(a, b) : a, b \in A, (a \div b) \in Integer\}$

$$R = \{(1,1)(2,2)(3,3)(4,4)(2,1)(3,1)(4,1)(4,2)\}$$

Note :- *Every identify relation is reflexive but every reflexive relation is not identify relation.*

Gateway Classes : 7455 9612 64

5. Irreflexive Relation:- *A Relation R on Set A is irreflexive if*

$$(a,\ a) \notin R \text{ for all } a \in A$$

Example:- *Let* $A = \{1, 2\}$ *and* $R = \{(1,\ 2), (2,\ 1)\}$

$a \not R a \quad \forall \ a \in A$

6. Non-reflexive Relation:- *A Relation R on a Set A is non-reflexive if R is neither reflexive nor irreflexive.*

Example:- *Let* $A = \{1, 2\}$ *and* $R = \{(1, 2), (2, 1)(1, 1)\}$

Gateway Classes : 7455 9612 84

Imp

7. Symmetric Relation:- A relation R on Set A is symmetric relation if a is related to b then b is also related to a.

Thus R is symmetric if $(a, b) \in R \Rightarrow (b, a) \in R$ for all $a, b \in A$

$aRb \Rightarrow bRa \quad \forall a, b \in A$

Example 8 :-Consider the relation R_1 and R_2 on set $A = \{2, 4, 5, 6\}$.

$R_1 = \{(2,4), (4,2), (4,5), (5,4), (6,6)\}$ and $R_2 = \{(2,4), (2,6), (6,2), (5,4), (4,5)\}$

For R_1

$(2,4) \in R_1 \Rightarrow (4,2) \in R_1$

$(4,5) \in R_1 \Rightarrow (5,4) \in R_1$

$(6,6) \in R_1 \Rightarrow (6,6) \in R_1$

Hence R_1 is symmetric

For R_2

$(2,4) \in R_2 \Rightarrow (4,2) \notin R_2$

Hence R_2 is not symmetric

Gateway Classes : 7455961284

8.Asymmetric Relation:- *A Relation R on set A is asymmetric if* $(a, b) \in R$ *then* $(b, a) \notin R$ *for* $a \neq b$.

Example:- *Let* $A = \{1, 2, 3\}$ *and* $R = \{(1, 2), (1, 1), (2, 3), (3, 1)\}$ *is a asymmetric relation.*

$(1,2) \in R \Rightarrow (2,1) \notin R$

$(2,3) \in R \Rightarrow (3,2) \notin R$

$(3,1) \in R \Rightarrow (1,3) \notin R$

Hence R is Asymmetric Relation

9. Antisymmetric Relation:- *A Relation R is said to be antisymmetric if* $(a, b) \in R$ *and* $(b, a) \in R$ *for* $a = b$

Example:- *The relation R =* $\{(1, 1), (3, 3)\}$ *is Antisymmetric Relation on* $A = \{1, 2, 3\}$

$A \times A = \{(1,1)(1,2)(1,3)(2,1)(2,2)(2,3)(3,1)(3,2)(3,3)\}$

$R = \{(1,1)(3,3)\}$

Gateway Classes : 7455 9612 64

10.Transitive Relation:- The Relation R on set A is transitive relation if a is related to b, b is related to c then a is also related to c

Imp

Thus R is transitive if $(a,b) \in R$ and $(b, c) \in R \Rightarrow (a, c) \in R$ forall $a, b, c \in A$

Thus R is transitive if aRb and $bRc \Rightarrow aRc \ \forall \ a, b, c \in A$

Example:- Consider a relation R on set $A = \{1, 2, 3, 4\}$ and $R = \{(a, b) : a < b\}$

$A \times A = \{(1,1)(1,2)(1,3)(1,4)(2,1)(2,2)(2,3)(2,4)(3,1)(3,2)(3,3)(3,4)$
$(4,1)(4,2)(4,3)(4,4)\}$

$R = \{(1,2)(1,3)(1,4)(2,3)(2,4)(3,4)\}$

$(1,2) \in R$ and $(2,3) \in R \Rightarrow (1,3) \in R$

$(1,2) \in R$ and $(2,4) \in R \Rightarrow (1,4) \in R$

$(1,3) \in R$ and $(3,4) \in R \Rightarrow (1,4) \in R$

$(2,3) \in R$ and $(3,4) \in R \Rightarrow (2,4) \in R$

Hence R is transitive

Gateway Classes 7455 961204

V. Imp (7) Marks

Equivalence Relation:- *A Relation R on set A is said to be an equivalence relation if and only if it is Reflexive, Symmetric and transitive simultaneously.*

NOTE: *A relation R in a set A is called*

(i) Reflexive if $(a, a) \in R \quad for\ all\ a \in A$

(ii) Symmetric if $(a, b) \in R \Rightarrow (b, a) \in R \quad fo\ all \quad a, b \in A$

(iii) Transitive if $(a, b) \in R \ and \ (b, c) \in R \Rightarrow (a, c) \in R \quad for\ all \quad a, b, c \in A$

Gateway Classes : 7455 9612 84

GATEWAY CLASSES

AKTU Full Courses (Paid)

Download Gateway Classes Application

From Google Play store

All Subjects

Link in Description

Gateway Classes : 7455961284

Thank You

DISCRETE STRUCTURES & THEORY OF LOGICS
(Discrete Mathematics)

UNIT -1 : (i) Set Theory & Relations

(ii) POSET & Lattices

Lec-8

Today's Target

- *Equivalence relation*
- *PYQ*
- *DPP*

Gateway Classes : 7455 9012 84

Course Details(Paid) : All Subjects

1	Recorded Video Lectures (100 % Syllabus Coverage)
2	Pdf Notes
3	Lecture wise DPP
4	Unit wise set of PYQs

Paid Courses are available in Gateway Classes Application

Link in Description

Gateway Classes : 7455 961284

Equality of Relation
Or
Equivalence Relation

Let A be a non-empty set and R be a relation defined on set A, then R is said to be equivalence relation if it is

(i) Reflexive: $(a, a) \in R \ \forall \ a \in A$

$$aRa \ \forall \ a \in A$$

(ii) Symmetric: $\text{If } (a, b) \in R \Rightarrow (b, a) \in R \ (\forall) \ a, b \in A$ (For all)

$$aRb \Rightarrow bRa \ \forall \ a, b \in A$$

(iii) Transitive: $\text{If } (a, b) \in R \text{ and } (b, c) \in R \Rightarrow (a, c) \in R \ \forall \ a, b, c \in A$

$$aRb \text{ and } bRc \Rightarrow aRc \ \forall \ a, b, c \in A$$

GW GATEWAY CLASSES
Gateway Classes 7455 9612 84

Equivalence class

Let R be an equivalence relation on a non-empty set A. The equivalence class of an element $a \in A$ is the set of elements of A which are related to 'a'

It is denoted by [a] or $\bar{a}$

$$[a] = equivalence\ class\ of\ a$$

$$[a] = \{x : x \in A, x\ R\ a\}$$

Gateway Classes : 7455 9612 84

Q.1. Let $A = \{a, b, c\}$ and let $R = \{(a, a), (b, b), (c, c), (a, b), (b, a)\}$ Where R is clearly an equivalence relation. Find the equivalence classes of the elements of A.

Equivalence classes

$[a] = \{a, b\}$

$[b] = \{a, b\}$

$[c] = \{c\}$

Gateway Classes : 7455 9612 84

Q.2. *Let $A = \{0, 1, 2, 3, 4\}$. Show that the relation $R = \{(0,0), (0,4), (1,1), (1,3), (2,2), (3,1), (3,3), (4,0), (4,4)\}$ is an equivalence relation. Also find the distinct equivalence Classes of R.*

$A = \{0, 1, 2, 3, 4\}$

$R = \{(0,0), (1,1), (2,2), (3,3), (4,4), (0,4), (1,3), (3,1), (4,0)\}$

Reflexive:

Since $(0,0) \in R$
$(1,1) \in R$
$(2,2) \in R$
$(3,3) \in R$
$(4,4) \in R$

$\Rightarrow (a, a) \in R \ \forall \ a \in A$

$\Rightarrow a R a \ \forall \ a \in A$

So, R is Reflexive

Symmetric

Since $(0,4) \in R \Rightarrow (4,0) \in R$

$(1,3) \in R \Rightarrow (3,1) \in R$

$(3,1) \in R \Rightarrow (1,3) \in R$

$(4,0) \in R \Rightarrow (0,4) \in R$

Gateway Classes: 7455 9612 84

$\Rightarrow (a,b) \in R \Rightarrow (b,a) \ \forall \ a,b \in A$

$\Rightarrow aRb \Rightarrow bRa \ \forall \ a,b \in A$

so, R is symmetric

Transitive:

$(0,4) \in R$ and $(4,0) \in R \Rightarrow (0,0) \in R$

$(0,4) \in R$ and $(4,4) \in R \Rightarrow (0,4) \in R$

$(1,3) \in R$ and $(3,1) \in R \Rightarrow (1,1) \in R$

$(1,3) \in R$ and $(3,3) \in R \Rightarrow (1,3) \in R$

$(4,0) \in R$ and $(4,4) \in R \Rightarrow (4,4) \in R$

$\Rightarrow (a,b) \in R$ and $(b,c) \in R \Rightarrow$

$(a,c) \in R \ \forall \ a,b,c \in A$

$\Rightarrow aRb$ and $bRc \Rightarrow aRc \ \forall \ a,b,c \in A$

so, R is transitive

Equivalence classes

$[0] = \{0,4\}$

$[1] = \{1,3\}$

$[2] = \{2\}$

$[3] = \{1,3\}$

$[4] = \{0,4\}$

Gateway Classes : 7455 9612 84

Q.3 *Let $X = \{1, 2, 3, \ldots . 7\}$ and $R = \{(x, y): x - y$ is divisible by $3\}$. Show that R is an equivalence relation .*

R is an equivalence Relation if

(i) Reflexive : For any $x \in X$

$x - x = 0$, which is divisible by 3

$\Rightarrow (x, x) \in R \ \forall \ x \in X$

so, R is Reflexive

(ii) Symmetric : Let $x, y \in X$, such that

$(x, y) \in R$

$\Rightarrow x - y$ is divisible by 3

$\Rightarrow x - y = 3n \quad , n \in I$

$\Rightarrow -(x - y) = -3n$

$\Rightarrow y - x = -3n$

$\Rightarrow (y, x) \in R$

So, R is symmetric

GatewayClasses : 7455961284

Transitive :

Let $x, y, z \in X$ such that

$(x, y) \in R$ and $(y, z) \in R$

$\Rightarrow (x-y)$ is divisible by 3 and $(y-z)$ is divisible by 3

$\Rightarrow x - y = 3n_1$ and $y - z = 3n_2$

$\Rightarrow x - \cancel{y} + \cancel{y} - z = 3n_1 + 3n_2$

$\Rightarrow x - z = 3(n_1 + n_2)$

$\Rightarrow (x, z) \in R$

So, R is transitive

Hence

R is an equivalence Relation

Gateway Classes : 7455 9612 84

Q4: Let $A = \{1, 2, 3, 4, 6, 7, 8, 9\}$ and let R be the relation on $A \times A$ defined by $(a, b)\ R\ (c, d)$ if $a + d = b + c$ for all $(a, b),\ (c, d) \in A \times A$. Prove that

(a) R is an equivalence relation .

(b) Find $[(2, 5)]$.

R is an equivalence Relation if

(1) Reflexive: For any $(a, a) \in A \times A$

$(a, a)\ R\ (a, a)$ if $a + a = a + a$ (true)

So, R is Reflexive

Symmetric:

Let $(a, b), (c, d) \in A \times A$ such that $(a, b)\ R\ (c, d)$

$\Rightarrow a + d = b + c$

$\Rightarrow b + c = a + d$

$\Rightarrow c + b = d + a$

$\Rightarrow (c, d)\ R\ (a, b)$

So, R is symmetric

Gateway Classes 7455 961284

Transitive:

Let $(a,b), (c,d), (e,f) \in A \times A$ such that

$(a,b) R (c,d)$ and $(c,d) R (e,f)$

$\Rightarrow a+d = b+c$ and $c+f = d+e$

$\Rightarrow a+\cancel{d}+\cancel{c}+f = b+\cancel{c}+\cancel{d}+e$

$\Rightarrow a+f = b+e$

$\Rightarrow (a,b) R (e,f)$

$(a,b) R (c,d)$

$a+d = b+c$

So, R is transitive

Hence

R is an equivalence Relation

Equivalence class

$[(2,5)] = \{(1,4)(2,5)(3,6)$

$(4,7)(5,8)(6,9)\}$

Gateway Classes : 7455 961284

Q.5: Let $A = R \times R$ (R be a set of Real numbers and define the following relation on A

$(a, b)\ R\ (c, d) \Leftrightarrow a^2 + b^2 = c^2 + d^2$. Verify that R is an equivalence relation

R is an equivalence Relation if

(I) Reflexive : For any $(a, a) \in A$

$(a, a) R (a, a)$ if $a^2 + a^2 = a^2 + a^2$ (True)

So, R is Reflexive

(II) Symmetric

Let $(a, b), (c, d) \in A$ such that

$(a, b) R (c, d)$

$\Rightarrow a^2 + b^2 = c^2 + d^2$

$\Rightarrow c^2 + d^2 = a^2 + b^2$

$\Rightarrow (c, d) R (a, b)$

So, R is symmetric

(III) Transitive :

Let $(a, b) (c, d) (e, f) \in A$

such that

Gateway Classes : 7259612184

$(a,b)\,R\,(c,d)$ and $(c,d)\,R\,(e,f)$

$\Rightarrow a^2+b^2=c^2+d^2$ and $c^2+d^2=e^2+f^2$

$\Rightarrow a^2+b^2+c^2+d^2=c^2+d^2+e^2+f^2$

$\Rightarrow a^2+b^2=e^2+f^2$

$\Rightarrow (a,b)\,R\,(e,f)$

So, R is transitive

Hence, R is an equivalence Relation

Gateway Classes : 7455 9612 84

NOTE: Two integers a and b are said to be congruence modulo m if a-b is divisible by m and we write

$a \equiv b \pmod{m}$

Thus, $a \equiv b \pmod{m} \iff$ (a-b) is divisible by m

Q.6: Let A be the set of all integers and a relation R is defined as

$R = \{(x, y) : x \equiv y \pmod{m}, m \text{ divide } (x - y)\}$ where m is a positive integer.

Prove that R is an equivalence relation.

R is an equivalence Relation if

(I) Reflexive: For any $x \in A$

$x - x = 0$ which is divisible by m

$\Rightarrow x \equiv x \pmod{m}$

$\Rightarrow x R x \ \forall \ x \in A$

So, R is Reflexive

(II) Symmetric:

Let $x, y \in A$ such that

$(x, y) \in R$

$\Rightarrow x - y$ is divisible by m

$\Rightarrow x \equiv y \pmod{m}$

Gateway Classes : 7453961284

$\Rightarrow x-y = mn \quad n \in I$

$\Rightarrow -(x-y) = -mn$

$\Rightarrow (y-x) = -mn$

$\Rightarrow (y,x) \in R$

$\Rightarrow y \equiv x \pmod m$

So, R is symmetric

Transitive

Let $x, y, z \in A$ such that

$(x,y) \in R$ and $(y,z) \in R$

$(x-y)$ is divisible by m and $(y-z)$ is divisible by m

$x \equiv y \pmod m$ and $y \equiv z \pmod m$

$\Rightarrow x-y = mn_1$ and $y-z = mn_2$

$\Rightarrow x-y+y-z = mn_1+mn_2$

$\Rightarrow x-z = m(n_1+n_2)$

$\Rightarrow (x-z)$ is divisible by m

Gateway Classes 7455961284

$\Rightarrow x \equiv z \pmod{m}$

$\Rightarrow (x, z) \in R$

So, R is transitive

Hence

R is an equivalence Relation

Gateway Classes : 7455 9612 84

DPP- 6 to 8

Topic : Equivalence Relation

Gateway Classes 7455 961284

Q.1. *Let* $N = \{1, 2, 3, - - - - -\}$ and a relation is defined in N X N as follows: (a, b) is related to (c, d) iff $\mathbf{ad} = \mathbf{bc}$. *Then show that whether R is an equivalence relation or not.*

Q.2. *Let R be a binary relation defined as* $R = \{(a, b) \in R^2 : (a - b) \leq 3\}$

Determine whether R is an equivalence relation.

Q.3. *Let S be the set of all points in a plane. Let R be a relation such that for any two points a and b*

$(a,\ b) \in R$ *is not an equivalence relation.*

AKTU Full Courses (Paid)

Download Gateway Classes Application

From Google Play store

All Subjects

Link in Description

Gateway Classes : 7455961284

Thank You

DISCRETE STRUCTURES & THEORY OF LOGICS
(Discrete Mathematics)

UNIT -1 : (i) Set Theory & Relations

(ii) POSET & Lattices

Lec-9

Today's Target

- *Important Theorem on Equivalence relation*
- *Composition of relation*
- *PYQ*
- *DPP*

Gateway Classes : 7455 9612 84

Course Details(Paid) : All Subjects

1	Recorded Video Lectures (100 % Syllabus Coverage)
2	Pdf Notes
3	Lecture wise DPP
4	Unit wise set of PYQs

Paid Courses are available in Gateway Classes Application

Link in Description

Gateway Classes : 7455 961 284

Equivalence Relation

Let A be a non-empty set and R be a relation defined on set A, then R is said to be equivalence relation if it is

(i) Reflexive: $(a, a) \in R \ \forall \ a \in A$

i.e $aRa \ \forall \ a \in A$

(ii) Symmetric: $(a, b) \in R \Rightarrow (b, a) \in R \ \forall \ a, b \in A$

i.e $aRb \Rightarrow bRa \ \forall \ a, b \in A$

(iii) Transitive: $(a, b) \in R$ and $(b, c) \in R \Rightarrow (a, c) \in R \ \forall \ a, b, c \in A$

i.e aRb and $bRc \Rightarrow aRc \ \forall \ a, b, c \in A$

Gateway Classes : 7455 9612 82

Theorem-1

If R is an equivalence relation on A, then prove that R^{-1} is also an equivalence relation on A .

Given

since R is an equivalence relation

(i) R is Reflexive

(ii) R is symmetric

(iii) R is transitive

To Prove

R^{-1} is also an equivalence relation

Proof :

(i) Reflexive : For any $n \in A$

$(n, n) \in R$ { $\because$ R is reflexive }

$\Rightarrow (n, n) \in R^{-1}$ $\forall$ $n \in A$

So, R^{-1} is reflexive

Gateway Classes : 7455961284

(II) Symmetric :

Let $x, y \in A$ such that

$(x, y) \in R \Rightarrow (y, x) \in R$

{ $\because$ R is symmetric }

$\Rightarrow (y, x) \in R^{-1} \Rightarrow (x, y) \in R^{-1}$

$\Rightarrow (x, y) \in R^{-1} \Rightarrow (y, x) \in R^{-1}$

So, R^{-1} symmetric

Transitive : Let $x, y, z \in A$ such that

$(x, y) \in R$ and $(y, z) \in R \Rightarrow (x, z) \in R$

{ $\because$ R is transitive }

$(y, x) \in R^{-1}$ and $(z, y) \in R^{-1} \Rightarrow (z, x) \in R^{-1}$

$(z, y) \in R^{-1}$ and $(y, x) \in R^{-1} \Rightarrow (z, x) \in R^{-1}$

So, R^{-1} is transitive Relation

Hence

R^{-1} is also an equivalence Relation

Gateway Classes : 7455 961 284

Theorem-2

If R_1 and R_2 are two equivalence relations on set A, then prove that $R_1 \cap R_2$ is also an equivalence relation.

Given

R_1 and R_2 are equivalence relation

(i) R_1 and R_2 are reflexive

(ii) R_1 and R_2 are symmetric

(iii) R_1 and R_2 are transitive

To Prove

$R_1 \cap R_2$ is also an equivalence Relation

Proof

(i) Reflexive For any $x \in A$

$(x, x) \in R_1$ { $\because$ R_1 is Reflexive }

$(x, x) \in R_2$ { $\because$ R_2 " }

$\Rightarrow (x, x) \in R_1 \cap R_2 \; \forall \; x \in A$

So, R is reflexive

Gateway Classes : 7455961284

(II) Symmetric: Let $x, y \in A$ such that

$(x, y) \in R_1 \Rightarrow (y, x) \in R_1$ {R is symmetric}

$(x, y) \in R_2 \Rightarrow (y, x) \in R_2$ {$\because$ " }

$\Rightarrow (x, y) \in R_1 \cap R_2 \Rightarrow (y, x) \in R_1 \cap R_2 \; \forall \; x, y \in A$

So, $R_1 \cap R_2$ is symmetric

(III) Transitive:

Let $x, y, z \in A$ such that

$(x, y) \in R_1$ and $(y, z) \in R_1 \Rightarrow (x, z) \in R_1$

$(x, y) \in R_2$ and $(y, z) \in R_2 \Rightarrow (x, z) \in R_2$

{$\because$ R_1 and R_2 are transitive}

$\Rightarrow (x, y) \in R_1 \cap R_2$ and $(y, z) \in R_1 \cap R_2$

$\Rightarrow (x, z) \in R_1 \cap R_2$

So, $R_1 \cap R_2$ is transitive

Hence, $R_1 \cap R_2$ is also an equivalence relation

Gateway Classes : 7455961284

Theorem-3

If R_1 and R_2 are two equivalence relations on set A, then prove that $R_1 \cup R_2$ is not necessarily an equivalence relation.

Example: *Let $A = \{1, 2, 3\}$ and Equivalence relation. $R_1 = \{(1,1)(2,2)(3,3)\}, (1,2), (2,1)$ and $R_2 = \{(1,1)(2,2)(3,3)(2,3), (3,2)\}$, Show that $R_1 \cup R_2$ is not an equivalence relation.*

Proof : $R_1 \cup R_2 = \{(1,1)(2,2)(3,3)(1,2)(2,1)(2,3)(3,2)\}$

Since $(3,2) \in R_1 \cup R_2$ and $(2,1) \in (R_1 \cup R_2)$

But $(3,1) \notin R_1 \cup R_2$

So, $R_1 \cup R_2$ is not an equivalence Relation

Gateway Classes : 7455 961284

Composition of Relations
or
Composite Relation

Let A, B and C be three non-empty sets.

Let R is a relation from A to B i.e $R \subseteq A \times B$

and S is a relation from B to C. i.e $S \subseteq B \times C$

Then the composition of Relation of R and S denoted by SoR, is a relation from A to C and and defined as

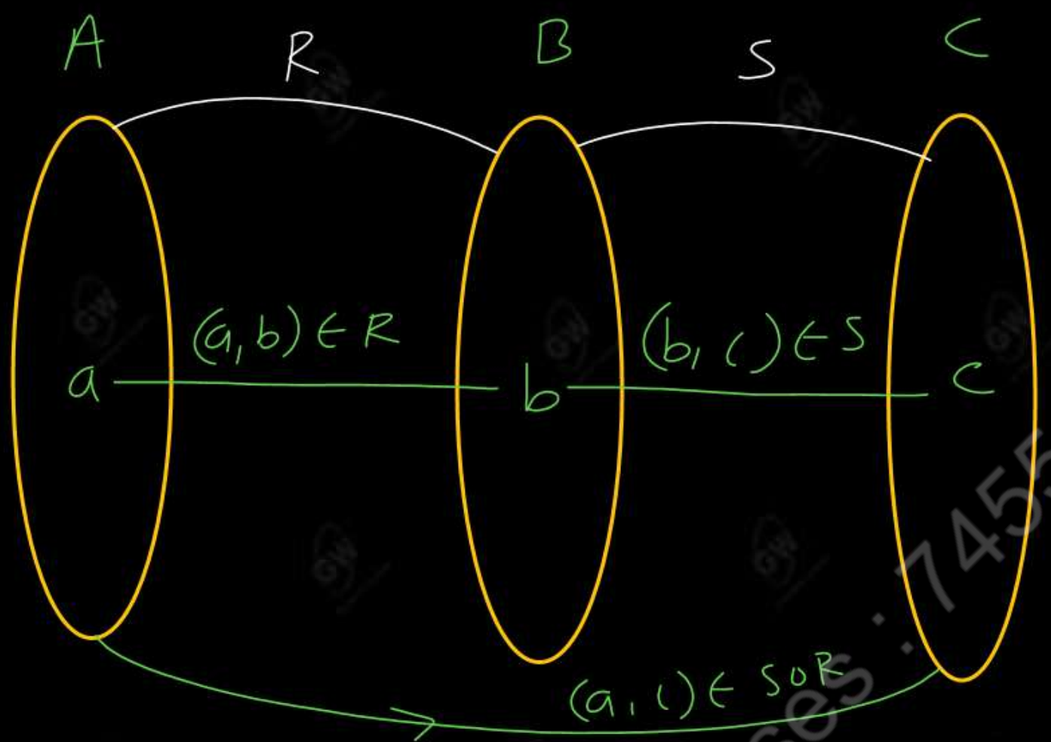
$$SoR = \{(a, c) \in A \times C : for\ some\ b \in B\ such\ that\ (a, b) \in R\ and\ (b, c) \in S\}$$

Where $a \in A,\ b \in B,\ c \in C$

Hence we can say

$$(a, b) \in R,\ (b, c) \in S \Rightarrow (a, c) \in SoR$$

Gateway Classes: 7455 9612 84



Note: (i) *If R and S are relation from A to B and C to D, then SoR is not defined unless B=C*

(ii)The composition of R with R : $RoR = R^2$

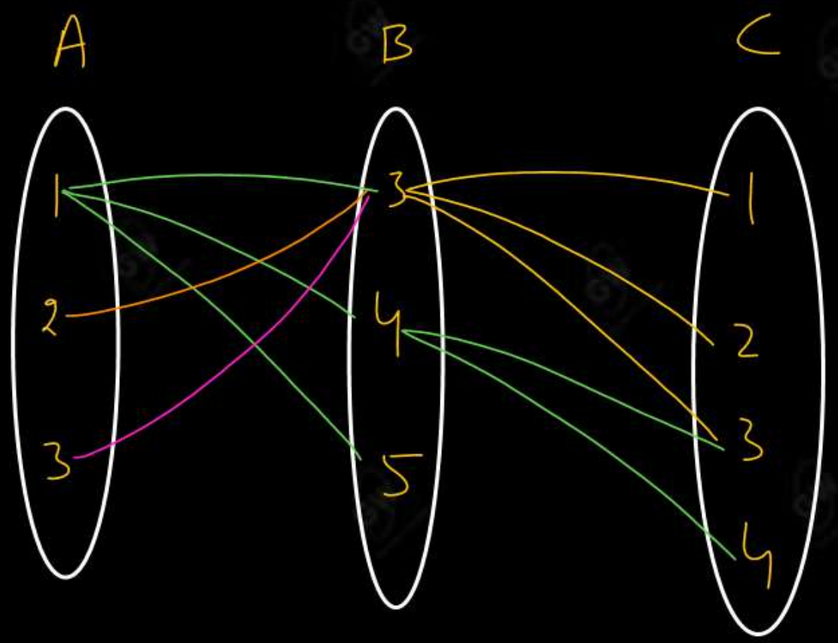
(iii) The composition of R^2 *with R :* $R^2oR = R^3 = RoRoR$

(iv) The composition of R *with* SoR : $Ro\,(SoR)$

Gateway Classes : 7455 9612 84

Q.1. Let $A = \{1, 2, 3\}$ $B = \{3, 4, 5\}$, $C = \{1, 2, 3, 4\}$ and Let $R = \{(1,3)(1,4)(1,5)(2,3)(3,3)\}$

$S = \{(3,1)(3,2)(3,3)(4,3)(4,4)\}$. compute $S \circ R$, $R \circ S$



$S \circ R = \{(1,1)(1,2)(1,3)(1,4)(2,1)$
$(2,2)(2,3)(3,1)(3,2)(3,3)\}$

$R = \{(1,3)(1,4)(1,5)(2,3)(3,3)\}$

$S = \{(3,1)(3,2)(3,3)(4,3)(4,4)\}$

$S \circ R = \{(1,1)(1,2)(1,3)(1,4)(2,1)(2,2)$
$(2,3)(3,1)(3,2)(3,3)\}$

Gateway Classes : 7455961284

Q.2 Let $R = \{(1,2)(3,4)(2,2)\}$ and $S = \{(4,2)(2,5)(3,1)(1,3)\}$

Find (i) SoR (ii) RoS (iii) RoR (iv) $RoRoR$ (v) $Ro(SoR)$ (vi) $(RoS)oR$

(i) SoR

$R = \{(1,2)(3,4)(2,2)\}$

$S = \{(4,2)(2,5)(3,1)(1,3)\}$

$SoR = \{(1,5)(3,2)(2,5)\}$

(ii) RoS

$S = \{(4,2)(2,5)(3,1)(1,3)$

$R = \{(1,2)(3,4)(2,2)\}$

$RoS = \{(4,2)(3,2)(1,4)\}$

Gateway Classes :7455 9612 84

(III) $R \circ R$

$R = \{(1,2)(3,4)(2,2)\}$

$R = \{(1,2)(3,4)(2,2)\}$

$R \circ R = \{(1,2)(2,2)\}$

$R^2 = \{(1,2)(2,2)\}$

(IV) $R \circ R \circ R = R^2 \circ R$

$R = \{(1,2)(3,4)(2,2)\}$

$R^2 = \{(1,2)(2,2)\}$

$R \circ R \circ R = \{(1,2)(2,2)\}$

$R^3 = \{(1,2)(2,2)\}$

(V) $R \circ (S \circ R)$

$S \circ R = \{\quad\}$

$R = \{\quad\}$

(VI) $(R \circ S) \circ R$

$R = \{\quad\}$

$R \circ S = \{\quad\}$

Gateway Classes: 7455 9612 84

Theorem:- *Let R be a relation from the set A to the Set B and S be a relation from the Set B to Set C, then*

$(SOR)^{-1} = R^{-1} O S^{-1}$ *(Reversal law in composite relation)*

Proof Let $(c, a) \in (SOR)^{-1}$

$\Rightarrow (a, c) \in SOR$

$\Rightarrow$ For $b \in B$ such that

$\Rightarrow (a, b) \in R$ and $(b, c) \in S$

$\Rightarrow (b, a) \in R^{-1}$ and $(c, b) \in S^{-1}$

$\Rightarrow (c, b) \in S^{-1}$ and $(b, a) \in R^{-1}$

$\Rightarrow (c, a) \in R^{-1} o S^{-1}$

Hence

$$(SOR)^{-1} = R^{-1} o S^{-1}$$

Hence proved

GW GATEWAY CLASSES
Gateway Classes : 7455 9612 84

Q.3. Let $A = \{a, b\}$

$R = \{a, a\}(b, a)(b, b)\}$

$S = \{a, b\}(b, a)(b, b)\}$

Then verify $(SOR)^{-1} = R^{-1} O S^{-1}$

LHS

$SOR = \{(a,b)(b,b)(b,a)\}$

$(SOR)^{-1} = \{(b,a)(b,b)(a,b)\}$ ← ①

RHS

$S^{-1} = \{(b,a)(a,b)(b,b)\}$

$R^{-1} = \{(a,a)(a,b)(b,b)\}$

$R^{-1} o S^{-1} = \{(b,a)(b,b)(a,b)\}$ ← ②

From ① and ②

$$(SOR)^{-1} = R^{-1} o S^{-1}$$

Gateway Classes : 7455961284

Discrete Mathematics by Gulshan Sir

DPP- 9

Topic : Composition of relation

Q.1. Let $R = \{(1,1)(2,1)(3,\ 2),(4,3)\}$, compute R^2, R^3, R^4 .

Q.2. Let $A = \{1,2,3\}$ $B = \{p,q,r\}$ and $C = \{x,y,z\}$

and let $R = \{(1,p)(1,r)(2,q)(3,q)\}$

and $S = \{(p,y)(q,x)(r,z)\}$ compute SoR

Gateway Classes : 7455 9612 84

GATEWAY CLASSES

AKTU Full Courses (Paid)

Download Gateway Classes Application

From Google Play store

All Subjects

Link in Description

Gateway Classes : 7455961284

Thank You

DISCRETE STRUCTURES & THEORY OF LOGICS (BCS303)

UNIT -1 : (i) Set Theory & Relations

(ii) POSET & Lattices

Lec-10

Today's Target

- *Total Number of relation*
- Partial Order Relation

Gateway Classes : 7455 96 12 84

Course Details(Paid) : All Subjects

1	Recorded Video Lectures (100 % Syllabus Coverage)
2	Pdf Notes
3	Lecture wise DPP
4	Unit wise set of PYQs

Paid Courses are available in Gateway Classes Application

Link in Description

Gateway Classes : 7455 961 284

$A \times A = n^2$

Total Number of relations

Let A be a set with n elements i.e n(A) = n, then the total number of

(i) Relation $= 2^{n^2}$

(ii) Reflexive relation $= 2^{n^2-n}$ or $2^{n(n-1)}$

(iii) Irreflexive relation $= 2^{n^2-n}$ or $2^{n(n-1)}$

(iii) Symmetric relation $= 2^{\frac{n(n+1)}{2}}$

(iv) Asymmetric relation $= 3^{\frac{n(n-1)}{2}}$

(v) Anti – symmetric relation $= 2^n \times 3^{\frac{n(n-1)}{2}}$

(vi) Reflexive and symmetric $= 2^{\frac{n^2-n}{2}}$

$A = \{1, 2, 3\}$

$A \times A = \{(1,1)\ (1,2)\ (1,3)$

$(2,1)\ (2,2)\ (2,3)$

$(3,1)\ (3,2)\ (3,3)\}$

$n^2 - n$

Gateway Classes : 7455 961284

PARTIAL ORDER RELATION

A relation R on set A is called partial order relation If it is

(i) Reflexive : $(a, a) \in R \quad \forall \quad a \in A$

(ii) Anti symmetric : $(a, b) \in R \text{ and } (b, a) \in R \Rightarrow a = b \quad \forall \quad a, b \in A$

(iii) Transitive : $(a, b) \in R \text{ and } (b, c) \in R \Rightarrow (a, c) \in R \quad \forall \quad a, b, c \in A$

Gateway Classes : 7455 9612 84

Q.1. *Identify the partial order relation from the following Relation on a set A* $= \{1, 2, 3, 4\}$

(i) $R_1 = \phi$ ✗

(ii) $R_2 = A \times A$ ✗

- Reflexive ✓
- Anti-symmetric ✗ $(2,1) \in R$ and $(1,2)$ but $2 \neq 1$
- Transitive ✓

(iii) $R_3 = \{(1,1)(2,2)(3,3)(4,4)(1,2)(2,3)(1,3)\}$ ✓

- R ✓
- A ✓
- T ✓

(iv) $R_4 = \{(1,1)(2,2)(3,3)(4,4)\}$ ✓

- R ✓
- A ✓
- T ✓

(v) $R_5 = \{(1,1)(2,2)(3,3)(4,4)(1,2)(2,1)\}$ ✗

- R ✓
- A ✗

Gateway Classes : 7455 961284

Q.2. ***Let N be the set of natural numbers, prove that the relation of " less than or equal to" is a partial order relation on N .***

$R = \{(a,b) : a \le b \text{ and } a, b \in N\}$

R is a partial order relation if

(I) Reflexive : For Any $a \in N$

since $a \le a$ (True)

$(a,a) \in R \ \forall \ a \in N$

So, R is a reflexive Relation

(II) Anti-Symmetric

Let $a, b \in N$ such that

$(a,b) \in R$ and $(b,a) \in R$

$\Rightarrow a \le b$ and $b \le a$

$\Rightarrow a = b$

So, R is Anti-symmetric Relation

Gateway Classes : 7456 9612 82

(III) Transitive: Let $a, b, c \in N$ such that

$(a, b) \in R$ and $(b, c) \in R$

$\Rightarrow a \leq b$ and $b \leq c$

$\Rightarrow a \leq c$

$\Rightarrow (a, c) \in R \; \forall \; a, b, c \in N$

= So, it Transitive

Hence, given relation is a partial order relation

Gateway Classes : 7455 962 84

Q.3. *Which of the following is not a partial order relation ?*

(i) $R_1 = \{(a, b) : a, b \in z, a < b\}$

(ii) $R_2 = \{(a, b) : a, b \in z, a \leq b\}$

(iii) $R_3 = \{(a, b) : a, b \in z, \frac{b}{a} \leq z\}$

(i) $R_1 = \{(a, b) : a, b \in Z, a < b\}$

Reflexive : For any $a \in Z$

$a < a$ (False)

$\Rightarrow (a, a) \notin R$

$\Rightarrow$ So, R is not reflexive

Hence, it is not a partial order relation

Gateway Classes : 7455 9612 84

GATEWAY CLASSES

GATEWAY CLASSES

Gateway Classes : : 7455 9612 84

Discrete Mathematics by Gulshan Sir

DPP- 10

Topic : Partial Order Relation

Q.1. Let $A = \{1, 2, 3\}$. Show that $R = \{(1,1)(2,2)(3,3)(1,2)(2,3)(1,3)\}$ is a partial order relation.

Q.2. Show that the relation $R = \{(x, y): x \geq y\}$ where x , $y \in z$ is a partial order relation

Gateway Classes : 7455 9612 84

GATEWAY CLASSES

GATEWAY CLASSES

AKTU Full Courses (Paid)

Download Gateway Classes Application

From Google Play store

All Subjects

Link in Description

Gateway Classes : 7455 961 284

Thank You

$\supseteq \subset \not\supseteq$

DISCRETE STRUCTURES & THEORY OF LOGICS (BCS303)

UNIT -1 : (i) Set Theory & Relations

(ii) POSET & Lattices

Lec-11

Today's Target

- *Digraphs of Relation*
- Matrix Representation of Relation

Gateway Classes : 7455 96 12 84

Course Details(Paid) : All Subjects

1	Recorded Video Lectures (100 % Syllabus Coverage)
2	Pdf Notes
3	Lecture wise DPP
4	Unit wise set of PYQs

Paid Courses are available in Gateway Classes Application

Link in Description

Gateway Classes : 7455 961 284

Digraphs of Relation

OR

Directed Graph of Relation

If A is finite set and R is a relation on A, then we can also represent R pictorially as

(i) Draw a small circle for each element of A and label the circle with corresponding elements of A , these circles are called nodes or vertices

(ii) Draw an arrow, called an edge , from vertex

$a_i \ to \ a_j \Leftrightarrow a_i \ R \ a_j$

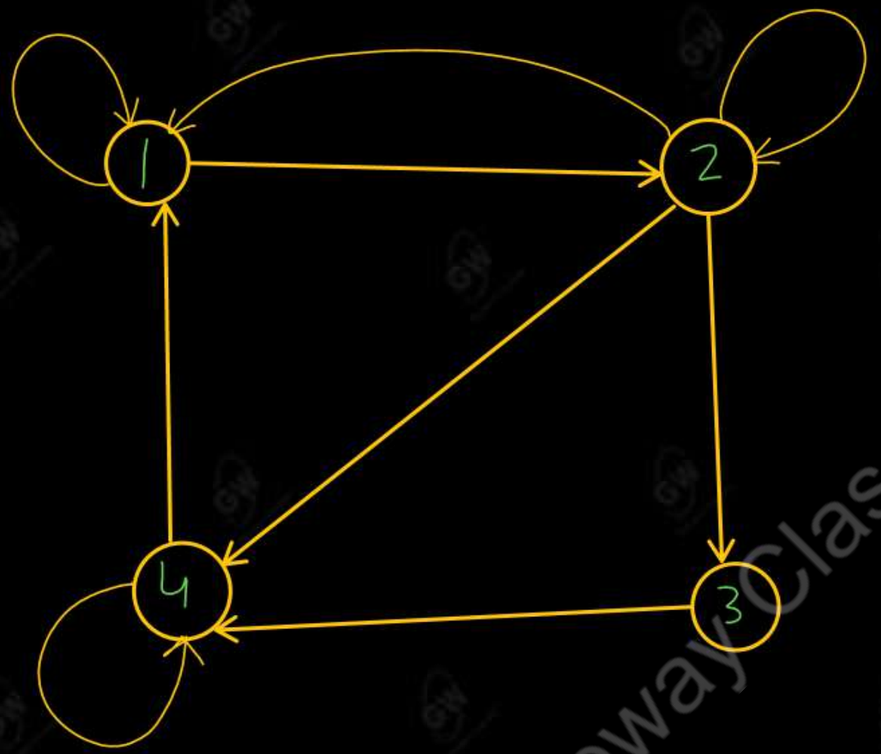
The resulting pictorial representation of R is called a Directed graph or Digraph of R.

Gateway Classes : 7455962284

Q.1- Let $A = \{1, 2, 3, 4, \}$ $R = \{(1,1), (1,2), (2,1), (2,2), (2,3), (2,4), (3,4), (4,1), (4,4)\}$

Construct the digraph of R

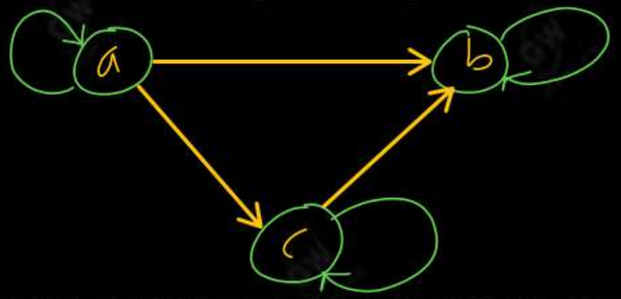
solution



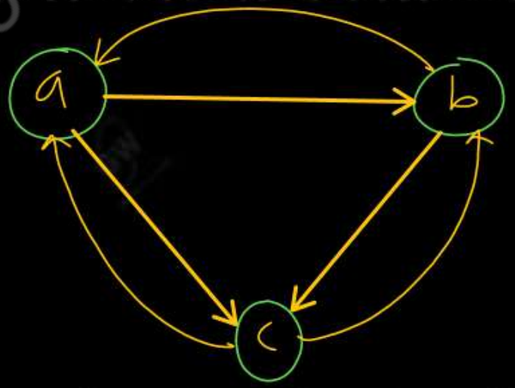
It is digraph of the given Relation

Gateway Classes: 7455 9612 84

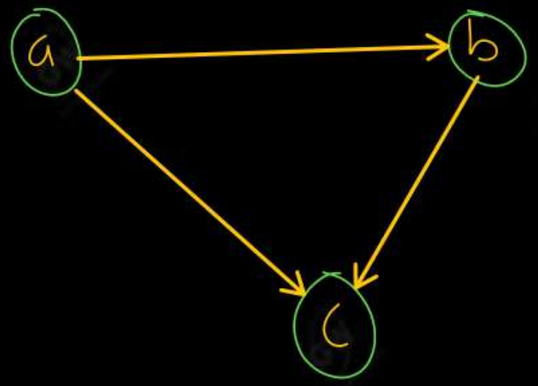
(1) Reflexive - A Relation R is reflexive if there is a loop at every Vertex of the directed graph



(2) Symmetric - A Relation R is symmetric, if for every edge between distinct vertices in its digraph there is an edge in the opposite direction.



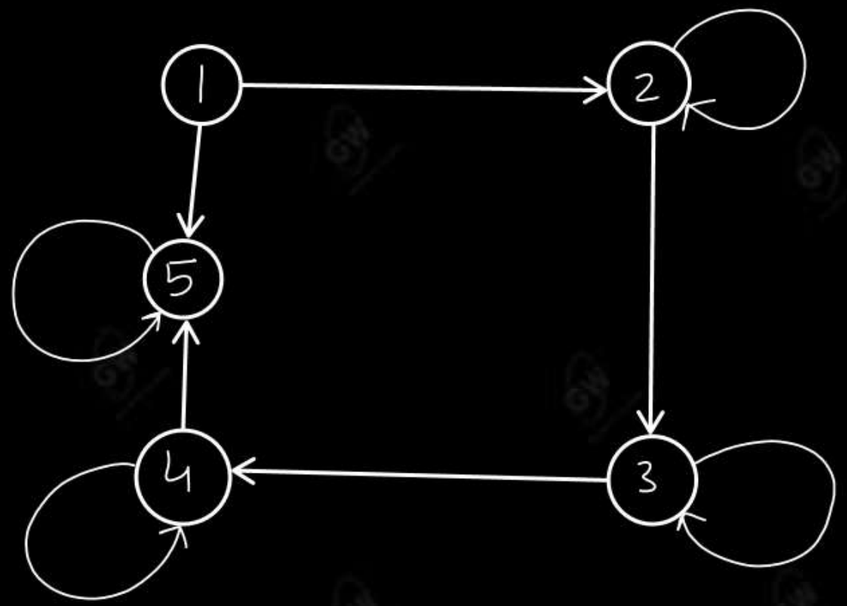
(3) Transitive - A Relation is transitive if there is an edge from vertex a to vertex b and from vertex b to vertex c, then there is an edge from vertex a to vertex c.



Gateway Classes : 7455 9612 84

Q.2- Find the relation by the following figures

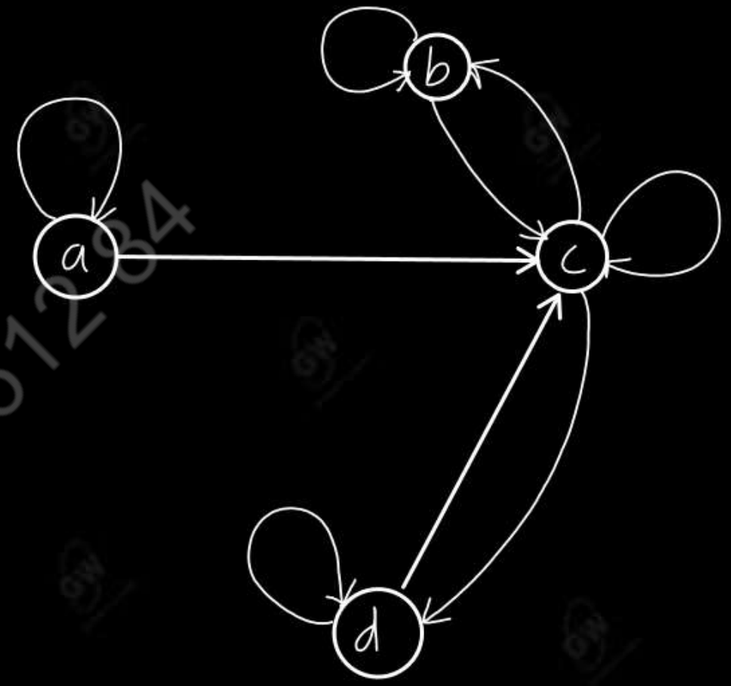
(i)



Solution

$R = \{(1,2), (2,2), (3,3), (3,4), (4,4), (4,5), (5,5), (1,5)\}$

(ii)



Solution

$R = \{(a,a), (a,c), (c,c), (c,b), (b,c), (b,b), (c,d), (d,c), (d,d)\}$

Gateway Classes : 7455 961284

Matrix Representation of Relation

Let $A = \{a_1, a_2, a_3 \ ---- a_m\}$ and $b = \{b_1, b_2, b_3 \ ---- b_n\}$ are finite sets containing m and n elements respectively and let R be a relation from A to B

Then R can be represented by the m x n matrix

$$M_R = [m_{ij}]$$

Where

$$m_{ij} = \begin{cases} 1 & if \ (a_i, b_j) \in R \\ 0 & if \ (a_i, b_j) \notin R \end{cases}$$

The matrix M_R is called matrix of Relation R

Gateway Classes : 7455 9612 84

Q.3- *Let R be the relation from the set* $A = \{1, 3, 4, \}$ *on itself and defined by*

$R = \{(1,1),\ (1,3), (3,1)\ (3,3), (4,4)\}$ *then find relation*

matrix

Given $A = \{1, 3, 4\}$

$R = \{(1,1)(1,3)(3,1)(3,3)(4,4)\}$

$$M_R = \begin{array}{c} \\ 1 \\ 3 \\ 4 \end{array} \begin{array}{c} \begin{array}{ccc} 1 & 3 & 4 \end{array} \\ \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \end{array}$$

Gateway Classes : 7455 9612 84

Q.4- Let $A = \{a, b, c, d\}$ and let $M_R = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix}$ Find R

Solution

$A = \{a, b, c, d\}$

$$M_R = \begin{array}{c} \\ a \\ b \\ c \\ d \end{array} \begin{array}{c} \begin{array}{cccc} a & b & c & d \end{array} \\ \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix} \end{array}$$

$R = \{(a,a)(a,b)(b,c)(b,d)(c,c)$
$(c,d)(d,a)(d,c)\}$

Gateway Classes : 7455 961284

Q.5- *Let* $A = \{1, 2, 3, 4\}$ *and let* R *be a relation on* A *whose matrix is* $M_R = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix}$

show that R is transitive.

Sol $A = \{1, 2, 3, 4\}$

$$M_R = \begin{array}{c} \\ 1 \\ 2 \\ 3 \\ 4 \end{array} \begin{array}{c} \begin{array}{cccc} 1 & 2 & 3 & 4 \end{array} \\ \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix} \end{array}$$

First Method

$R = \{(1,1)(1,2)(1,3)(1,4)(3,1)(3,2)(3,3)(3,4)(4,2)\}$

Second Method

R is transitive if

$$M_R^2 + M_R = M_R$$

Gateway Classes 7455 9612 84

$$M_R^2 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

$$M_R^2 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$M_R^2 + M_R = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

$$M_R^2 + M_R = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix} = M_R$$

Gateway Classes : 7455 9612 84

Since

$$M_R^2 + M_R = M_R$$

Hence, R is a transitive Relation

Gateway Classes : 7455 9612 84

Q.6- *Let* $P = \{2, 3, 4, 5\}$. *Consider the relation R and S on P defined by*

$R = \{(2,2), (2,3), (2,4), (2,5), (3,4), (3,5), (4,5), (5,4\}$ *and*

$S = \{(2,3), (2,5), (3,4), (3,5), (4,2), (4,3), (4,5), (5,2), (5,5)\}$

Find the matrices of the above relations

Use matrices to find the composition relation of R and S

$M_R =$

	2	3	4	5
2	1	1	1	1
3	0	0	1	1
4	0	0	0	1
5	0	0	1	0

$M_S =$

	2	3	4	5
2	0	1	0	1
3	0	0	1	1
4	1	1	0	1
5	1	0	0	1

Gateway Classes : 7455 9012 84

$$SOR = M_R \times M_S$$

Gateway Classes : 7455 9612 84

Discrete Mathematics by Gulshan Sir

DPP- 11

Topic : Matrix Representation of Relation

Q.1. *Let* $A = \{a_1, a_2, a_3\}$ *and* $B = \{b_1, b_2, b_3, b_4\}$ *which ordered pairs are in the relation R represented by the matrix*

$$M_R = \begin{array}{c} \\ a_1 \\ a_2 \\ a_3 \end{array} \begin{array}{c} \begin{array}{cccc} b_1 & b_2 & b_3 & b_4 \end{array} \\ \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix} \end{array}$$

Q.2. *Let* $A = \{1, 2, 3, 4, 8\}$, $B = \{1, 4, 6, 9\}$. *Let a R b iff a divides b. Find the relation matrix*

Gateway Classes : 7455 9612 84

GATEWAY CLASSES

GATEWAY CLASSES

AKTU Full Courses (Paid)

Download Gateway Classes Application

From Google Play store

All Subjects

Link in Description

Gateway Classes : 7455961284

Thank You

GATEWAY CLASSES

GATEWAY CLASSES

DISCRETE STRUCTURES & THEORY OF LOGICS (BCS303)

UNIT -1 : (i) Set Theory & Relations

(ii) POSET & Lattices

Lec-12

Today's Target

- *Closure of Relation*
- Warshall's Algorithm
- Univ PYQs
- DPPs

Gateway Classes : 7455 962 284

Course Details(Paid) : All Subjects

1	Recorded Video Lectures (100 % Syllabus Coverage)
2	Pdf Notes
3	Lecture wise DPP
4	Unit wise set of PYQs

Paid Courses are available in Gateway Classes Application

Link in Description

Gateway Classes : 7455 961 284

Closure of Relation

Let R be a relation on a set A. R may or may not have some property P, such as reflexivity, symmetry or transitivity.

If we add some pairs then we have the desired property, the smallest relation S on A that contains R and posses the desired property P is called Closure of relation R with respect to that property.

$$R \subseteq S$$

Set A

↓

Relation R (does not satisfy some property P)

↓

Add some order pairs

↓

New Relation S

Where S is called closure of Relation

- Reflexive Property
- Symmetric property
- Transitive property

Gateway Classes 7455 96 12 84

Closer of Relation (S)

- Reflexive closure — R^r
- Symmetric closure — R^s
- Transitive closure — R^+

(i) Reflexive closure

The reflexive closure R^r of a relation R is the *smallest* reflexive relation that contains R as a subset.

Reflexive closure is obtained by taking union of R and I_A.

$$R^r = R \cup I_A$$

Where $I_A = \{(a, a) : a \in A\}$

= Identity Relation

$A = \{1, 2, 3\}$

$I_A = \{(1,1)(2,2)(3,3)\}$

Gateway Classes : 7455 9612 84

Q.1 Let $A = \{1, 2, 3, 4\}$ *and* $R = \{(1,1)(1,2)(2,3)\}$ be the relation on A. Find the reflexive closure of R.

Given $A = \{1, 2, 3, 4\}$

$R = \{(1,1), (1,2), (2,3)\}$

$I_A = \{(1,1), (2,2), (3,3), (4,4)\}$

$R^r = R \cup I_A$

$R^r = \{(1,1), (2,2), (3,3), (4,4), (1,2), (2,3)\}$

Gateway Classes : 7455 9612 84

(ii) Symmetric closure

The Symmetric closure R^s is the smallest symmetric relation that contains R as a subset.

The symmetric closure is obtained by taking the union of R and R^{-1}

$$R^s = R \cup R^{-1}$$

$$R^{-1} = \{(b,a) : (a,b) \in R\}$$

$$R = \{(1,2)(1,3)\}$$

$$R^{-1} = \{(2,1)(3,1)\}$$

Gateway Classes : 7455 9612 84

Q.2 If $R = \{(1,2)(4,3)(2,2)(2,1)(3,1)\}$ be a relation on A= $\{1,2,3,4\}$. Find the symmetric closure.

Given $A = \{1,2,3,4\}$

$R = \{(1,2),(4,3),(2,2),(2,1),(3,1)\}$

$R^{-1} = \{(2,1),(3,4),(2,2),(1,2),(1,3)\}$

$R^S = R \cup R^{-1}$

$$R^S = \{(1,2)(4,3)(2,2)(2,1)(3,1)(3,4)(1,3)\}$$

Gateway Classes : 7455 961284

Transitive closure

The relation obtained by adding the least number of ordered pairs to ensure transitivity is called the transitive closure of the relation.

The transitive closure of relation, R is denoted by R^+

Consider a relation, R on Set $A = \{a_1, a_2, a_3 \dots\dots\dots\dots a_m\}$

Then transitive closure of R

$$R^+ = R \cup R^2 \cup R^3 \cup \dots\dots \cup R^m$$

$R^2 = R \circ R$

$R^3 = R^2 \circ R$

$R^4 = R^3 \circ R$

$R^5 = R^4 \circ R$

Gateway Classes : 7455 961284

Q.3 Let $R=\{(1,2)(2,3)(3,1)\}$ defined on $A=\{1,2,3\}$ find the transitive closure of R .

Given

$A = \{1,2,3\}$

$R = \{(1,2)(2,3)(3,1)$

$R^+ = RUR^2UR^3$

$R^2 = R \circ R$

$R^2 = \{(1,2)(2,3)(3,1)\} \circ \{(1,2)(2,3)(3,1)\}$

$R^2 = \{(1,3),(2,1),(3,2)\}$

$R^3 = R^2 \circ R$

$R^3 = \{(1,2)(2,3)(3,1)\} \circ \{(1,3)(2,1)(3,2)\}$

$R^3 = \{(1,1)(2,2)(3,3)\}$

Gateway Classes : 7455 9612 84

$$R^+ = \{(1,2)(2,3)(3,1)\} \cup \{(1,3),(2,1),(3,2)\} \cup \{(1,1),(2,2),(3,3)\}$$

$$R^+ = \{(1,2)(2,3)(3,1)(1,3)(2,1)(3,2)(1,1)(2,2)(3,3)\}$$

Gateway Classes : 7455 9612 84

Warshall's Algorithm to find transitive Closure

A more efficient method to calculate transitive closure of relation

Step-1: Let n be the number of elements in a given Set A

To find transitive closure of Relation R on A, maximum n warshall sets can be find.

$W_0, W_1, W_2 \ldots\ldots\ldots\ldots W_n$ $(Here\ W_0 = M_R)$

Step-2: Procedure to calculate W_k from W_{K-1}

(i) Copy I from W_{K-1} in W_k with same position

(ii) In K^{th} row and K^{th} column of W_{K-1}, see K^{th} column and check the position of 1 and put below C

See K^{th} row and check the position of 1 and put below R

(iii) Mark entries in W_k as 1 for (C, R) of W_k if there are not already 1.

Step-3: Stop when W_n is obtained and it the required transitive closure.

Gateway Classes : 7455 961284

Q.4 Let $A = \{1, 2, 3, 4\}$ *and* $R = \{(1,4)(2,1)(2,3)(3,1)(3,4)(4,3)\}$ be a relation on A.

Find transitive closure using Warshall's Algorithm.

Given

$A = \{1, 2, 3, 4\}$

$R = \{(1,4)(2,1)(2,3)(3,1)(3,4)(4,3)\}$

$n(A) = 4$

$M_R =$

	1	2	3	4
1	0	0	0	1
2	1	0	1	0
3	1	0	0	1
4	0	0	1	0

$W_0 =$

	1	2	3	4
1	0	0	0	1
2	1	0	1	0
3	1	0	0	1
4	0	0	1	0

From First colum and First Row

$W_1 =$

	1	2	3	4
1	0	0	0	1
2	1	0	1	1
3	1	0	0	1
4	0	0	1	0

C: {2,3} R: {4}

$C \times R = \{(2,4)(3,4)\}$

Gateway Classes : 7455 961284

From second column and second Row

$$W_2 = \begin{array}{c} \\ 1 \\ 2 \\ 3 \\ 4 \end{array} \begin{array}{c} \begin{array}{cccc} 1 & 2 & 3 & 4 \end{array} \\ \begin{bmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \end{array}$$

C = { } R = {1, 3, 4}

CXR = { }

From Third column and third Row

$$W_3 = \begin{array}{c} \\ 1 \\ 2 \\ 3 \\ 4 \end{array} \begin{array}{c} \begin{array}{cccc} 1 & 2 & 3 & 4 \end{array} \\ \begin{bmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 1 \end{bmatrix} \end{array}$$

C = {2, 4} R = {1, 4}

CXR = {(2,1) (2,4) (4,1) (4,4)}

From Fourth column and Fourth Row

$$W_4 = \begin{array}{c} \\ 1 \\ 2 \\ 3 \\ 4 \end{array} \begin{array}{c} \begin{array}{cccc} 1 & 2 & 3 & 4 \end{array} \\ \begin{bmatrix} 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 \end{bmatrix} \end{array}$$

C = {1, 2, 3, 4} R = {1, 3, 4}

CXR = {(1,1) (1,3) (1,4) (2,1) (2,3) (2,4) (3,1) (3,3) (3,4) (4,1) (4,3) (4,4)}

Transitive closure

$$R^+ = \{(1,1)\ (1,3)\ (1,4)\ (2,1)\ (2,3)\ (2,4)\ (3,1)\ (3,3)\ (3,4)\ (4,1)\ (4,3)\}$$

Gateway Classes : 7455 9612 84

Discrete Mathematics by Gulshan Sir

DPP- 12

Topic : *Closure of Relation and* Warshall's Algorithm

Q.1. Let $A = \{1, 2, 3, 4\}$ *and* $R = \{(1, 2)(2, 3)(3, 4)\}$ *be a relation on set A. Find the transitive closure.*

Q.2. Let $A = \{4, 6, 8, 10\}$ *and* $R = \{(4, 4)(4, 10)(6, 6)(6, 8)(8, 10)\}$ is a relation on set A. Determine the transitive closure of R using Warshall's Algorithm.

Q.3. Let $R = \{(1, 2)(2, 3)(3, 1)\}$ defined on $A = \{1, 2, 3\}$ find the transitive closure of R using Warshall's Algorithm.

Gateway Classes : 7455 961284

GATEWAY CLASSES

GATEWAY CLASSES

AKTU Full Courses (Paid)

Download Gateway Classes Application

From Google Play store

All Subjects

Link in Description

Gateway Classes : 7455961284

Thank You

GATEWAY CLASSES

GATEWAY CLASSES

DISCRETE STRUCTURES & THEORY OF LOGICS (BCS303)

UNIT -1 : (i) Set Theory & Relations

(ii) POSET & Lattices

Today's Target

- Partial order relation
- Partial order set or POSET
- Graphical representation of POSET - Hasse Diagram.
- Hasse Diagram from Directed graph.
- Easiest way to draw Hasse diagram

Gateway Classes : 7455 961284

Course Details(Paid) : All Subjects

1	Recorded Video Lectures (100 % Syllabus Coverage)
2	Pdf Notes
3	Lecture wise DPP
4	Unit wise set of PYQs

Paid Courses are available in Gateway Classes Application

Link in Description

Gateway Classes : 7455 961 284

Partial order relation

A relation R on set A is called partial order relation if it is

(i) Reflexive $(a, a) \in R \quad \forall \; a \in A$

(ii) Anti-symmetric $(a, b) \in R \text{ and } (b, a) \in R \Rightarrow a = b \quad \forall \; a, b \in A$

(iii) Transitive $(a, b) \in R \text{ and } (b, c) \in R \Rightarrow (a, c) \in R \quad \forall \; a, b, c \in A$

Partial order set or POSET

If A is any non-empty set and R is a partial order relation on set A, then the order pair (A, R) is called Partial order set or POSET

POSET is denoted by (A, R) Or (A, ≤)

Gateway Classes : 7455 9612 84

Hasse Diagram

Relation

A partial order ≤ on a set A can be represented by means of a diagram known as Hasse Diagram of (A, ≤)

i.e Hasse diagram is a graphical representation of POSET

Hasse Diagram from directed graph or digraph

Step 1:- Start with directed Graph

Step 2:- Remove self loop

Reason : Self loops are quite obvious and they must be present for all vertices because R is reflexive, so we can remove them to simply diagram.

Gateway Classes : 7455 9612 84

Step 3:- Remove all transitive edges.

Reason: Transitive edges are quite obvious and they must be present because R is transitive. So, we can remove all transitive edges to Simply diagram.

Step 4:- Remove circle by dot

Step-5: Arrange each edge so that its initial vertex is below its terminal vertex i.e. all edges are pointing upwards.

Step-6: Remove all the arrows the directed graph.

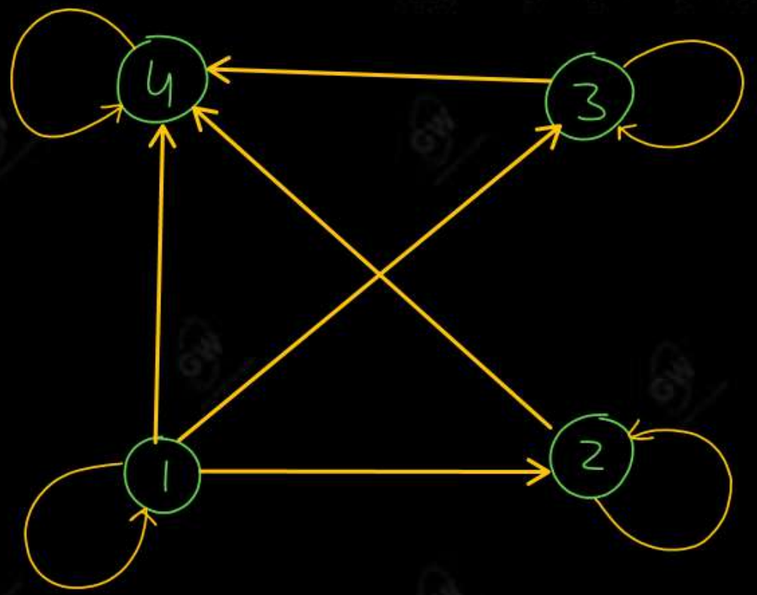
Reason: since all edges are pointing upwards, so there is no need to show the directions in the graph.

Gateway Classes : 7455 9612 84

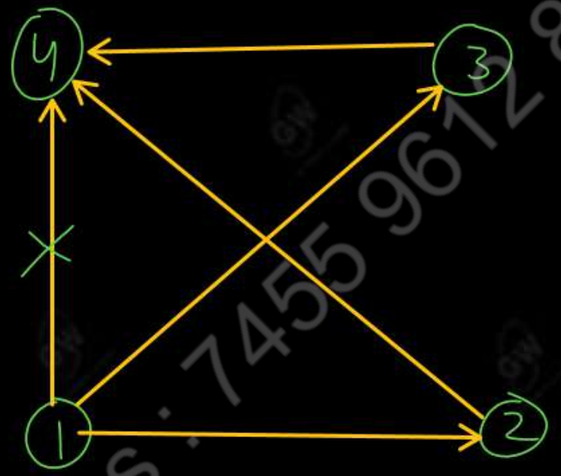
Q.1 Determine the Hasse diagram of the relation R on set $A = \{1, 2, 3, 4\}$

(A, R)

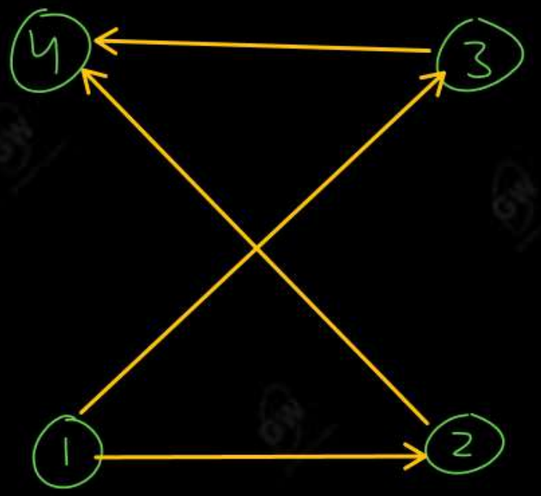
$$R = \{(1,1), (1,2), (2,2), (2,4), (1,3), (3,3), (3,4), (1,4), (4,4)\}$$



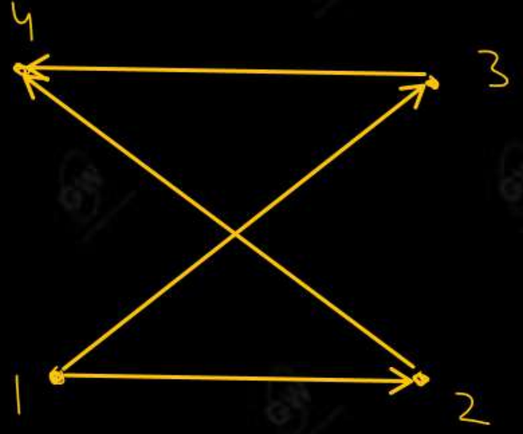
=



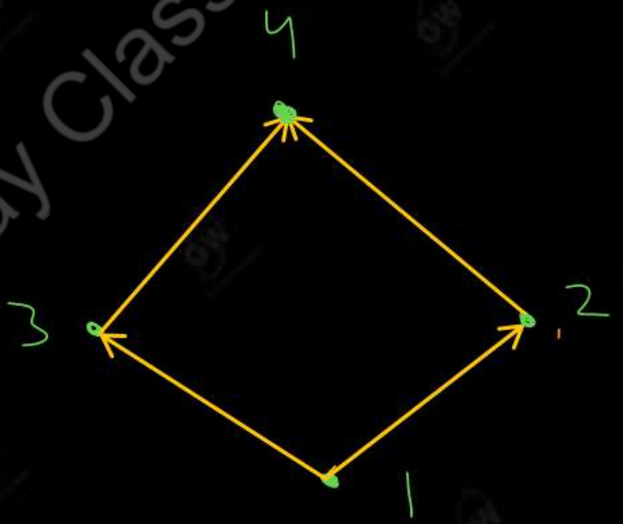
=



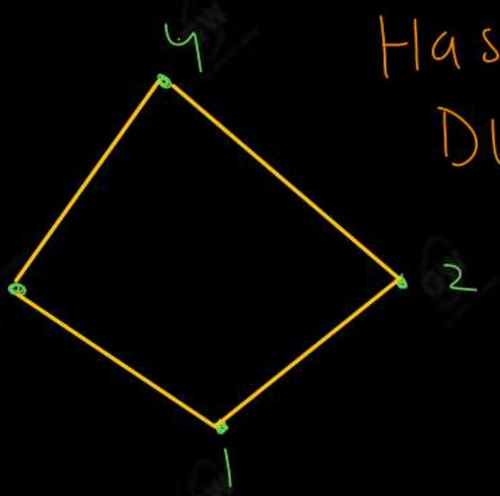
=



=



=



Hasse Diagram

Gateway Classes : 7455 9612 84

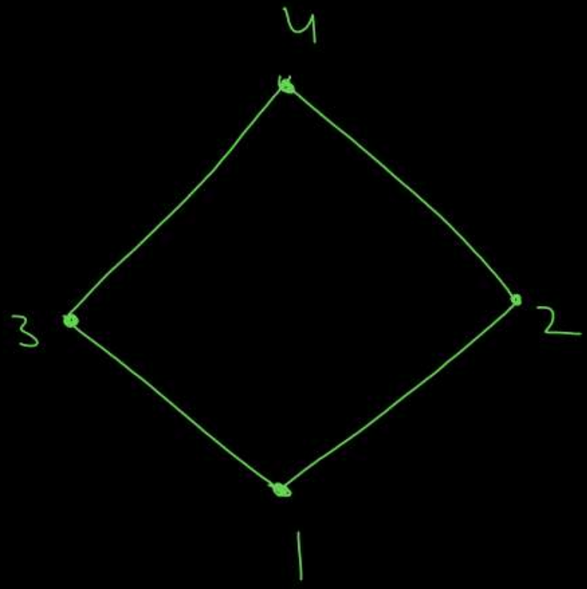
Easy Method

$R = \{(1,1)(1,2)(2,2)(2,4)(1,3)(3,3)(3,4)(1,4)(4,4)\}$

$R = \{(1,2)(2,4)(1,3)(3,4)(1,4)\}$

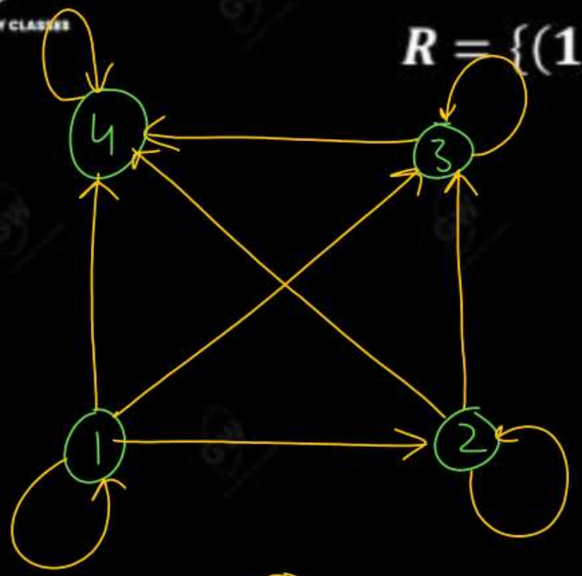
$R = \{(1,2)(2,4)(1,3)(3,4)\}$

Gateway Classes : 7455 9612 84

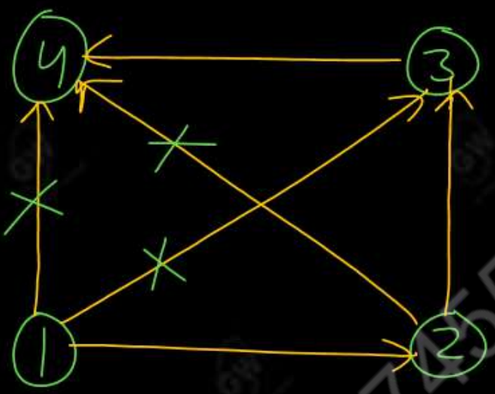


Q.2 *Determine the Hasse diagram of the relation R on set* $A = \{1, 2, 3, 4\}$ (A, R)

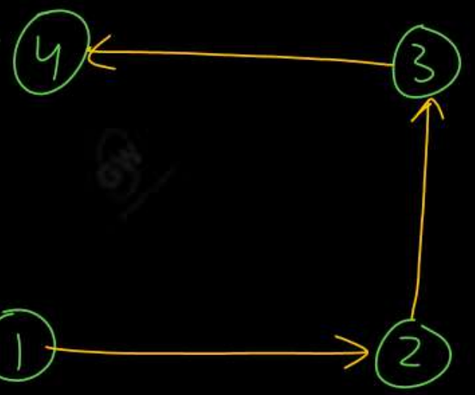
$$R = \{(1,1), (1,2), (1,3), (1,4), (2,2), (2,3), (2,4), (3,3), (3,4), (4,4)\}$$



=



=



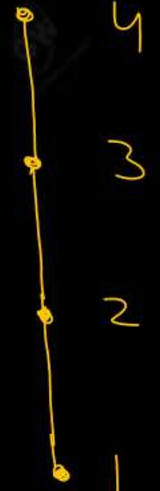
=



=



=



Hasse Diagram

Gateway Classes : 7455 961284

Easy Method

$$R = \{(1,2)\ (1,3)\ (1,4)\ (2,3)\ (2,4)\ (3,4)\}$$

(1,3), (1,4), (2,4) are marked with X

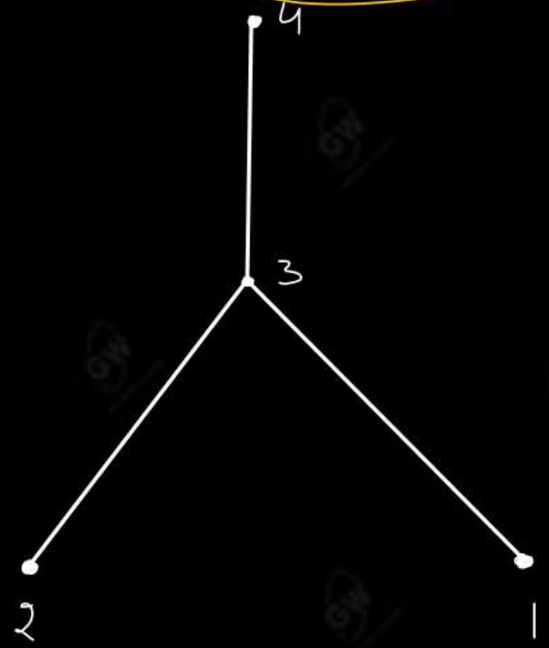
$$R = \{(1,2)\ (2,3)\ (3,4)\}$$



Hasse Diagram

Gateway Classes : 7455 9612 84

Q.3- Describe the order pairs in the relation determined by the Hasse diagram of a POSET (A, ≤) on the set $A = \{1, 2, 3, 4\}$



solution

From Hasse Diagram

(1, 3) (2, 3) (3, 4)

Reflexive order pair

(1, 1) (2, 2) (3, 3) (4, 4)

Transitive Pair

(2, 4) (1, 4)

$$R = \{(1,1)(2,2)(3,3)(4,4)(1,3)(2,3)(3,4)(2,4)(1,4)\}$$

Gateway Classes 7455 9012 84

Easiest Way to draw the Hasse diagram

Q.4 Draw the Hasse diagram for the following POSET

(i) $(A, \leq)$ where $A = \{1, 2, 3, 4, 5, 6\}$

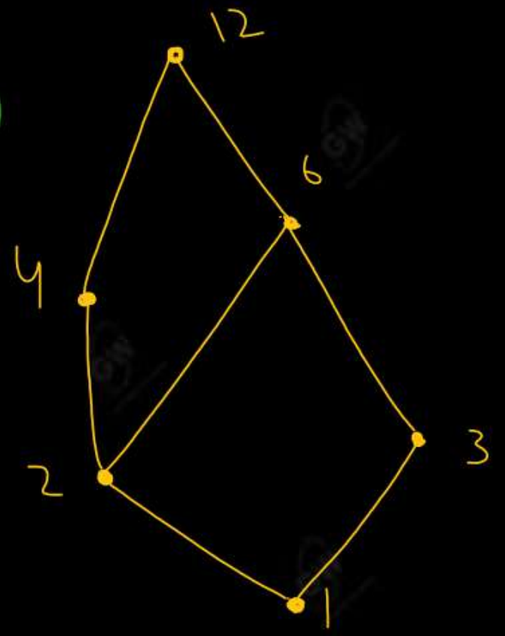
(ii) (A, I) where $A = \{1, 2, 3, 4, 6, 12\}$

(iii) $(A, \subseteq)$ where $A = \{\phi, \{1\}, \{2\}, \{1, 2\}\}$

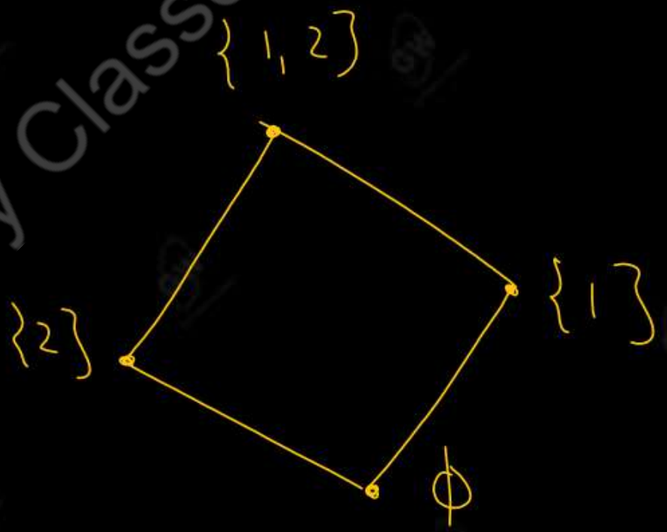
(i)



(ii)



(iii)



Gateway Classes : 7455 9612 84

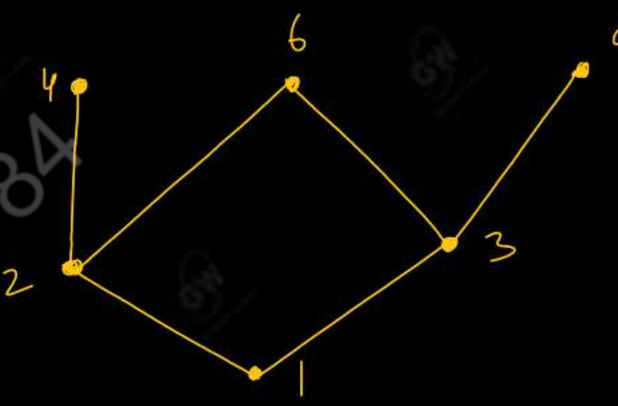
(iv) (A, I) where $A = \{1, 2, 3, 4, 6, 9\}$

(v) (A, I) where $A = \{2, 3, 5, 30, 60, 120, 180, 360\}$

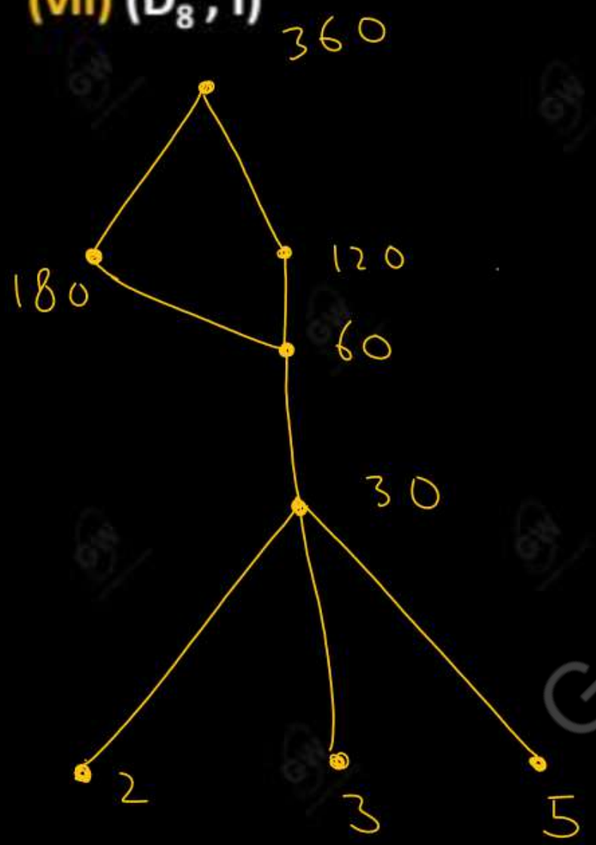
(vi) (A, I) where $A = \{1, 2, 3, 5, 6, 10, 15, 30\}$

(vii) (D_8, I)

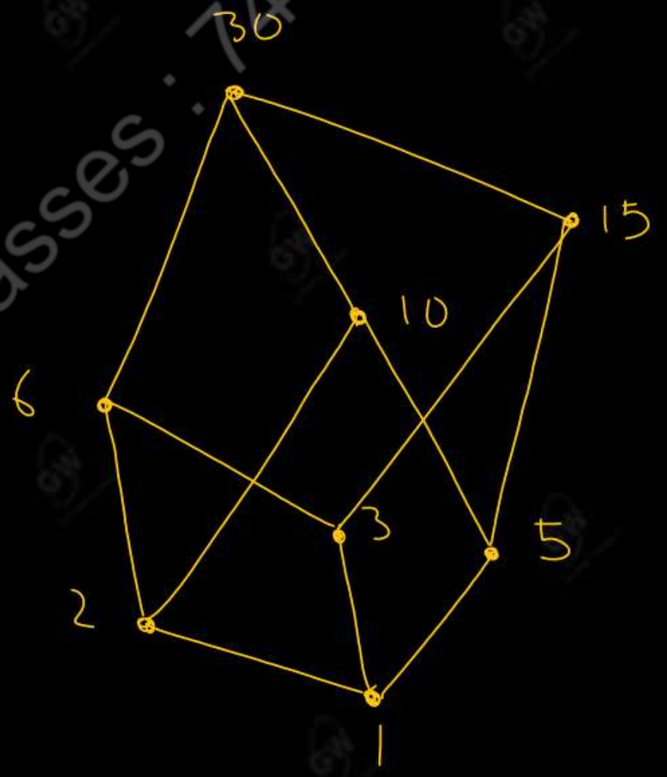
(IV)



(V)



(VI)



(VII) (D_8, I)

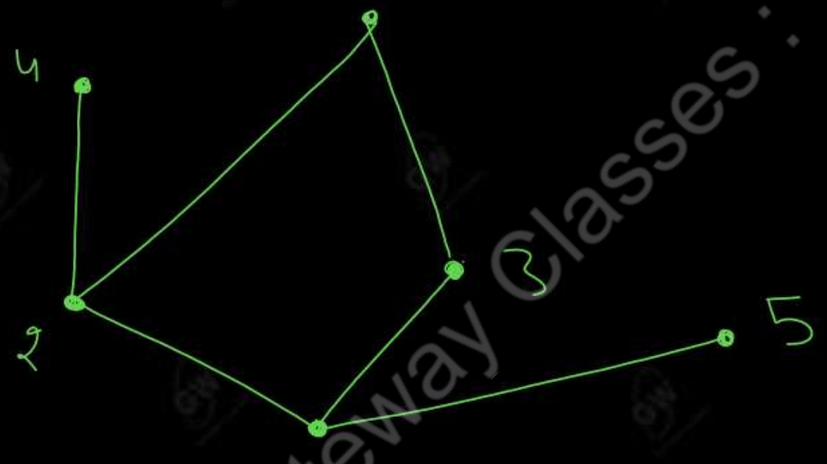
$A = \{1, 2, 4, 8\}$



Gateway Classes : 7455 9612 84

Q.5- Let $X = \{1, 2, 3, 4, 5, 6\}$, then I is a partial order relation on X. Draw the Hasse diagram of (X, I)

Hasse Diagram



Gateway Classes : 7455 9612 84

Q.6- Draw the Hasse diagram for the POSET $(P(s), \subseteq)$ where $P(s)$ is the power set on $A = \{a, b, c\}$

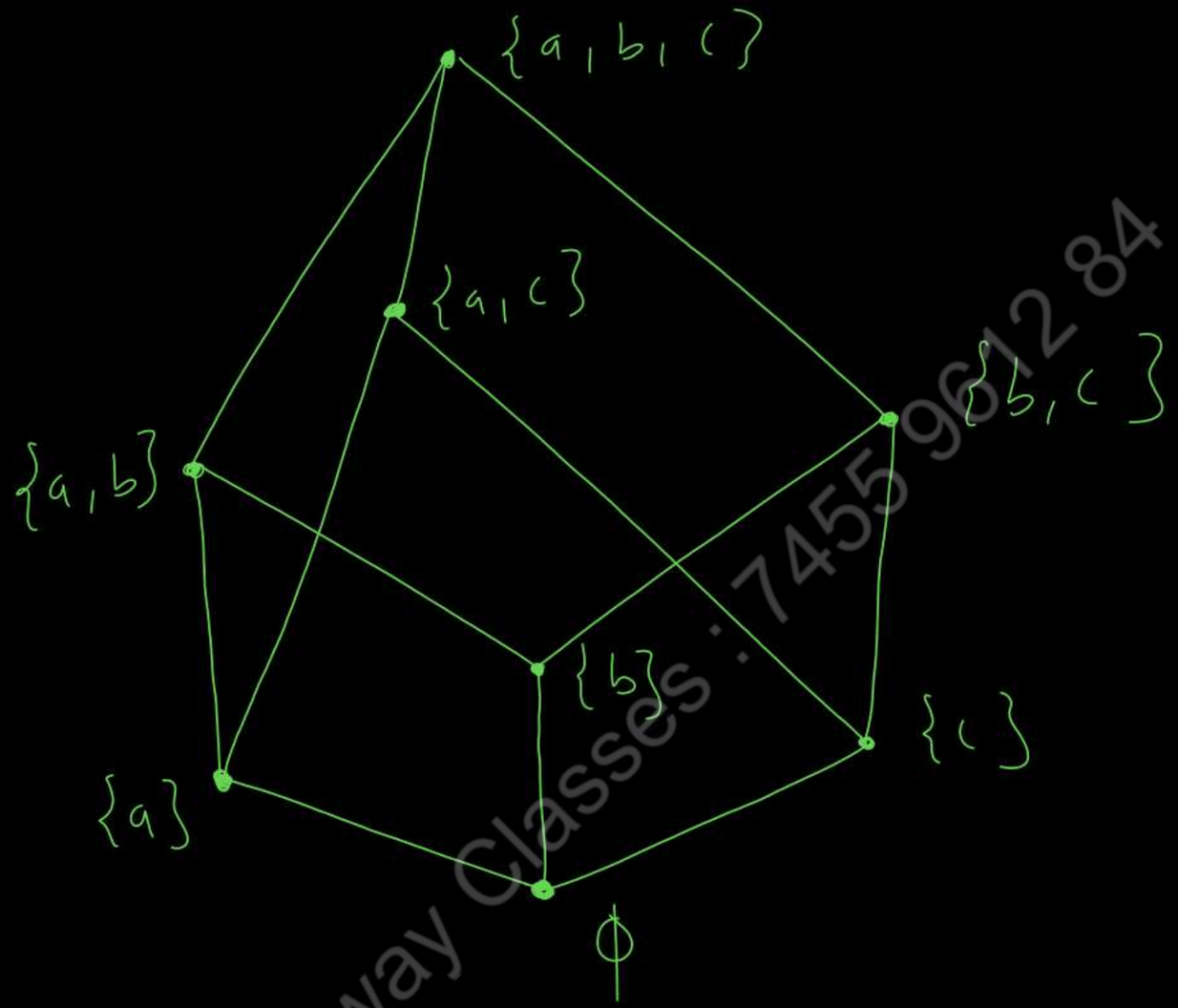
$A = \{a, b, c\}$

Subset of A

$\phi, \{a\}, \{b\}, \{c\}, \{a, b\}, \{b, c\}, \{a, c\}, \{a, b, c\}$

$P(s) = \{\phi, \{a\}, \{b\}, \{c\}, \{a, b\}, \{b, c\}, \{a, c\}, \{a, b, c\}\}$

Gateway Classes : 7455 9612 84



Gateway Classes : 7455 96 12 84

Relation

Q.7- Let $X = \{1, 3, 6, 12, 24, 36\}$ and the relation $\leq$ be such that $x \leq y$ if 'x *divides* y'

$x R y$

Draw the Hasse diagram.



Gateway Classes : 7455 9612 84

DPP- 13

Topic : Hasse's diagram

Q.1. Draw the Hasse's diagram of the POSET (L, $\subseteq$) Where

$L = \{s_0, s_1, s_2, s_3, s_4, s_5, s_6, s_7\}$ where the sets are given by

$S_0 = \{a, b, c, d, e, f\}$, $S_1 = \{a, b, c, d, e\}$, $S_2 = \{a, b, c, e, f\}$, $S_3 = \{a, b, c, e\}$, $S_4 = \{a, b, c\}$, $S_5 = \{a, b\}$,

$S_6 = \{a, c\}$, $S_7 = \{a\}$

Q.2. Let $A = \{1, 2, 3, 4, 6, 8, 9, 12, 18, 24\}$ be ordered by the relation 'a divides b' find the Hasse diagram.

Q.3 *Draw the Hasse diagram of the relation S defined as "divides" on set B where*

$B = \{2, 3, 4, 6, 12, 36, 48\}$

Q.4 *Draw Hasse diagram for the following relations on set*

$A = \{1, 2, 3, 4, 12\}$

$R = \{(1,1), (2,2), (3,3), (4,4), (12,12), (1,2), (4,12), (1,3), (1,4), (1,12), (2,4), (2,12), (3,12)\}$

Gateway Classes : 7455 9612 84

GATEWAY CLASSES

GATEWAY CLASSES

AKTU Full Courses (Paid)

Download Gateway Classes Application

From Google Play store

All Subjects

Link in Description

Gateway Classes : 7455961284

Thank You

GATEWAY CLASSES

GATEWAY CLASSES

DISCRETE STRUCTURES & THEORY OF LOGICS (BCS303)

UNIT -1 : (i) Set Theory & Relations

(ii) POSET & Lattices

Today's Target

- Maximal and Minimal elements in POSET
- Maximum and Minimum elements in POSET
- Upper Bound and lower Bound in Hasse diagram
- Least upper Bound and Greatest Lower Bound in Hasse diagram
- PYQ
- DPP

Gateway Classes: 7455 98128 4

Course Details(Paid) : All Subjects

1	Recorded Video Lectures (100 % Syllabus Coverage)
2	Pdf Notes
3	Lecture wise DPP
4	Unit wise set of PYQs

Paid Courses are available in Gateway Classes Application

Link in Description

Gateway Classes : 7455 961 284

Maximal and Minimal elements in POSET

Maximal element : *If in a POSET, an element is not related to any other element, then it is called maximal elements*

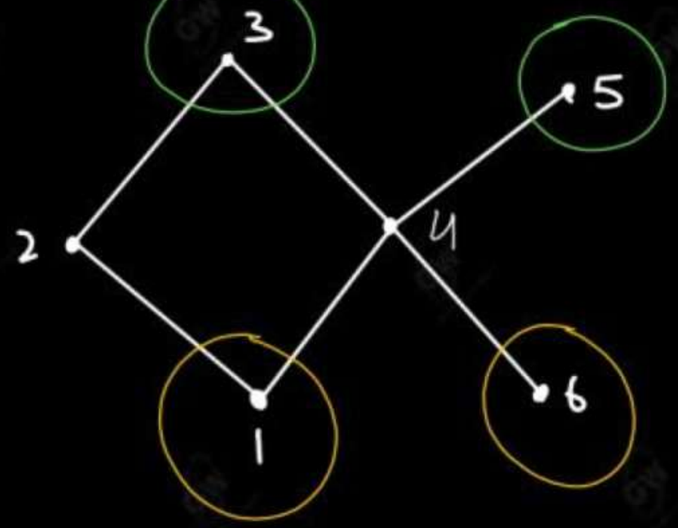
Green circle

Minimal elements : *If in a POSET, no element is related to any other element, then it is called minimal elements*

Yellow circle

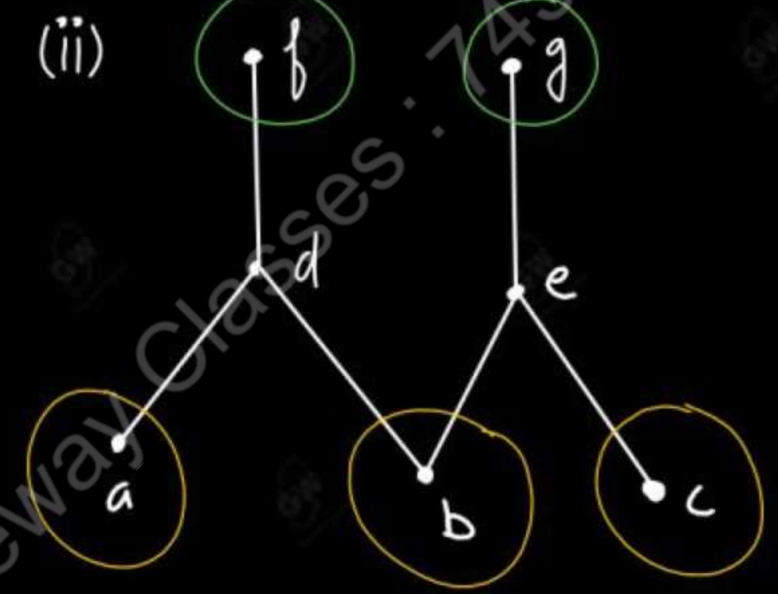
Q.1: *Find all the maximal and minimal elements of the POSET whose Hasse diagrams are given below.*

(i)

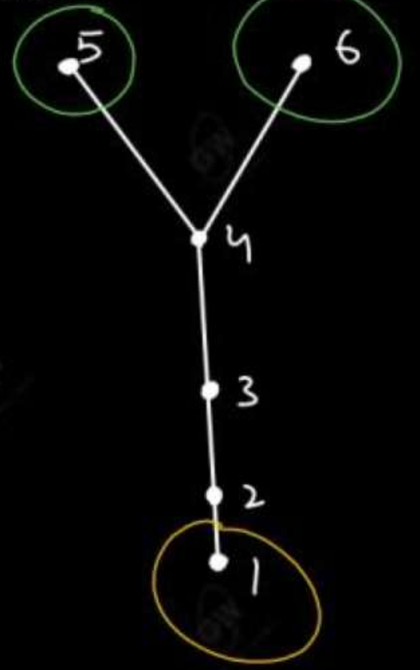


Maximal elements = 3, 5

(ii)



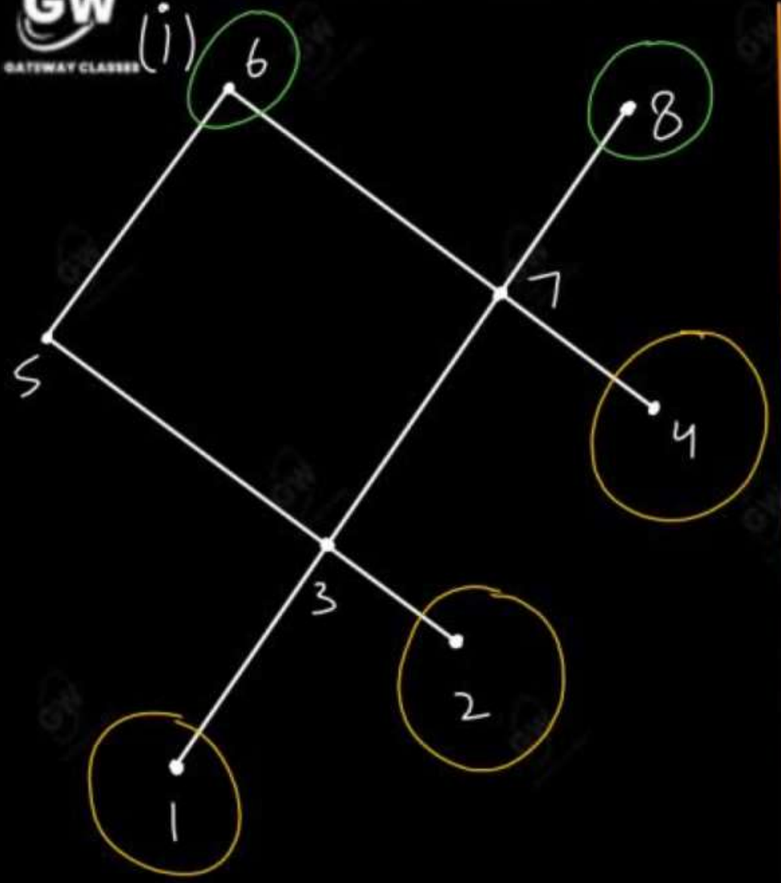
(iii)



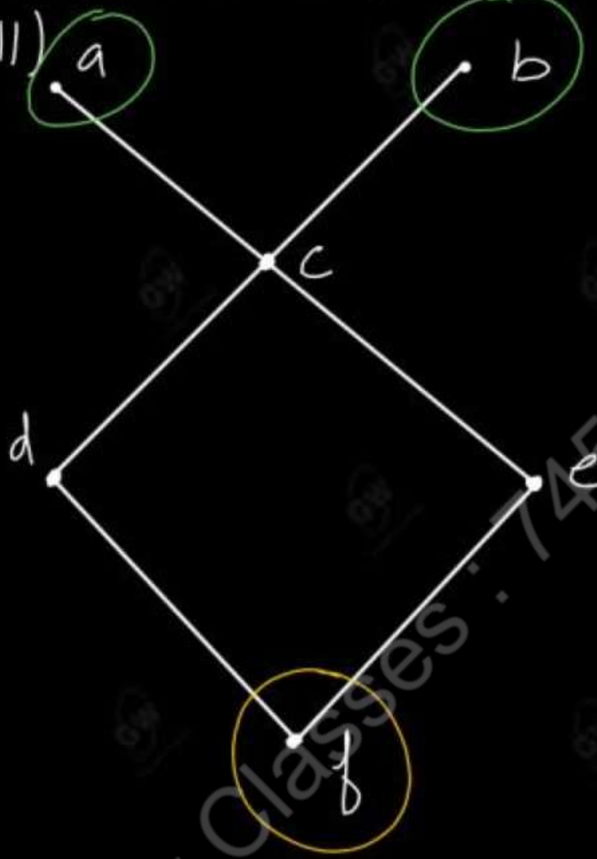
Gateway Classes : 7455961284

Q.2: *Find all the maximal and minimal elements of the POSET whose Hasse diagrams are given below.*

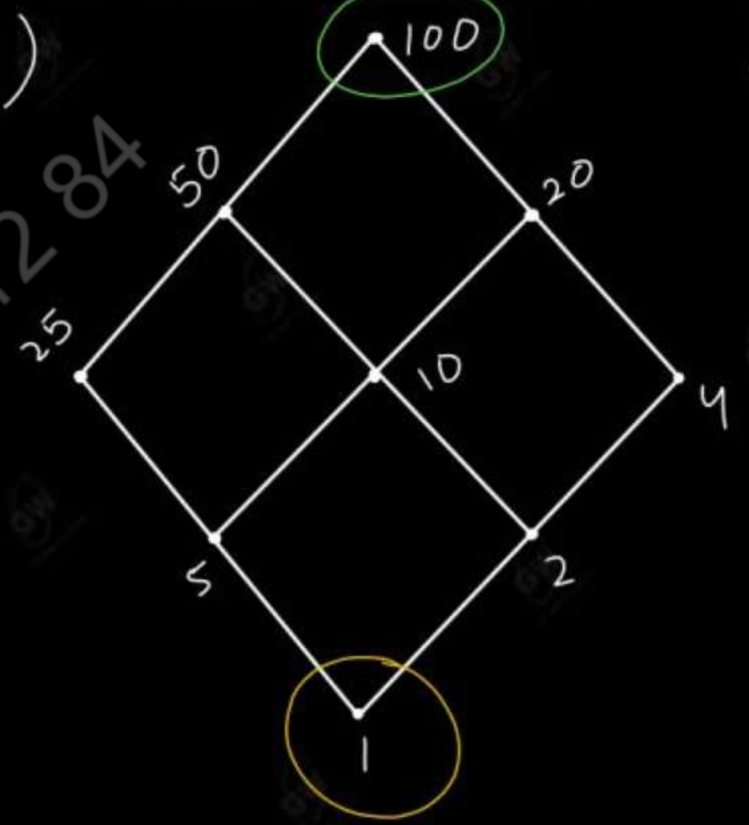
(i)



(ii)



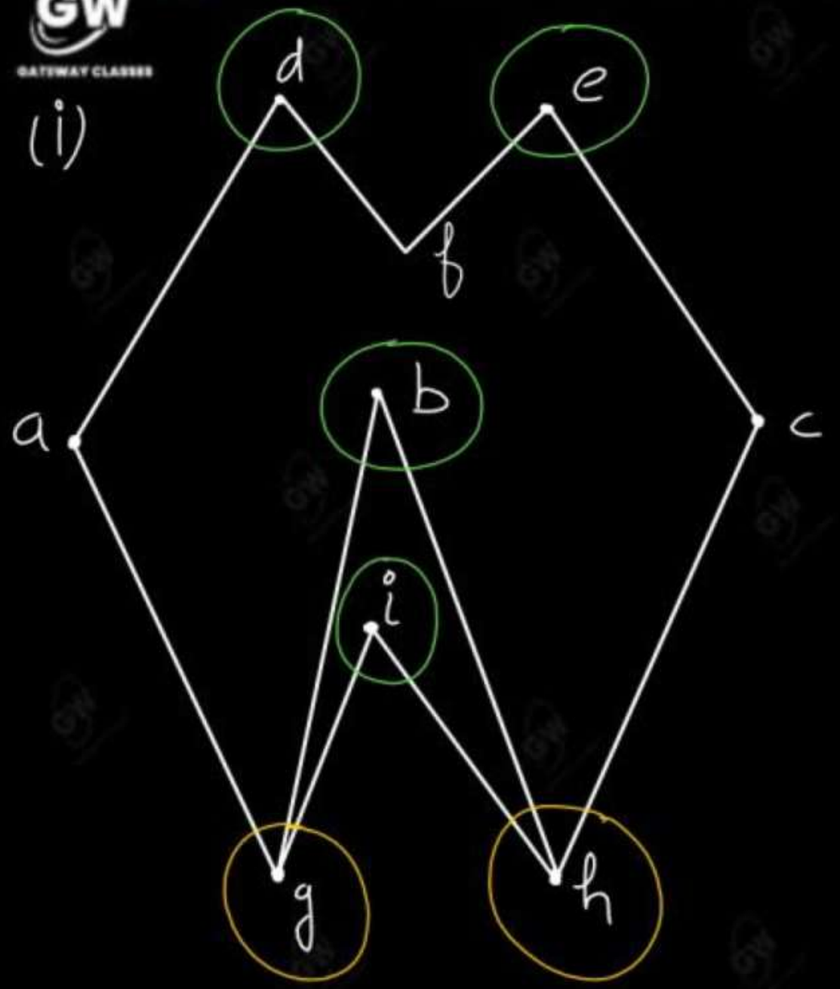
(iii)



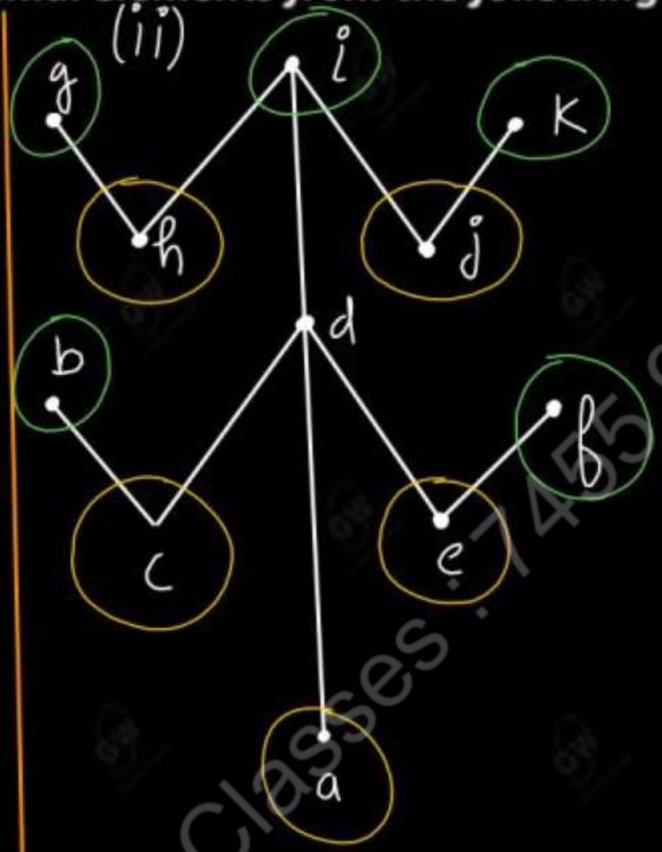
Gateway Classes : 7455 9612 84

Q.3:- ***Find all the maximal and minimal elements from the following Hasse diagrams.***

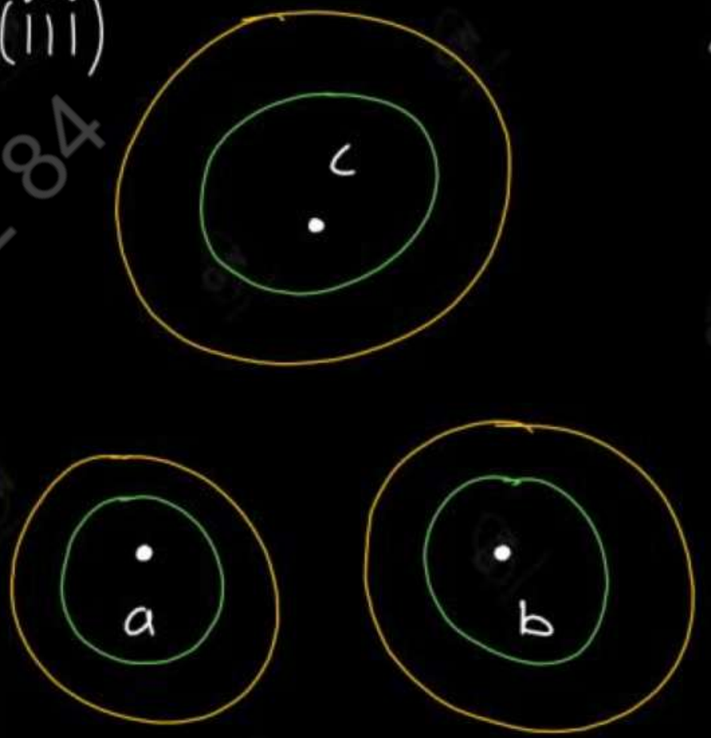
(i)



(ii)



(iii)



Maximal = a, b, c

Minimal = a, b, c

Gateway Classes : 7455 9612 84

Maximum and Minimum element

Maximum element / Greatest element: If it is maximal and every element is related to it

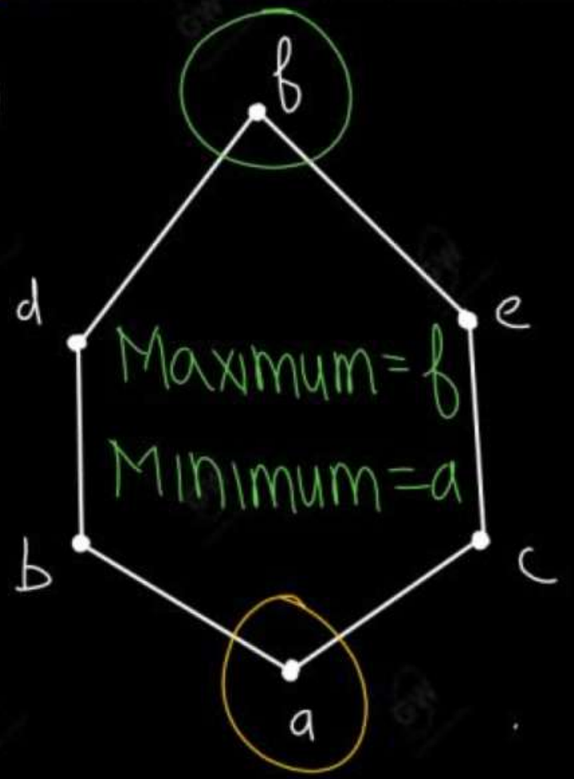
Minimum elements / Least element : If it is minimal and it is related to every element.

Maximum = Null

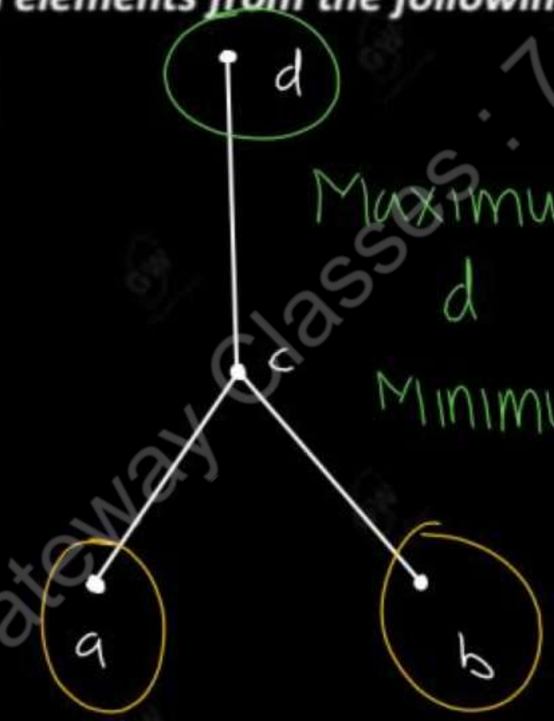
Minimum = Null

Q.4 : Find the maximum and minimum elements from the following Hasse diagrams

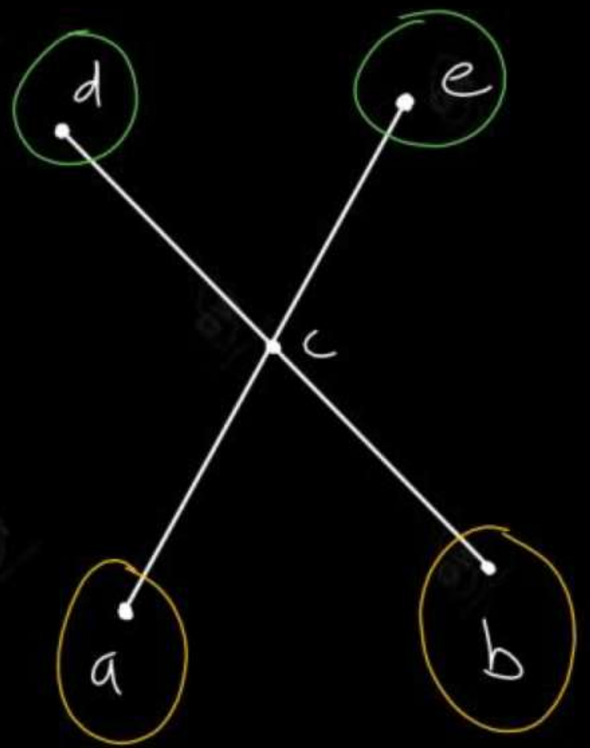
(i)



(ii)



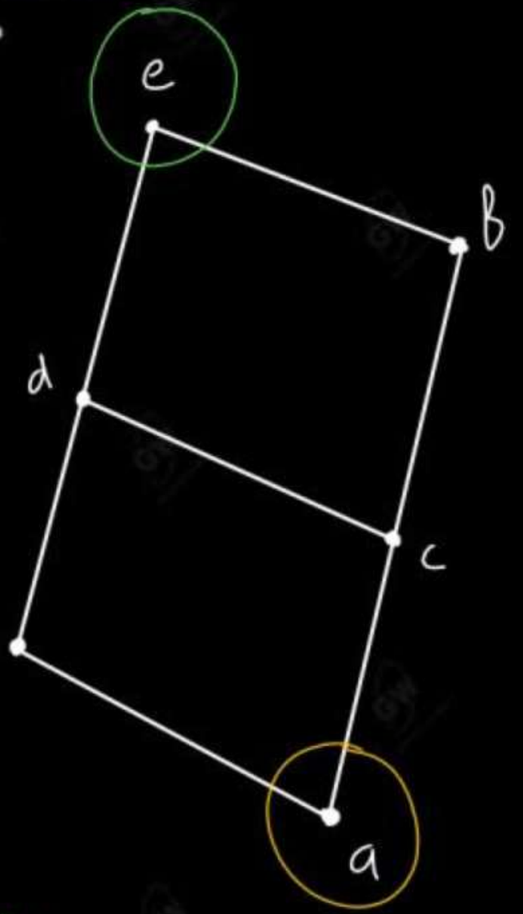
(iii)



Gateway Classes : 7455 961 284

Q.5 : Find the maximum and minimum elements from the following Hasse diagrams

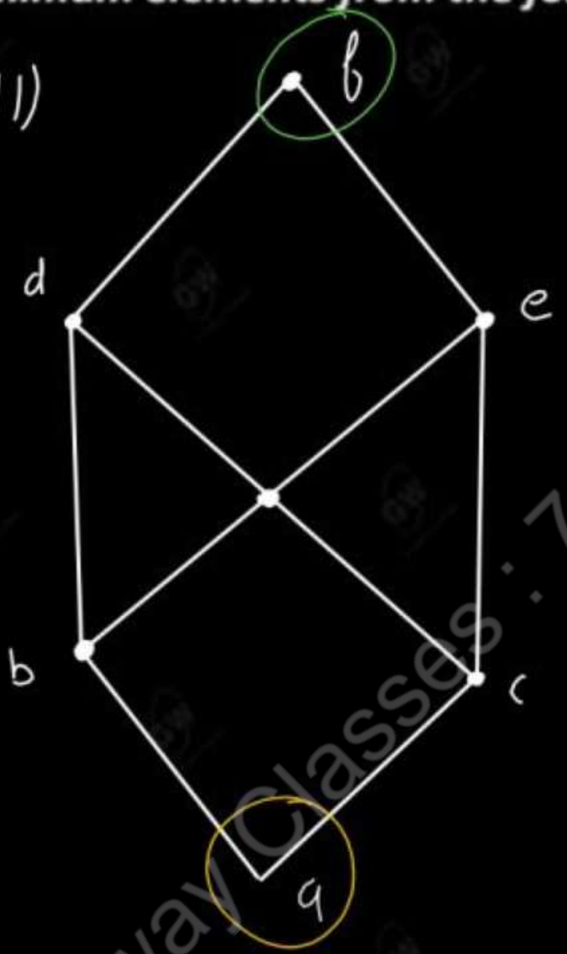
(i)



Maximum = e

Minimum = a

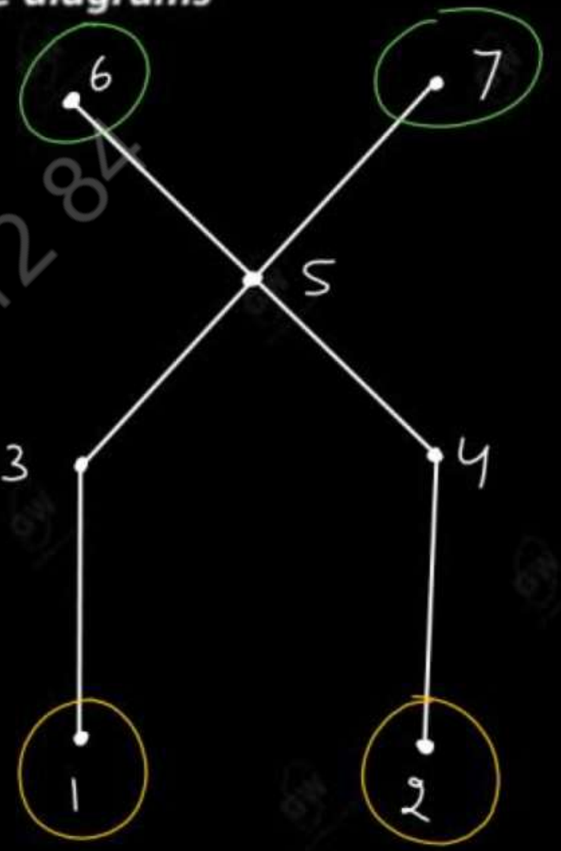
(ii)



Maximum = m

Minimum = a

(iii)



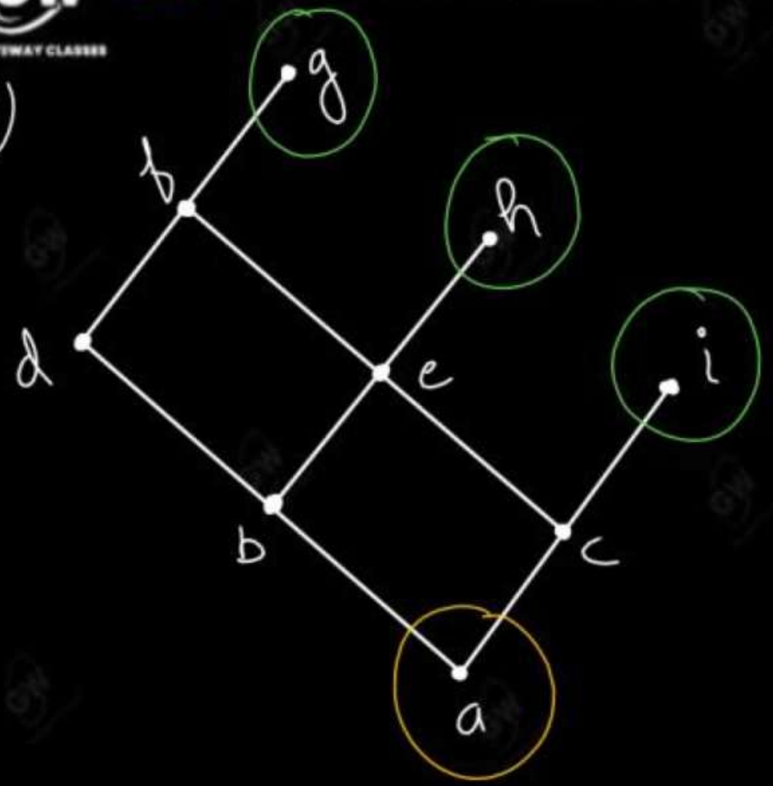
Maximum = X

Minimum = X

Gateway Classes : 7455 9612 84

Q.6 : Find the Greatest and Least elements of the following Hasse diagrams.

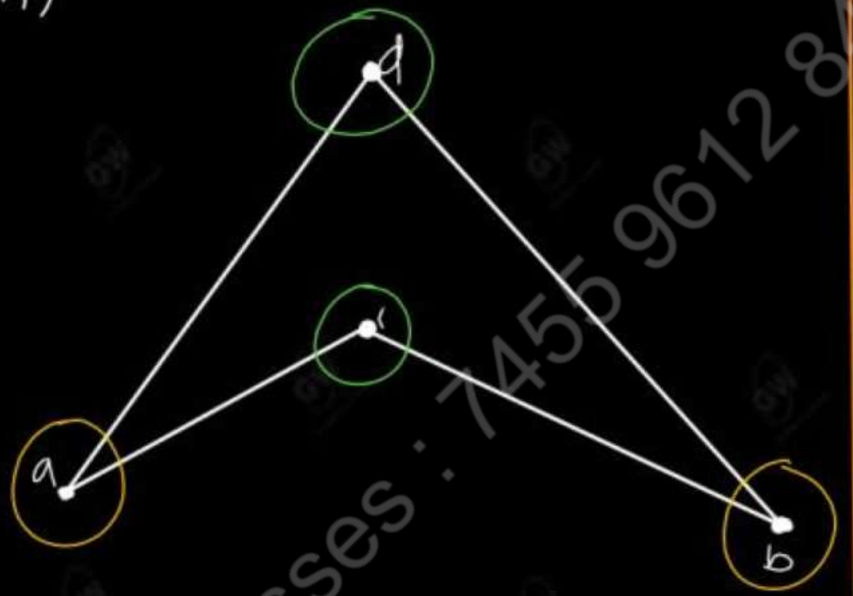
(i)



Maximum = X

Minimum = a

(ii)



Maximum = X

Minimum = X

(iii)



Maximum = X

Minimum = X

Gateway Classes : 7455 9612 84

Upper Bound ands Lower Bound

$B \subseteq A$

Upper Bound

Let $(A, \leq)$ be POSET and let B be a subset of A. An element $x \in A$ is called an upper bound of B if x succeeds every element of B. i. e

$$\text{i. e } a \leq x \quad \forall \quad a \in B$$

Lower Bound

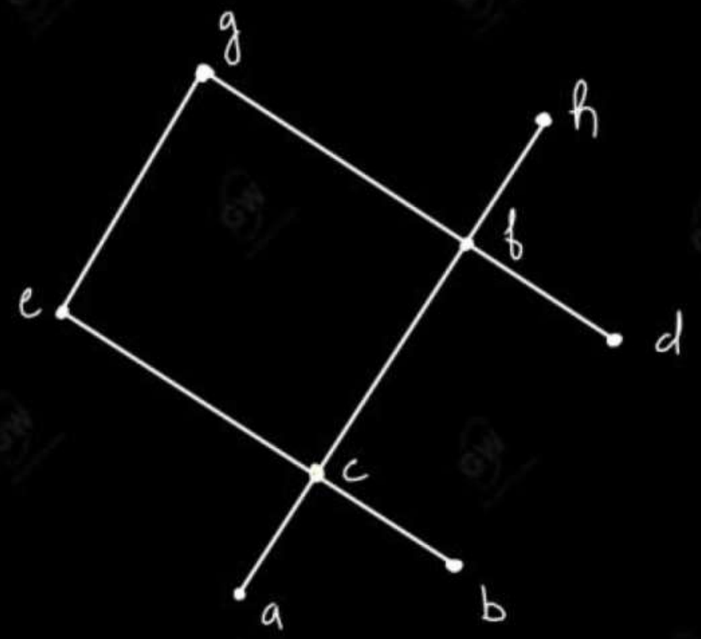
Let $(A, \leq)$ be a POSET and let B be a subset of A. An element $x \in A$ is called a Lower bound of B if x preceeds every element of B.

$$\text{i. e } x \leq a \quad \forall \quad a \in B$$

Gateway Classes : 7455 9612 84

Q.7 Find the upper Bound and Lower Bound of set B from the following

(i)



$B = \{e, c\}$

$LB = a, b, c$

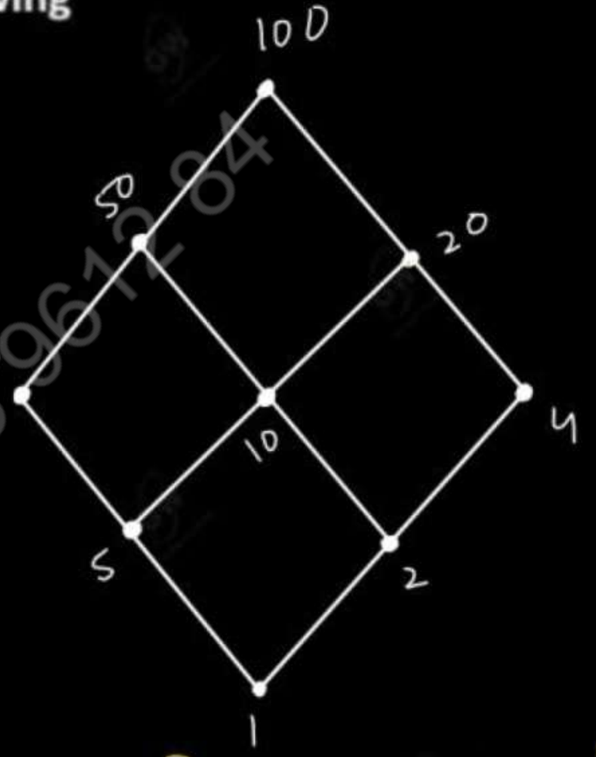
$UB = g$

$B = \{c, f, d\}$

$LB = \phi$

$UB = f, g, h$

(ii)



$B = \{5, 10\}$

$LB = 1, 5$

$UB = 10, 20, 50, 100$

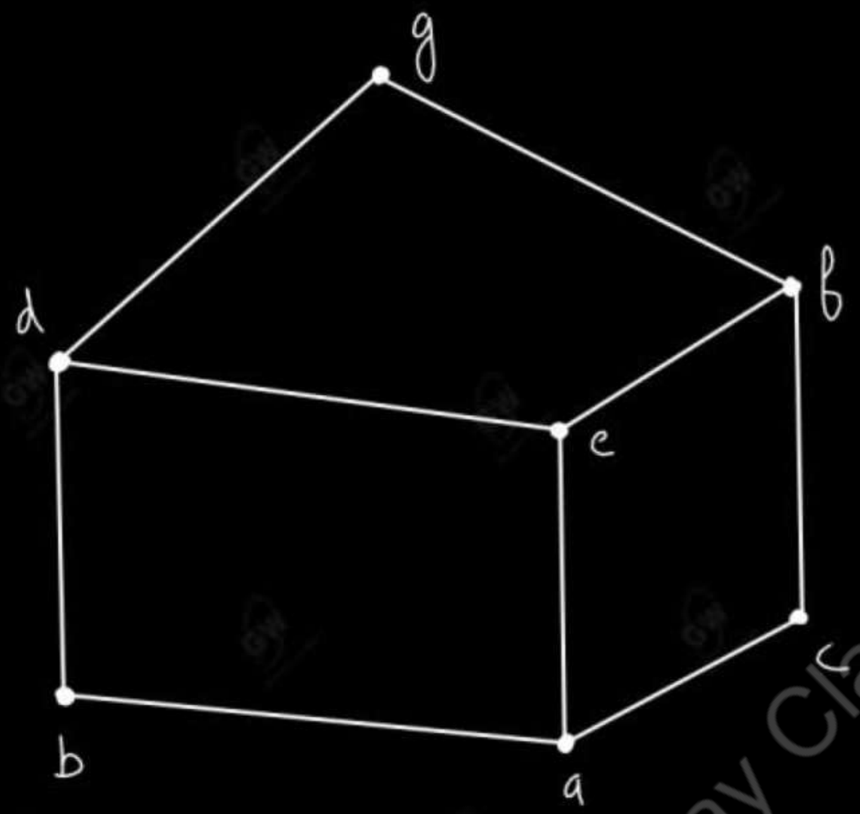
$B = \{5, 10, 2, 4\}$

$LB = 1$

$UB = 20, 100$

Gateway Classes : 7455961284

Q.8 Find the upper Bound and Lower Bound of set B from the following



$B = \{d, g\}$

$LB = b, a, \cancel{g}\ e, d$

$UB = g$

$B = \{e, f\}$

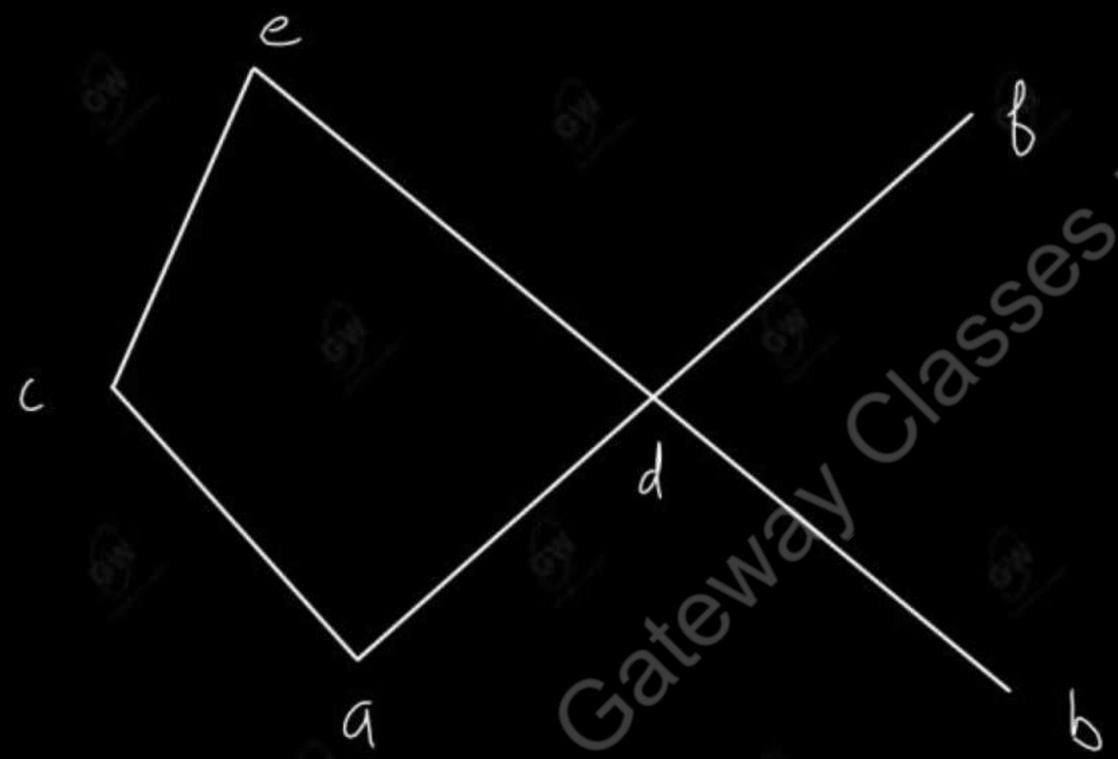
$LB = a, e$

$UB = g, f$

Gateway Classes : 7455961284

1. **Least Upper Bound / LUB/ Supremum/ Join/($\vee$ or $\cup$)** : Least (minimum) element in upper Bound
2. **Greatest Lower Bound/GLB/Infimum/Meet /($\wedge$ or $\cap$)** : Greatest (maximum) element in lower Bound

Q.9 **Find the Least upper Bound and Greatest Lower Bound of set B from the following**



$B = \{c, d\}$

UB = e

LUB = e

LB = a

GLB = a

$B = \{a, b\}$

UB = d, e, f

LUB = d

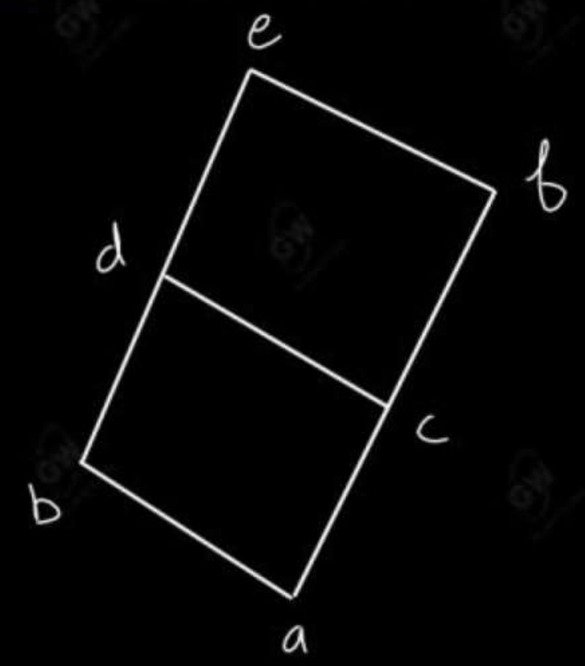
LB = ϕ

GLB = ϕ

Gateway Classes : 7455 9612 84

Q.10 Find the Least upper Bound and Greatest Lower Bound of set B from the following

(i)



$B = \{a, c\}$

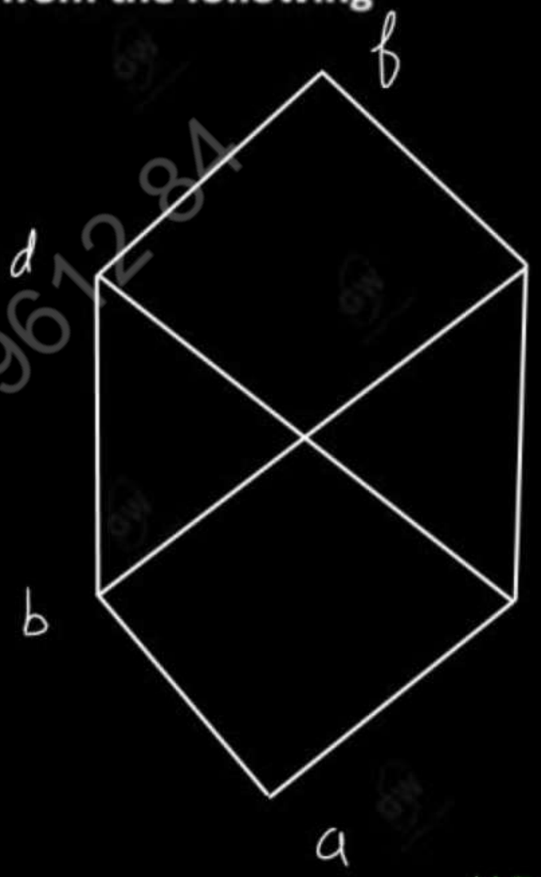
$UB = c, f, d, e$

$LUB = c$

$LB = a$

$GLB = a$

(ii)



$B = \{d, e\}$

$UB = f$

$LUB = f$

$LB = a, b, c$

$GLB = \phi$

$B = \{b, c\}$

$UB = d, e, f$

$LUB = \phi$

$LB = a$

$GLB = a$

Gateway Classes : 7455 961284

Discrete Mathematics by Gulshan Sir

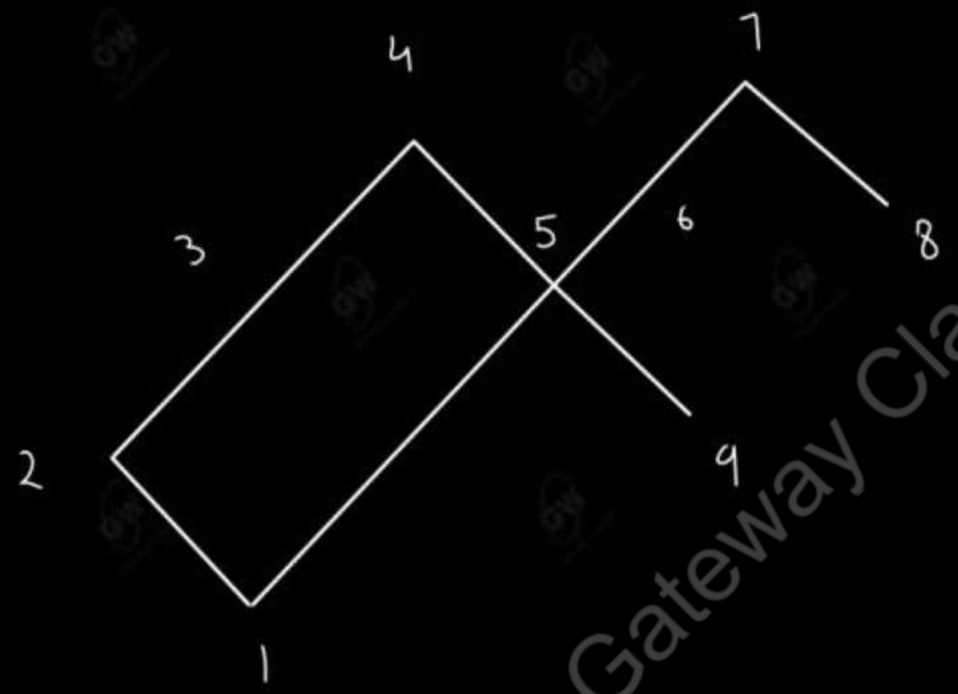
DPP- 10

Topic : Partial Order Relation

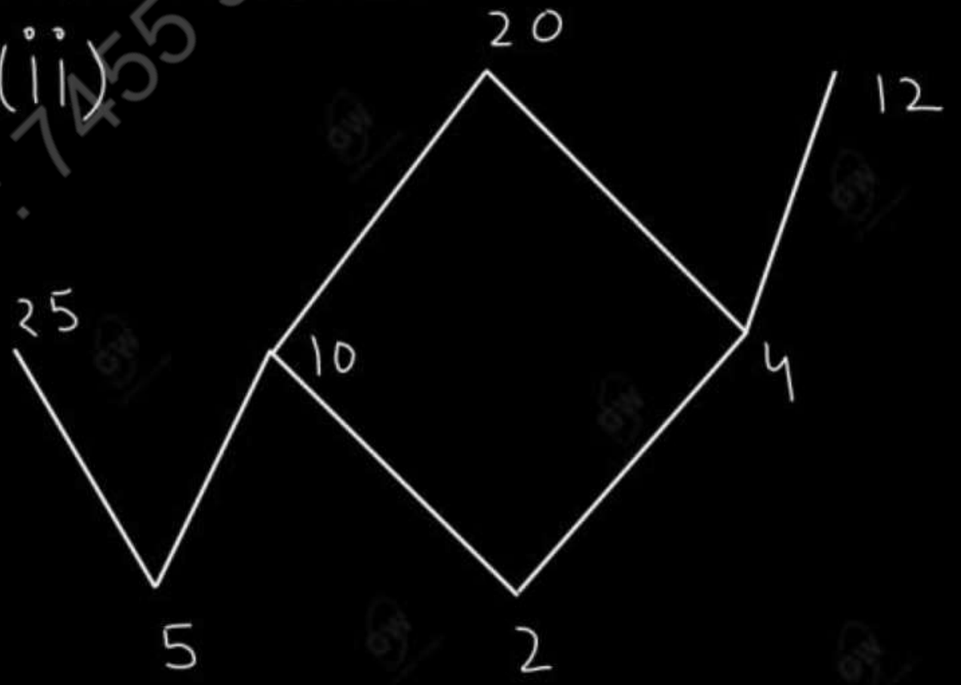
Gateway Classes : 7455 9612 84

Q.1. ***Find all the maximal and minimal elements from the following Hasse diagrams.***

(i)



(ii)



AKTU Full Courses (Paid)

Download Gateway Classes Application

From Google Play store

All Subjects

Link in Description

Gateway Classes : 7455961284

Thank You

DISCRETE STRUCTURES & THEORY OF LOGICS (BCS303)

UNIT -1 : (i) Set Theory & Relations

(ii) POSET & Lattice

Today's Target

- *LATTICE*
- *Properties of Lattice*
- *Distributive Lattice*
- *Bounded Lattice*
- *Complemented lattice*
- PYQ
- DPP

Gateway Classes : 7455 9612 84

Course Details(Paid) : All Subjects

1	Recorded Video Lectures (100 % Syllabus Coverage)
2	Pdf Notes
3	Lecture wise DPP
4	Unit wise set of PYQs

Paid Courses are available in Gateway Classes Application

Link in Description

Gateway Classes : 7455 961 284

Join Semi Lattice

A POSET $(L, \leq)$ is a join semi lattice if $\forall\, a, b \in L$, $a \vee b$ exist in L.

Note: $a \vee b = lub\{a, b\} = sup\{a, b\}$

$a \vee b = a\ join\ b$

lub – least upper bound & sup – supremum

OR

In a POSET if join exist for every pair of elements, then POSET is called Join Semi Lattice

$a \vee b$

Meet Semi Lattice

A POSET $(L, \leq)$ is a meet semi lattice if $\forall\, a, b \in L$, $a \wedge b$ exist in L.

Note: $a \wedge b = glb\{a, b\} = inf\{a, b\}$

$a \wedge b = a\ meet\ b$

glb – greatest lower bound & inf – infimum

OR

In a POSET if meet exist for every pair of elements, then POSET is called Join Semi Lattice

$a \wedge b$

Gateway Classes : 7455 9612 84

Lattice

A POSET $(L, \leq)$ is a lattice if $\forall\, a, b \in L$, $a \vee b$ and $a \wedge b$ exist in L.

Where $a \vee b = lub\{a, b\} = sup\{a, b\}$

$a \wedge b = glb\{a, b\} = inf\{a, b\}$

Note: $a \vee b = a\ join\ b$ and $a \wedge b = a\ meet\ b$

lub – least upper bound & sup – supremum

glb – greatest lower bound & inf – infimum

OR

In a POSET if join and meet exist for every pair of elements, then POSET is called Lattice

OR

A POSET $(L, \leq)$ is a lattice if it is :

(i) Join Semi Lattice

(ii) Meet Semi Lattice

Gateway Classes : 7455 9612 84

Q.1. *Determine whether the following Hasse diagram represent lattice or not*

(i)



lub table

$\vee$	a	b	c	d
a	a	b	c	d
b	b	b	c	d
c	c	c	c	d
d	d	d	d	d

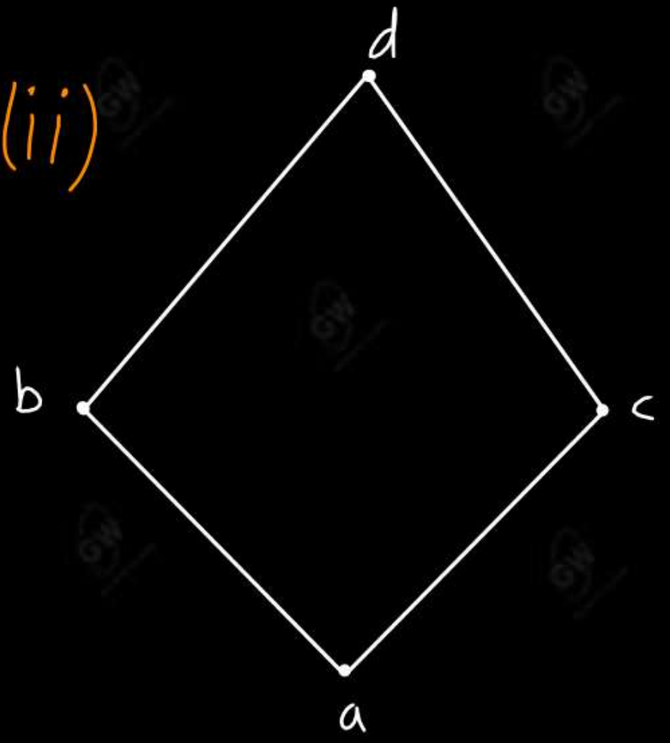
glb table

$\wedge$	a	b	c	d
a	a	a	a	a
b	a	b	b	b
c	a	b	c	c
d	a	b	c	d

Gateway Classes : 7455 9612 84

∴ represent a lattice

(ii)



lub table

$\vee$	a	b	c	d
a	a	b	c	d
b	b	b	d	d
c	c	d	c	d
d	d	d	d	d

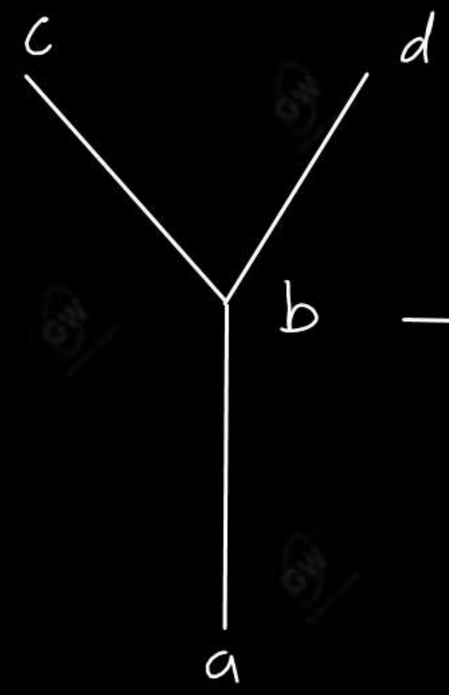
glb table

$\wedge$	a	b	c	d
a	a	a	a	a
b	a	b	a	b
c	a	a	c	c
d	a	b	c	d

It represent a lattice

Gateway Classes : 7455 9612 84

(iii)



lub table

$\vee$	a	b	c	d
a	a	b	c	d
b	b	b	d	d
c	c	d	c	—
d	d	d	—	d

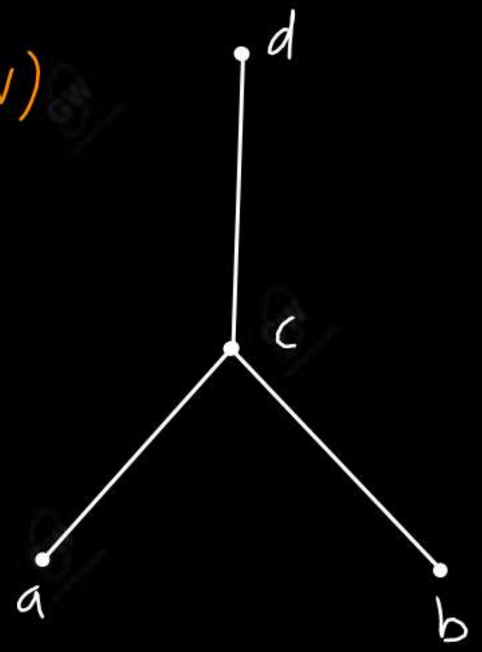
glb table

$\wedge$	a	b	c	d
a	a	a	a	a
b	a	b	b	b
c	a	b	c	b
d	a	b	b	d

It does not represent lattice

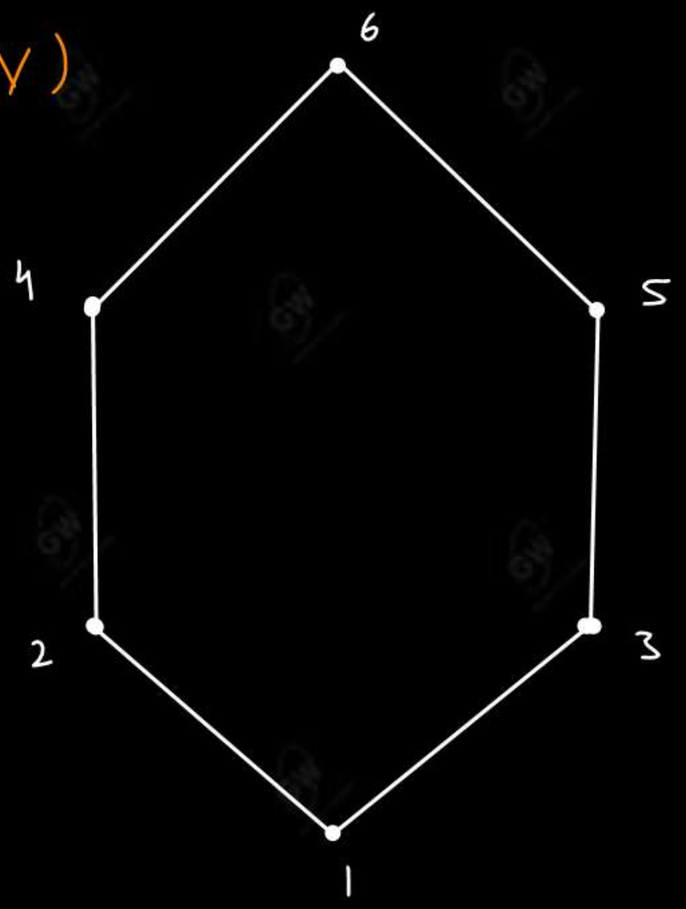
Gateway Classes : 7455 961284

(IV)



Gateway Classes : 7455 9612 84

(Y)



lub table

V	1	2	3	4	5	6
1	1	2	3	4	5	6
2	2	2	6	4	6	6
3	3	6	3	6	5	6
4	4	4	6	4	6	6
5	5	6	5	6	5	6
6	6	6	6	6	6	6

glb table = ?

It represent a lattice

Gateway Classes: 7455 961284

Properties of Lattice

Let (L, ≤) be a Lattice, then the following results holds

(i) Idempotent law : $\forall \ a \in L$

(i) $a \vee a = a$

(ii) $a \wedge a = a$

(ii) Commutative law :- $\forall\ a,\ b \in L$

(i) $a \vee b = b \vee a$

(ii) $a \wedge b = b \wedge a$

(iii) Associative law :- $\forall\ a,\ b, c \in L$

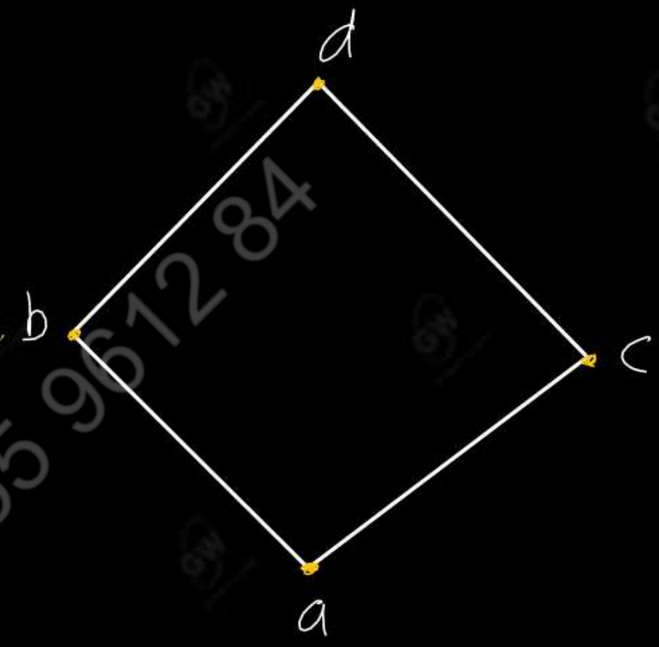
(i) $a \vee (b \vee c) = (a \vee b) \vee c$

(ii) $a \wedge (b \wedge c) = (a \wedge b) \wedge c$

(iv) Absorption law :- for all $a,\ b \in L$

(i) $a \vee (a \wedge b) = a$

(ii) $a \wedge (a \vee b) = a$



Gateway Classes : 7455 9612 84

Distributive Lattice

A lattice (L, ≤) is called distributive lattice if $\forall\ a,\ b, c\ \in L$, it satisfies the following properties

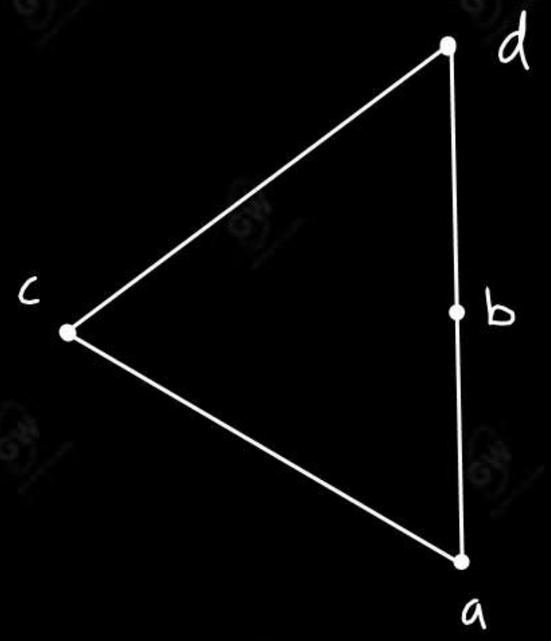
(i) $a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c)$

(ii) $a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c)$

- If the lattice L does not satisfy the above properties, the it is called a non-distributive lattice
- If the lattice L satisfy the above properties, the it is distributive lattice

Gateway Classes : 7455 9612 84

Q.2. Show that the given lattice is distributive



Given lattice is distributive If it satisfy the following properties

(i) $a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c)$

$a \vee a = b \wedge c$

$a = a$

(ii) $a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c)$

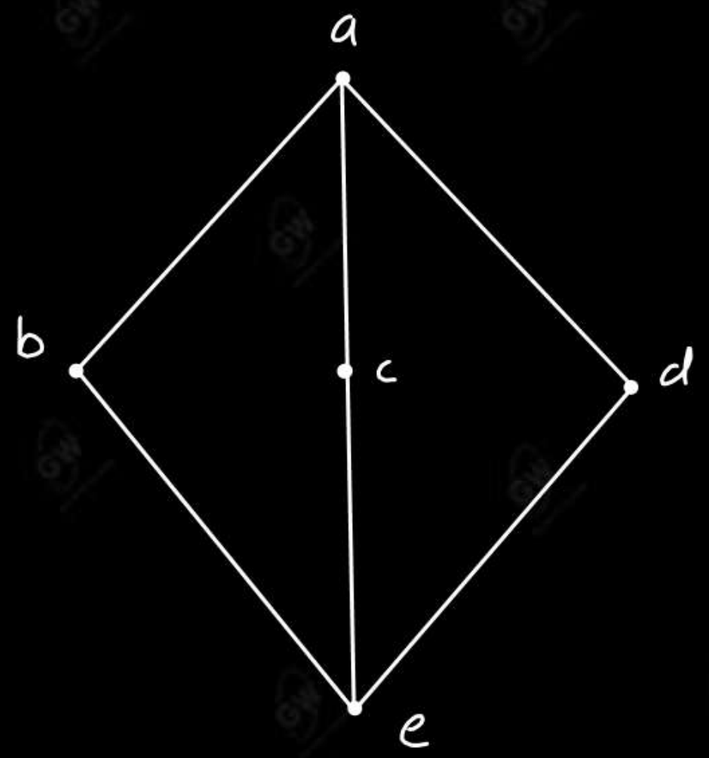
$a \wedge d = a \vee a$

$a = a$

Hence given lattice is distributive

Gateway Classes : 7455 961284

Q.3. Show that the given lattice is not distributive



For b, c, d

(i) $b \vee (c \wedge d) = (b \vee c) \wedge (b \vee d)$

$b \vee e = a \wedge a$

$b = a$

$LHS \neq RHS$

Hence given Lattice is not distributive

Gateway Classes : 7455 9612 84

Theorem :

Let $a, b, c \in L$ where $(L, \leq)$ is a distributive lattice then

$$a \vee b = a \vee c \text{ and } a \wedge b = a \wedge c \Rightarrow b = c$$

Given

$(L, \leq)$ is a distributive lattice

Where

$a \vee b = a \vee c$

$a \wedge b = a \wedge c$

To Prove

$b = c$

Proof

$b = b \vee (b \wedge a)$ By Absorption law

$= b \vee (a \wedge b)$ By commutative law

$= b \vee (a \wedge c)$ Given

$= (b \vee a) \wedge (b \vee c)$ By Distributive law

$= (a \vee b) \wedge (c \vee b)$ By commutative law

$= (a \vee c) \wedge (c \vee b)$ Given

$= (c \vee a) \wedge (c \vee b)$ By commutative law

Gateway Classes : 7455 9612 84

$= c \vee (a \wedge b)$ By distributive law

$= c \vee (a \wedge c)$ Given

$= c \vee (c \wedge a)$ By commutative law

$= c$ By Absorption law

$\therefore$ $\boxed{b = c}$ Hence proved

Gateway Classes: 7455 9612 84

Bounded Lattice :

A lattice $(L, \leq)$ is called a bounded lattice if it has a greatest element 1 and a least element 0.

Properties of Bounded Lattice :

If $(L, \leq)$ is a bounded lattice then for any element $a \in L$ we have

(i) $a \vee 1 = 1$

(ii) $a \wedge 1 = a$

(iii) $a \vee 0 = a$

(iv) $a \wedge 0 = 0$

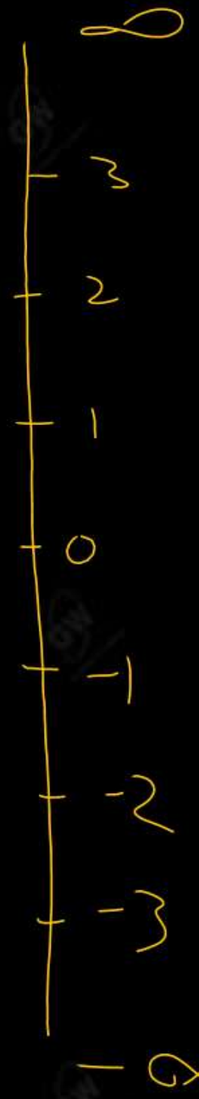
Note:- Every finite lattice is bounded

$A \cup U = U$

$A \cap U = A$

$A \cup \phi = A$

$A \cap \phi = \phi$



Infinite Lattice

Gateway Classes: 7455961284

Complement of an element in a lattice :

Let $(L, \leq)$ is a bounded lattice with least element 0 and greatest element 1, then $a' \in L$ is called a complement of $a \in L$ if

(i) $a \vee a' = 1$

(ii) $a \wedge a' = 0$

$$\therefore \ a' = a$$

Note:- (1) It is not necessary that every element $a \in L$ has a complement

(2) An element $a \in L$ have more than one complement

(3) $1' = 0$ and $0' = 1$

Complemented lattice :

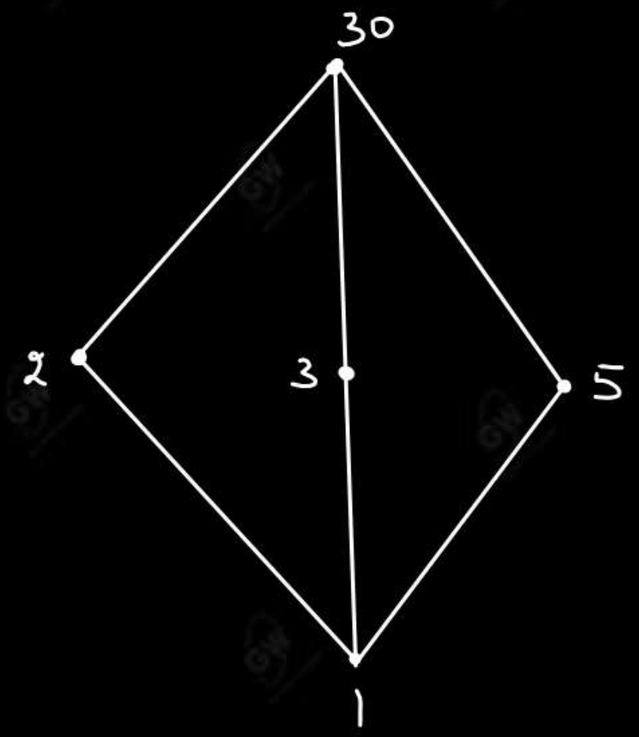
A Lattice $(L, \leq)$ is called a complemented lattice if

(i) if L is bounded

(ii) Every element $a \in L$ has a complement

Gateway Classes : 7455 9612 84

Q.4. *Let* $A = \{1, 2, 3, 5, 30\}$ *and* $a \leq b$ *iff a divided b . Then find the complement of 2.*



(i) $2 \vee 3 = 30$ $\quad 2 \wedge 3 = 1$

(ii) $2 \vee 5 = 30$ $\quad 2 \wedge 5 = 1$

Hence

3 and 5 are the complement of 2

$2' = 3$

$2' = 5$

Gateway Classes : 7455 9612 84

Theorem:

If a and b are the elements in a bounded distributive lattice L and if 'a' has complement a' then

(i) $(a')' = a$

(ii) $(a \vee b)' = a' \wedge b'$ **DE Morgan's law**

(iii) $(a \wedge b)' = a' \vee b'$ **DE Morgan's law**

(iv) $a \vee (a' \wedge b) = a \vee b$

(v) $a \wedge (a' \vee b) = a \wedge b$

a a'

$$a \vee a' = 1$$

(ii) complement of $a \vee b = a' \wedge b'$

By property of complement

$(a \vee b) \vee (a' \wedge b') = 1$ and $(a \vee b) \wedge (a' \wedge b') = 0$

$= [(a \vee b) \vee a'] \wedge [(a \vee b) \vee b']$ By Distributive law

$= [(b \vee a) \vee a'] \wedge [a \vee (b \vee b')]$ By commutative law

$= [b \vee (a \vee a')] \wedge [a \vee 1]$

Gateway Classes : 7455 961284

$= (b \vee 1) \wedge (a \vee 1)$

$= 1 \wedge 1$

$= 1$

and

$(a \vee b) \wedge (a' \wedge b') = 0$

$= (a \vee b) \wedge (a' \wedge b')$

$= (a' \wedge b') \wedge (a \vee b)$ By commutative law

$= [(a' \wedge b') \wedge a] \vee [(a' \wedge b') \wedge b]$ By Distributive law

$= [(b' \wedge a') \wedge a] \vee [a' \wedge (b' \wedge b)]$ By commutative law

$= [b' \wedge (a' \wedge a)] \vee [a' \wedge 0]$

$= (b' \wedge 0) \vee (a' \wedge 0)$

$= 0 \vee 0$

$= 0$

LHS = RHS

Hence

$(a \vee b)' = a' \wedge b'$

Gateway Classes : 7455961284

(iv) $a \vee (a' \wedge b) = a \vee b$

LHS

$= a \vee (a' \wedge b)$

$= (a \vee a') \wedge (a \vee b)$ By Distributive law

$= 1 \wedge (a \vee b)$

$= a \vee b$

$=$ RHS

Hence proved

(v) $a \wedge (a' \vee b) = a \wedge b$

LHS

$= a \wedge (a' \vee b)$

$= (a \wedge a') \vee (a \wedge b)$ By Distributive law

$= 0 \vee (a \wedge b)$

$= a \wedge b$

$=$ RHS

Hence proved

Gateway Classes: 7455 9612 84

THEOREM

Show that in a distributive Lattice, if an element has a complement then this complement is unique.

OR

Prove that for a bounded lattice L, the complements are unique if they exist.

Let $(L, \leq)$ be a bounded distributive lattice

Let $a \in L$ having two complements b and c

Then show that

$$b = c$$

Since b and c be the complement of a then

$a \vee b = 1$ $\quad a \wedge b = 0$

$a \vee c = 1$ $\quad a \wedge c = 0$

Proof

$b = (b \wedge 1)$

$b = b \wedge (a \vee c)$ Given

$b = (b \wedge a) \vee (b \wedge c)$ Distributive law

$b = (a \wedge b) \vee (b \wedge c)$ commutative law

$b = 0 \vee (b \wedge c)$ Given

Gateway Classes : 7455 961 284

$b = (a \wedge c) \vee (b \wedge c)$

$b = (a \vee b) \wedge c$

$b = 1 \wedge c$

$\boxed{b = c}$

Hence proved

Gateway Classes : 7455 9612 84

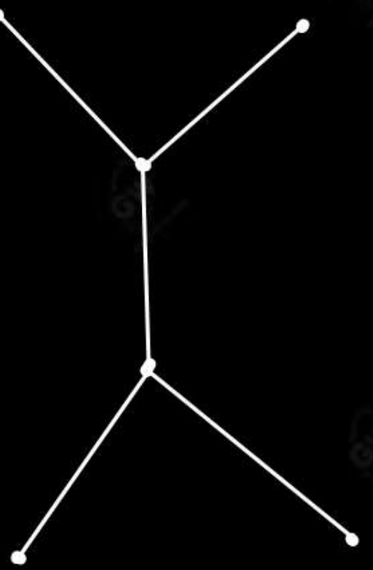
Discrete Mathematics by Gulshan Sir

DPP- 10

Topic : *LATTICE*

Q.1. ***Determine whether the following Hasse diagram represent lattice or not***

(I)



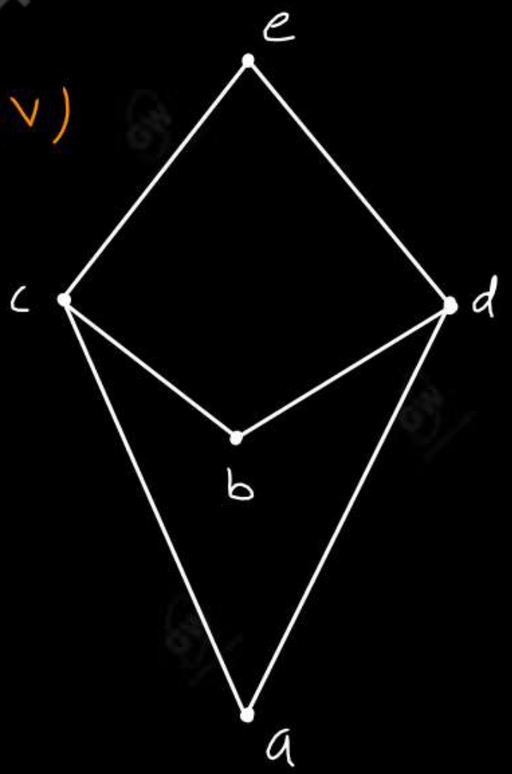
(II)



(III)

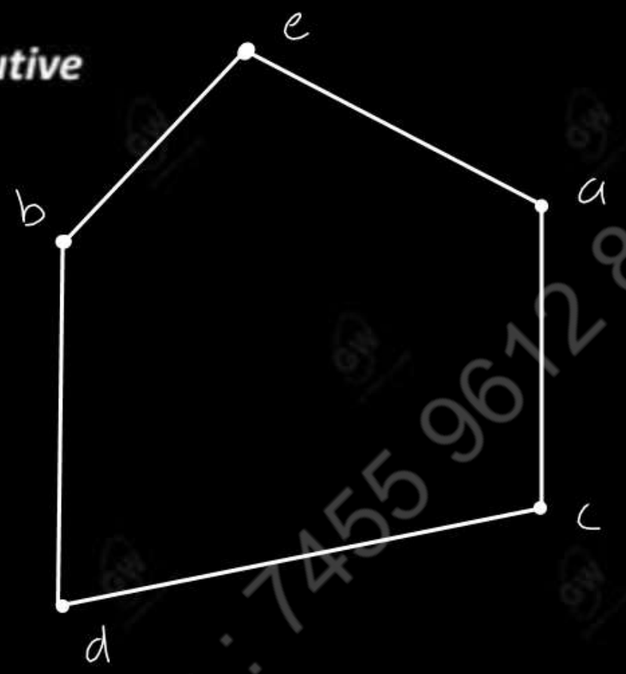


(IV)

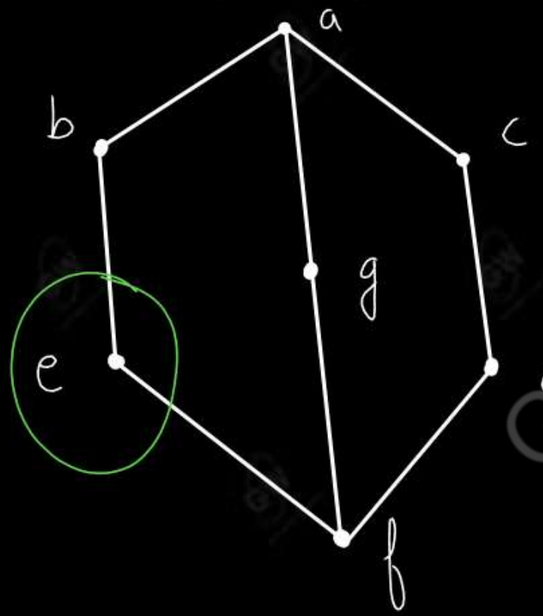


Gateway Classes : 7455 9612 84

Q.2. *Show that the given lattice is not distributive*



Q.3. *In the Lattice defined by the Hasse Diagram. How many complements does the element 'e' have ?*



Gateway Classes : 7455 9612 84

GATEWAY CLASSES

AKTU Full Courses (Paid)

Download Gateway Classes Application

From Google Play store

All Subjects

Link in Description

Gateway Classes : 7455961284

Thank You

GATEWAY CLASSES

DISCRETE STRUCTURES & THEORY OF LOGICS (BCS303)

UNIT -1 : (i) Set Theory & Relations

(ii) POSET & Lattices

Today's Target

- *Modular LATTICE*
- *Complete Lattice*
- *Sub Lattice*
- *Isomorphic Lattice*
- PYQ
- DPP

Gateway Classes : 7455 9612 84

Course Details(Paid) : All Subjects

1	Recorded Video Lectures (100 % Syllabus Coverage)
2	Pdf Notes
3	Lecture wise DPP
4	Unit wise set of PYQs

Paid Courses are available in Gateway Classes Application

Link in Description

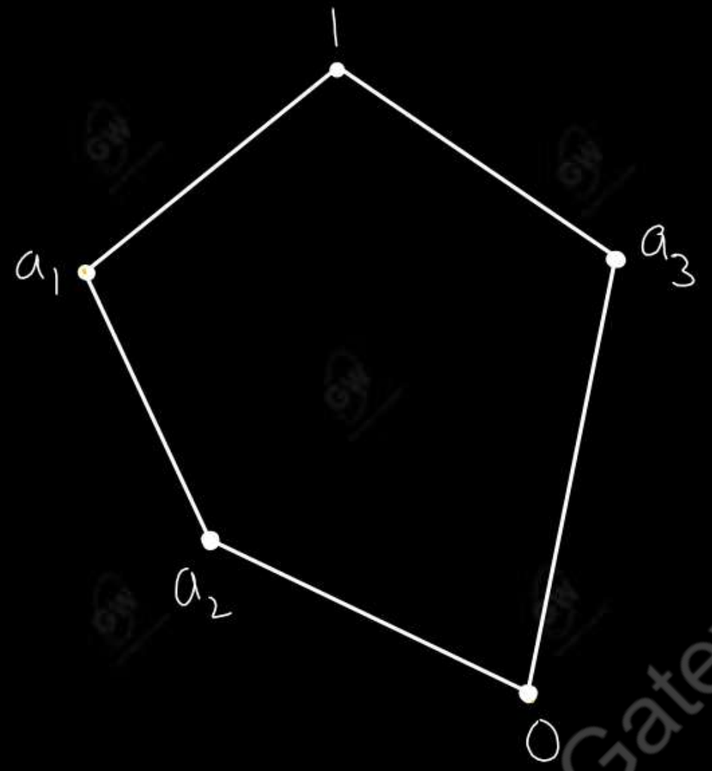
Gateway Classes : 7455 961 284

Modular Lattice

A lattice $(L, \leq)$ is said to be modular lattice if $\forall\, a, b, c \in L$

$$a \vee (b \wedge c) = (a \vee b) \wedge c \quad \text{whenever } a \leq c$$

Q.1. Show that that pentagonal lattice given below is not modular



consider a, b, c in such a way that $a \leq c$

- ✓ $a = a_2$
- $b = a_3$
- ✓ $c = a_1$

Now

$$a \vee (b \wedge c) = a_2 \vee (a_3 \wedge a_1)$$

$$= a_2 \vee 0$$

$$= a_2$$

Gateway Classes : 7455 9612 84

$$(a \vee b) \wedge c = (a_2 \vee a_3) \wedge a_1$$

$$= 1 \wedge a_1$$

$$= a_1$$

$$\therefore \boxed{A \vee (b \wedge c) \neq (a \vee b) \wedge c}$$

Hence, given lattice is not a modular lattice

Gateway Classes : 7455 9612 84

Theorem

Prove that every distributive lattice is modular.

Let $(L, \leq)$ be a distributive lattice and $a, b, c \in L$ such that

$a \leq c$ then

$$a \vee c = c \quad \text{---} \quad (1)$$

By Distributive law

$$a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c)$$

By using (1)

$$\boxed{a \vee (b \wedge c) = (a \vee b) \wedge c}$$

Hence, every distributive lattice is modular

Note - Modular lattice may or may not a distributive lattice

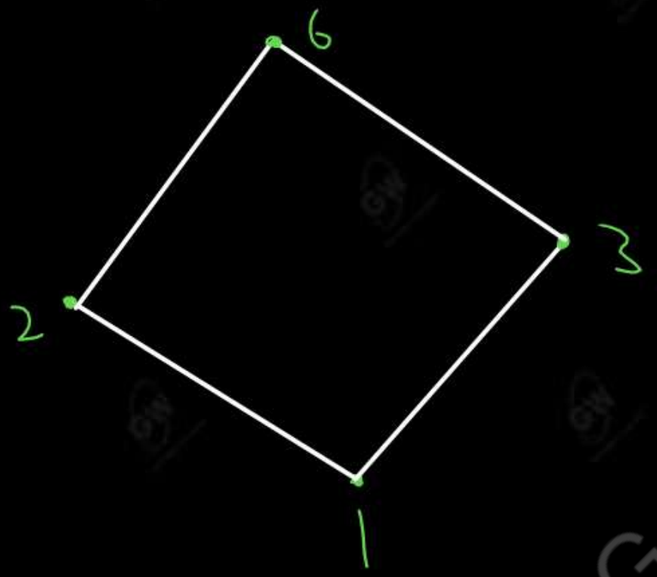
Gateway Classes: 7455961284

Complete lattice :-

A Lattice $(L, \leq)$ is complete, if every non-empty subset of L has a least upper bound (sup) and greatest lower bound (inf) in L.

Example : Show that $(D_6, |)$ is a complete lattice

$D_6 = \{1, 2, 3, 6\}$



Subsets

$A_1 = \{1, 2\}$

$A_2 = \{1, 2, 6\}$

$A_3 = \{2, 6\}$

$A_4 = \{3, 6\}$

Hence, the given lattice is a complete lattice

Gateway Classes : 7455 9612 84

D_n – Lattice

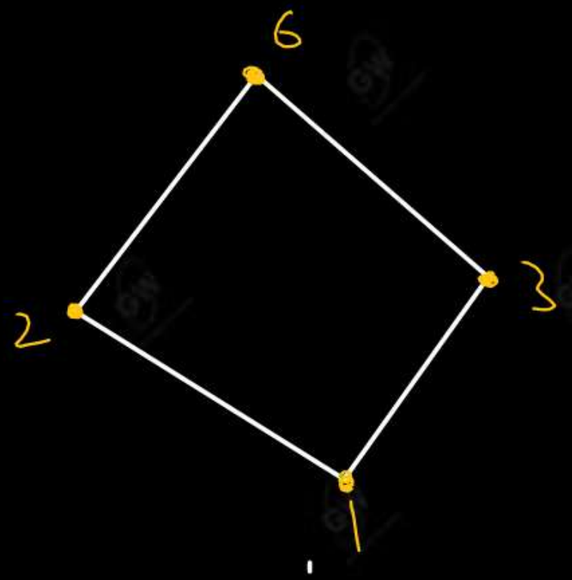
Let n be a positive integer and let D_n be the set of all +ve divisors on n

Then D_n is a lattice under the relation of Divisibility

It is denoted by (D_n, I)

Q.1. *Prove that (D_6, I) is a lattice .*

$D_6 = \{1, 2, 3, 6\}$



lub table

V	1	2	3	6
1	1	2	3	6
2	2	2	6	6
3	3	6	3	6
6	6	6	6	6

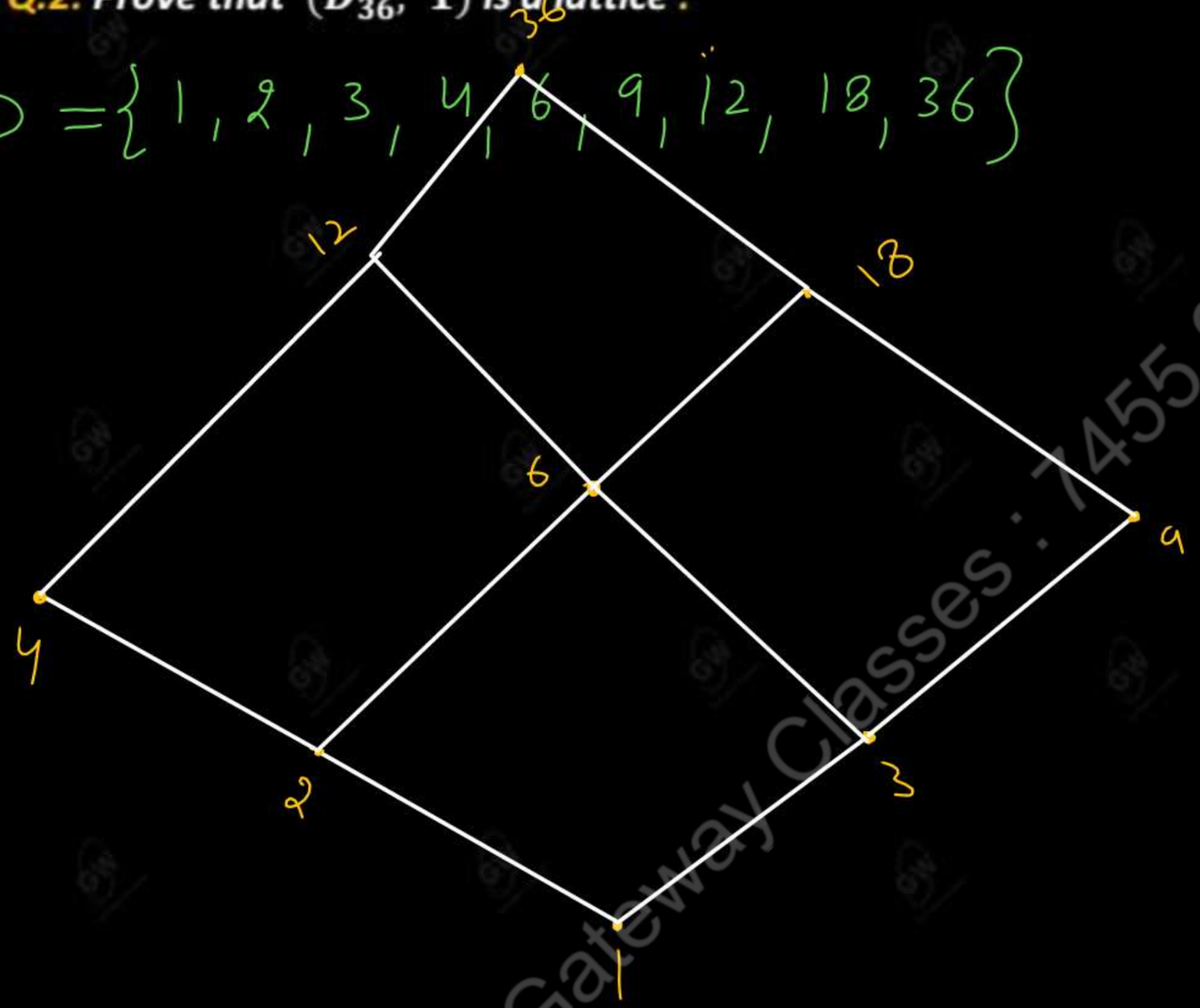
glb table

Λ	1	2	3	6
1	1	1	1	1
2	1	2	1	2
3	1	1	3	3
6	1	2	3	6

Gateway Classes : 7455 9612 84

Q.2: *Prove that* $(D_{36}, 1)$ *is a lattice .*

$D = \{1, 2, 3, 4, 6, 9, 12, 18, 36\}$



lub table

V	1	2	3	4	6	9	12	18	36
1									
2									
3									
4									
6									
9									
12									
18									
36									

Gateway Classes : 7455 961284

glb table

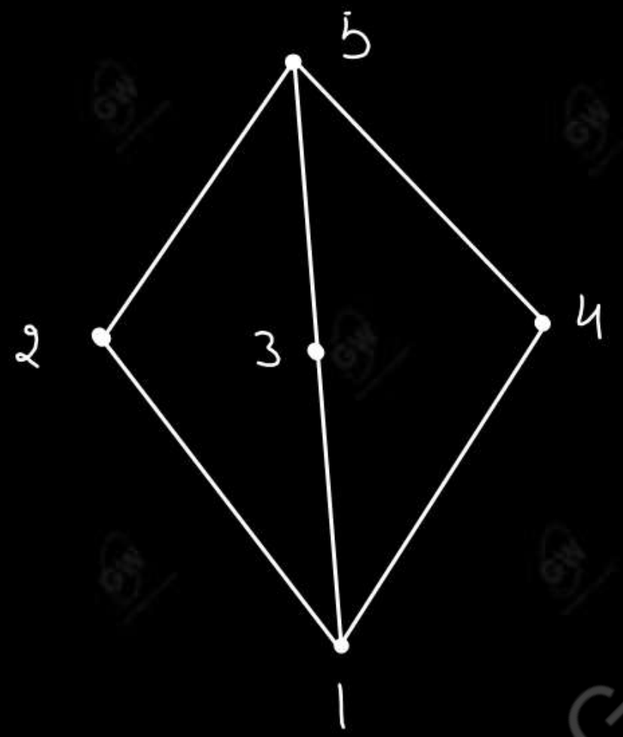


Gateway Classes : 7455 9612 84

Sublattice :-

A non- empty subset M of Lattice $(L, \leq)$ is said to be a sub-lattice of L if M is closed with respect to join $(\vee)$ and meet $(\wedge)$ i.e $a, b \in M \Rightarrow a \vee b \in M$ and $a \wedge b \in M$

Q.3. *Consider the lattice $L = \{1, 2, 3, 4, 5\}$ given below Determine all sub lattices with three or more elements.*



$L_1 = \{1, 2, 5\}$

$L_2 = \{1, 3, 5\}$

$L_3 = \{1, 4, 5\}$

$L_4 = \{1, 2, 5, 3\}$

$L_5 = \{1, 4, 5, 3\}$

$L_6 = \{1, 2, 5, 4\}$

Gateway Classes: 7455 961284

Isomorphic Lattice

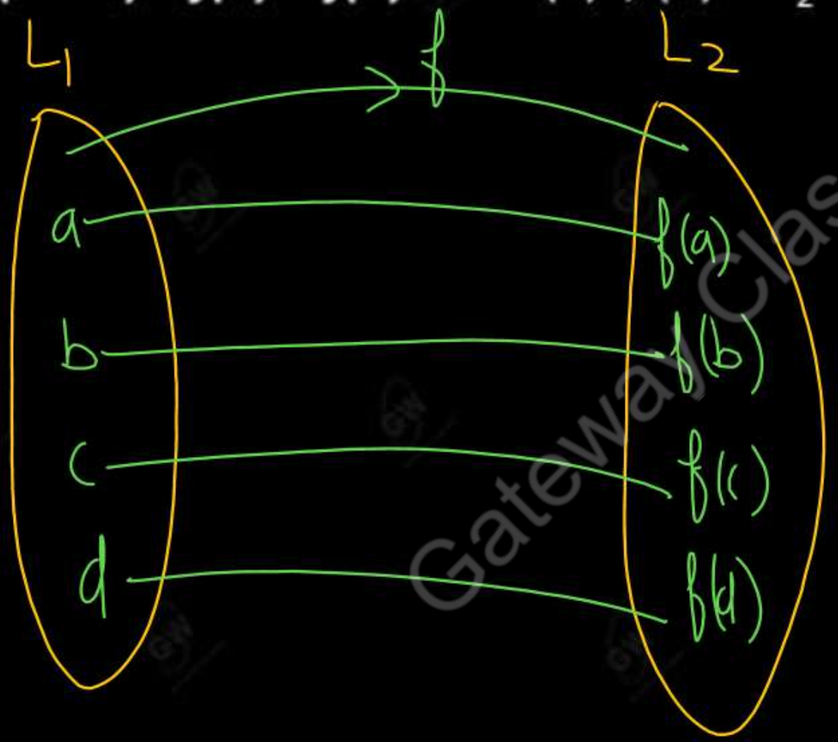
$f: L_1 \longrightarrow L_2$

Two lattice L_1 and L_2 are isomorphic if

(i) There exist one-to-one correspondence between vertices and edges (Bijection)

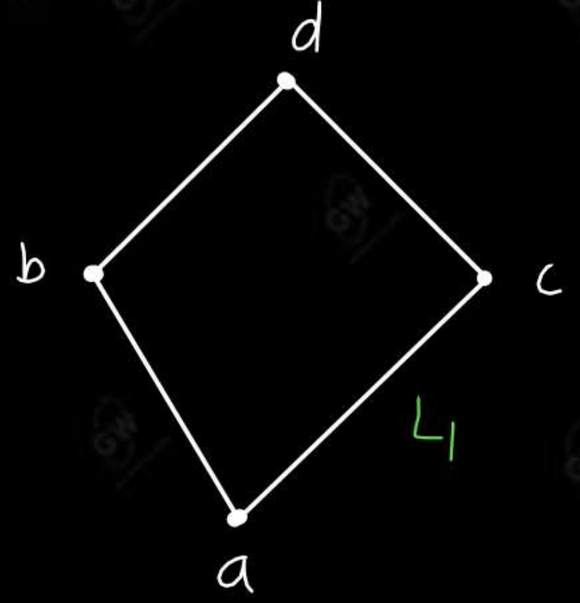
(ii) $f(a \vee b) = f(a) \vee f(b) \quad \forall\, a, b \in L_1$

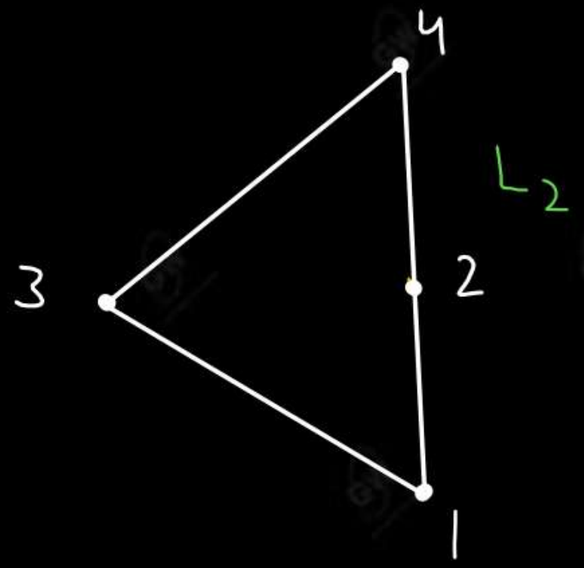
$f(a \wedge b) = f(a) \wedge f(b) \quad \forall\, f(a), f(b) \in L_2$



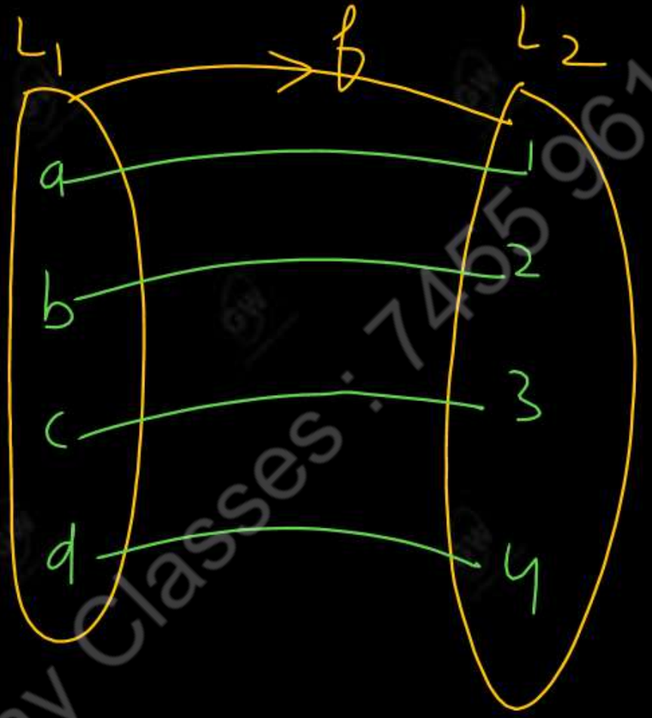
Gateway Classes : 7455 9612 84

Q.4. *Determine whether the lattice given below are isomorphic or not.*





$f : L_1 \longrightarrow L_2$



such that

$f(a) = 1$

$f(b) = 2$

$f(c) = 3$

$f(d) = 4$

Gateway Classes : 7455 9612 84

$f(a \vee b) = f(b) = 2$ — ①

$f(a) \vee f(b) = 1 \vee 2 = 2$ — ②

From ① and ②

$$\boxed{f(a \vee b) = f(a) \vee f(b)}$$

$f(a \wedge b) = f(a) = 1$ — ③

$f(a) \wedge f(b) = 1 \wedge 2 = 1$ — ④

From ③ and ④

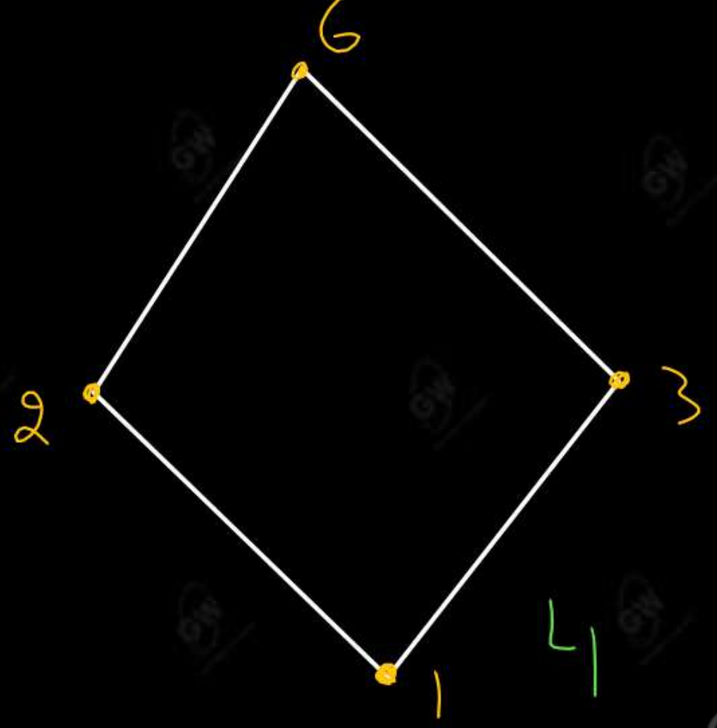
$$\boxed{f(a \wedge b) = f(a) \wedge f(b)}$$

Hence L_1 and L_2 are isomorphic lattices

GW GATEWAY CLASSES
Gateway Classes : 7455961284

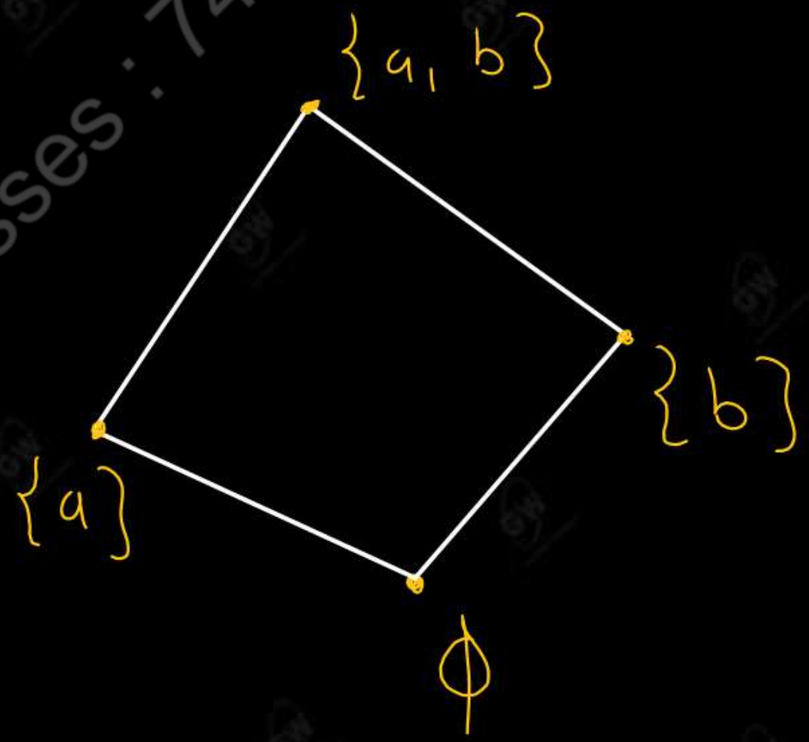
Q.5. Let L_1 be the lattice defined as D_6 and L_2 be the lattice $(P(S), \subseteq)$, where P(S) is the power set defined on set S = {a, b} justify that the two lattices are isomorphic.

$D_6 = \{1, 2, 3, 6\}$

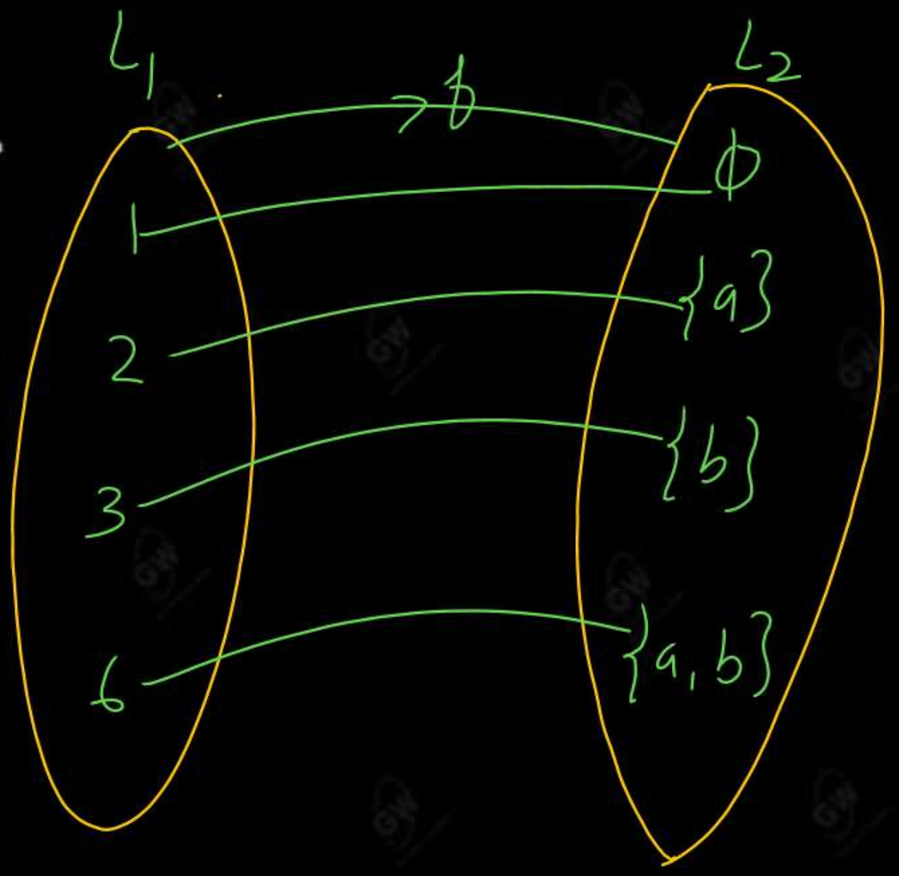


$L_2 = (P(S), \subseteq)$

$P(S) = \{\phi, \{a\}, \{b\}, \{a, b\}\}$



Gateway Classes : 7455 96 12 84



$f(1 \vee 2) = f(2) = \{a\}$

$f(1) \vee f(2) = \phi \vee \{a\} = \{a\}$

$\therefore$ $\boxed{f(1 \vee 2) = f(1) \vee f(2)}$

$f(1 \wedge 2) = f(1) = \phi$

$f(1) \wedge f(2) = \phi \wedge \{a\} = \phi$

$\boxed{f(1 \wedge 2) = f(1) \wedge f(2)}$

Gateway Classes: 7455961284

Discrete Mathematics by Gulshan Sir

DPP- 16

Topic : *LATTICE*

Q.1. *Prove that* $(D_{20}, 1)$ *is a lattice*

Q.2. *Prove that* $(D_{30}, 1)$ *is a lattice*

Q.3. *Determine all the sub lattices of* D_{30} *that contain four elements.*

Gateway Classes : 7455 9612 84

GATEWAY CLASSES

AKTU Full Courses (Paid)

Download Gateway Classes Application

From Google Play store

All Subjects

Link in Description

Gateway Classes : 7455961284

Thank You

GATEWAY CLASSES